Improving variable selection properties with data integration and transfer learning

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Abstract

Sparse high-dimensional signal recovery is only possible under certain conditions on the number of parameters, sample size, signal strength and underlying sparsity. We study how these mathematical limits can be pushed by leveraging external information on the likelihood of variables to be associated with the outcome. As an important particular case, we consider a transfer learning framework where the set of variables selected in source datasets is transferred to guide variable selection in a target dataset. We introduce a family of penalized linear regression methods motivated by connections to Bayesian variable selection where the penalties depend on the external information. We show that they attain variable selection consistency where otherwise it would not be possible, or attain it at a faster rate. We first precisely quantify the gains that are achievable for ideal penalties set by an oracle. Subsequently, we propose computationally fast algorithms for the incorporation of external information and transfer learning that do not require an oracle and are inspired by empirical Bayes techniques. We prove that our algorithms have better selection properties than selection methods not leveraging external information. We illustrate our algorithms on simulated data and an example where findings from mice experiments are transferred to learn gene expression patterns in humans.

Keywords: high-dimensional statistics, variable selection, ℓ_0 penalty, empirical Bayes, transfer learning.

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♡ [David: General comment: JASA is less theory-oriented and more methods-oriented than AOS or Bernoulli. In the latter it's ok to spent a substantial part of the paper explaining the reasoning that leads to the main conclusions, but at JASA the focus is more on the main conclusions and less on the exact proof techniques / chain of reasoning. That is, a JASA paper should make clear claims that don't require the reader to become intimately familiar with the proof details or arguments. Some parts of the paper feel a bit too "technical" at the moment. I didn't do much to edit these parts yet, instead I focused on clarifications and shortening things a bit.]

♡ [David: The new intro is very good and I like the way how you emphasized transfer learning and outlined an associated algorithm (with some caveats explained below). This said, it feels like the intro could be streamlined. It goes a bit back and forth, certain narratives/ideas appear in multiple times scattered across the intro instead of being fully outlined in a single place. I can help polish this once Paul and Piotr have gone through another iteration.]

1 Introduction

In a data-rich world, basing statistical inference on only a single source of information increasingly seems like a missed opportunity, and many have answered statistical questions by leveraging multiple datasets. When integrating multiple sources, there is often heterogeneity (or non-exchangeability) in variables in the sense that one has information, external to the dataset of interest, on a varying likelihood that each variable is truly associated with an outcome. This may occur when working with *multimodal* data, i.e. featuring different types of variables **?**, where one block of variables may be expected to be less sparse than another, e.g. clinical vs genomic markers in a biological context. Another example is when meta-data is available on the variables themselves, e.g functional annotations on genes. A setting of particular interest is transfer learning which can broadly be defined as the problem of improving learning for given *target* task and dataset by using knowledge acquired from similar *source* tasks and datasets. In variable selection, transfer learning refers to recognizing that a set of variables being identified in the source data provides external information on their likelihood of being truly associated with the outcome in the target dataset. We refer to this task as structural transfer learning, in contrast to parameter estimation and prediction tasks more commonly considered in transfer learning.

In high-dimensional settings, leveraging external information is particularly attractive. From a theoretical perspective, the inherent scarcity of information provided by a single dataset implies that strong assumptions are required to carry out accurate inference. In structural learning problems such as variable selection, those assumptions regard sparsity and signal strength and, although mathematically necessary, they can be too strong in practice (see Giannone et al. (2021) for a discussion in the context of social sciences). From a practical perspective, numerous works observed improved inference when employing external information to guide structural learning (Boulesteix et al. 2017, Stingo et al. 2011, Chen et al. 2021). Despite this empirical success, a theoretical framework explaining precisely why and how leveraging external information may deliver improved structural learning is not currently available. In this work, we investigate how, in linear regression, integrating external information and structural transfer learning, allows pushing the mathematical conditions under which consistently recovering the true subset of variables associated with the outcome is possible, and improving the corresponding rates. Consistency is understood here as recovering the true subset with probability going to 1 as the number of samples n and possibly the dimension p go to infinity.

To make ideas concrete, consider variable selection in the Gaussian linear regression

$$\mathbf{y} \;=\; \mathbf{X}\boldsymbol{\beta}^* + \epsilon, \qquad \epsilon \sim N(0, \sigma^2 I_n), \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{n\times p}$, $\mathbf{y} \in \mathbb{R}^n$, $\sigma > 0$, and $\boldsymbol{\beta} \in \mathbb{R}^p$ are the data-generating parameters. Our results can be easily extended to non-linear regression, where $\mathbf{X}$ is given by a suitable basis such as splines. Penalized log-likelihood methods for variable selection are based on optimizing the log-likelihood plus a penalty term driven by the ℓ_q "norm" of the estimated $\hat{\boldsymbol{\beta}}$ for $q \in [0,1]$ (Tibshirani 1996, Bertsimas et al. 2016). Here we focus on ℓ_0 penalties because they possess theoretically superior properties for model recovery than other types of penalties (Wainwright 2010, Gao & Aragam 2025), and our goal is to study how external information can improve upon these best known properties. Computational advances have also made ℓ_0 problems much more tractable: optimization methods can solve them exactly for p in the thousands (Bertsimas & Van Parys 2020) and Markov Chain Monte Carlo (MCMC) have linear cost in p under sparsity conditions (Yang et al. 2016, Zhou et al. 2022). There is also a direct connection between ℓ_0 penalties and Bayesian variable selection based on posterior model probabilities (Schwarz 1978, Chen & Chen 2008, Rossell 2022). Under mild regularity conditions, a choice of ℓ_0 penalty corresponds to a choice of prior probability for variable inclusion. This connection converts ℓ_0 penalization into a natural and intuitive way to encode external information on the likelihood of variables to be truly associated with an outcome.

We consider a variable selection framework where ℓ_0 penalties may depend on vector of external information $\boldsymbol{z} = (z_1, \ldots, z_p) \in \mathbb{R}^p$ that partitions the p variables into b blocks, denoted by $B_j \subset \{1, \ldots, p\}$, $j = 1, \ldots, b$. For example, in structural transfer learning, z may be an indicator of whether variables were found to be promising in related source datasets. When $\boldsymbol{z}$ is informative for variable selection, key characteristics such as the proportion of truly active variables may vary across blocks, hence one allows the ℓ_0 penalty to differ across blocks. We derive the variable selection properties of such informed ℓ_0 penalties in general and in the context of structural transfer learning algorithm, and compare them to standard ℓ_0 penalties.

Before proceeding, we review some selected literature. External information has been used to guide variable selection in many applications. For instance, Stingo et al. (2011) and Cassese et al. (2014) proposed Bayesian variable selection for gene expression where prior inclusion probabilities depend on biological knowledge and meta-variables. Chen et al. (2021) predicted disease outcomes with ℓ_1 penalties that depend on functional annotation categories. A series of works has discussed methodology, but not theory, for external-information dependent penalization or prior inclusion probabilities. In the Bayesian literature, van de Wiel et al. (2019) suggested an empirical Bayes approach to regress prior inclusion probabilities on external data. Velten & Huber (2021) proposed a variational Bayes approximation for regression with external information-dependent prior inclusion probabilities. Recently, Busatto & van de Wiel (2023) considered a horseshoe prior that depends on external information. In the frequentist literature, in multi-omics data Boulesteix et al. (2017) let the ℓ_1 penalty depend on the modality that each variable belongs to. Zeng et al. (2020) and van Nee et al. (2023) proposed empirical Bayes approaches to setting externally-informed ℓ_1 and elastic-net penalties respectively. See van de Wiel & van Wieringen (2024) for a review.

The literature on the statistical properties of transfer in linear regression has focused on the transfer of parameter estimates unlike the transfer of the set of selected variables that we consider here. Jiang et al. (2016) and Li et al. (2021) showed improvements in convergence rates in prediction and estimation with ℓ_1 penalties on the discrepancy with source estimates. In a Bayesian setting, Lai et al. (2024) showed improvements in posterior contraction rates of parameters with horseshoe priors centered at a weighted average of source-specific estimates.

The paper is organized as follows. In Section 2 we introduce externally-informed ℓ_0 penalties, discuss their connection to Bayesian variable selection.

Section 3 studies the properties of our informed ℓ_0 penalties. We first show that an oracle, knowing the true pattern of sparsity and signal strength, may take advantage of the external information so that variable selection consistency is either attained where otherwise it would not be possible, or is attained at a faster rate. These oracle results describe the maximum gains achievable by leveraging external information.

In Section 4, we propose a, two-step, data-based, empirical Bayes procedure to set informed ℓ_0 penalties. We show that the procedure improves variable selection properties without requiring an oracle.

In Section 5, we study a transfer learning algorithm based on the empirical Bayes procedure of Section 4. We show that our algorithm has better selection properties than any algorithm ♠ Paul: "any" sounds a lot but David wrote that that does not transfer information, and that it is robust to so-called *negative transfer*, a situation where transfer learning worsens the performance in the target. To our knowledge, this is the first theoretical analysis of transfer learning for variable selection, as opposed to parameter estimation.

Finally, Section 6 provides numerical illustrations using simulations and a gene expression study where one wishes to transfer findings from mouse models to humans.

Before proceeding, we lay out notation. The parameters of interest in the target regression are denoted by $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\boldsymbol{\beta}^*$ denotes their true values. The set of variable indices is $V = \{1, \ldots, p\}$. For any $A \subseteq V$ and any vector $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{x}_A$ denotes the subvector of $\mathbf{x}$ with entries corresponding to indices in A. For any matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, $\mathbf{X}_A$ denotes the submatrix of $\mathbf{X}$ obtained by selecting the *columns* with indices in A. $S = \{i \in V : \ \beta_i^* \neq 0\}$ denotes the true support of $\boldsymbol{\beta}^*$, its size is $s = |S|$, and hence $p - s$ is the number of truly inactive parameters. The set of candidate models is denoted by $\mathcal{M}$ and is given by subsets $M \subseteq V$. Denote by $\beta^*_{\min} = \min_{i \in S} |\beta_i^*|$ the smallest true signal. The set V is partitioned into disjoint blocks $B_j \subseteq V$ for $j = 1, \ldots, b$, where b is fixed. $S_j = \{i \in B_j : \ \beta_i^* \neq 0\}$ denotes the set of active parameters in block B_j, $s_j = |S_j|$ its size, $p_j - s_j$ the number of inactive parameters in the block, and $\beta^*_{\min,j} = \min_{i \in S_j} |\beta_i^*|$. We denote $\mathfrak{B} = \{\boldsymbol{\beta} \in \mathbb{R}^p :$

for all $j \in \{1, \ldots, b\}$ and $i \in B_j$, $|\beta_i| \geq \beta^*_{\min,j}$ if $i \in S_j$ and $\beta_i = 0$ otherwise$\}$ the set of possible data-generating $\boldsymbol{\beta}^*$.

Given sequences $f(n) > 0$ and $g(n) > 0$, $f(n) = O(g(n))$ means that there exists a constant $c < \infty$ such that $f(n) \leq cg(n)$ for all $n \geq n_0$ and some fixed n_0, $f(n) = o(g(n))$ means that $\lim_{n\to\infty} f(n)/g(n) = 0$, and $f(n) = \Theta(g(n))$ means that $f(n) = O(g(n))$ and $g(n) = O(f(n))$. For any set A, A^C denotes the complement of A.

2 Externally-informed penalties and transfer learning

We describe externally-informed ℓ_0 penalization and their connection to Bayesian variable selection (Section 2.1) We discuss next general assumptions used across results (Section 2.2).

2.1 Externally-informed penalization

Consider a set of candidate models $\mathcal{M}$ given by subsets $M \subseteq V$ such that $|M| \leq n$ and their parameter spaces

$$\mathcal{L}_M \ := \ \{\beta \in \mathbb{R}^p : \beta_i = 0 \text{ if } i \notin M\}.$$

A standard ℓ_0 procedure with penalty $\eta(M) = \kappa|M|$ for some $\kappa > 0$ selects the model

$$\hat{S}^{sd} = \arg\max_{M\in\mathcal{M}} \left\{ \max_{\beta\in\mathcal{L}_M} \ell(\mathbf{y};\boldsymbol{\beta}) \ - \ \kappa|M| \right\}, \tag{2}$$

where $\ell(\mathbf{y};\boldsymbol{\beta})$ is the log-likelihood function. ♠ Paul: I added $|M| \leq n$ for all $M \in \mathcal{M}$

A popular Bayesian variable selection framework are discrete spike-and-slab priors that feature, for each variable i, a binary indicator $m_i = I(\beta_i \neq 0)$ for the inclusion of variable i in the model. Each model $M \in \mathcal{M}$ corresponds then to a particular value of $(m_1, \ldots, m_p)$. The joint prior on parameters and models is

$$p(\boldsymbol{\beta}, M \mid \theta) \ = \ p(\boldsymbol{\beta} \mid M)p(M \mid \theta) \tag{3}$$

where $p(\boldsymbol{\beta} \mid M)$ is a prior on regression coefficients given the model, and $p(M \mid \theta)$ the prior probability of model M. Specifically, assume that variable inclusions are independent a priori and Bernoulli distributed with inclusion probability θ,

$$p(M \mid \theta) \propto \prod_{i=1}^{p} \text{Bern}\left(m_i; \theta\right). \tag{4}$$

Posterior model probabilities are $p(M \mid \mathbf{y}, \theta) \propto p(\mathbf{y} \mid M) p(M \mid \theta)$, where $p(\mathbf{y} \mid M)$ is the so-called marginal likelihood of model M. The BIC approximation (Schwarz 1978) to $p(M \mid \mathbf{y}, \theta)$ gives that, for a wide family of priors $p(\boldsymbol{\beta} \mid M)$,

$$\ln p(M \mid \mathbf{y}, \theta) \ \approx \ \ln p(\mathbf{y} \mid \tilde{\boldsymbol{\beta}}^{(M)}) - \left(\frac{1}{2}\ln(n) + \ln\left(1/\theta - 1\right)\right)|M| + c_M, \tag{5}$$

where $\tilde{\boldsymbol{\beta}}^{(M)}$ is the maximum likelihood estimator (MLE) under model M, and c_M a constant that may depend on M. Neglecting c_M, simple algebra shows that the ℓ_0 selector in (2) with $\kappa = \frac{1}{2}\ln(n) + \ln(1/\theta - 1)$ approximately maximizes $p(M \mid \mathbf{y}, \theta)$, giving a correspondence between a choice of prior inclusion probability θ and a choice of ℓ_0 penalty κ.

Now, assume one has external information $\boldsymbol{z} = (z_1, \ldots, z_p)$ on the likelihood of variables to be truly associated with $\mathbf{y}$. In a Bayesian spike-and-slab regression, this prior information could be naturally reflected by setting prior inclusion probabilities that are a function of $\boldsymbol{z}$, that is replacing θ in (4) by $\theta(z_i)$, $i = 1, \ldots, p$. In particular, in our case where $\boldsymbol{z}$ induces a partition of the set of variables in blocks $B_1, \ldots, B_b$, one would then get block-dependent prior inclusion probabilities $\theta(z_i) = \theta_j$ for all $i \in B_j$,

$$p(M \mid \boldsymbol{\theta}) \propto \prod_{i=1}^{p} \text{Bern}\left(m_i; \theta_j\right) I(i \in B_j), \tag{6}$$

The BIC approximation for such model prior gives

$$\ln p(M \mid \mathbf{y}, \theta) \ \approx \ \ln p(\mathbf{y} \mid \tilde{\boldsymbol{\beta}}^{(M)}) - \sum_{j=1}^{b}\left(\frac{1}{2}\ln(n) + \ln\left(1/\theta_j - 1\right)\right)|M_j| + c_M, \tag{7}$$

where $M_j \subseteq B_j$ are the selected parameters in block j and a model is denoted by $M = M_1 \cup \cdots \cup M_b$.

In this work, we study the ℓ_0 selector analog to maximizing $p(M \mid \mathbf{y}, \theta)$ in (7). That is, an informed ℓ_0 selector that penalizes differently the blocks $B_1, \ldots, B_b$,

$$\hat{S}^{ei} \in \arg\max_{M \in \mathcal{M}} \left\{ \max_{\beta \in \mathcal{L}_M} \ell(\mathbf{y}; \boldsymbol{\beta}) - \sum_{j=1}^{b} \kappa_j |M_j| \right\}, \tag{8}$$

where $\kappa_1 > 0, \ldots, \kappa_b > 0$. Under the BIC approximation in (7), $\hat{S}^{ei}$ corresponds to the posterior mode with block prior inclusion probabilities $\theta_j = (e^{\kappa_j - \ln(n)/2} + 1)^{-1}$. In the case that $b = 1$, $\hat{S}^{ei}$ recovers $\hat{S}^{sd}$.

2.2 General assumptions of theoretical results

We study the properties of $\hat{S}^{ei}$ when the number of samples n goes to infinity, and then assume throughout

(A1) $n \to \infty$.

Although not explicitly denoted, p, s and $\beta^*_{\min}$ are functions of n, and so are s_j, p_j and $\beta^*_{\min,j}$. In particular the block numbers of inactive and active variables, $p_j - s_j \geq 1$ and $s_j \geq 1$, can either be fixed or diverge. We make no general assumption about the asymptotic regime linking n and p, but the main interest is in $n = o(p)$ settings.

We assume that $\mathcal{M}$ is a well-specified set of candidate models, that is $S \in \mathcal{M}$, and that one constrains attention to models with full-rank design, such that for any model $M \in \mathcal{M}$, $\text{rank}(\mathbf{X}_M^\top \mathbf{X}_M) = |M|$. This assumption implies $s \leq n$. We also assume $\hat{S}^{ei}$ is uniquely defined and that error variance σ^2 is known. To simplify notation, without loss of generality, we set $\sigma^2 = 1$. Non-uniqueness of $\hat{S}^{ei}$, non-full rank models, and unknown error variance can be accommodated, though, at the expense of a slightly more involved treatment.

♠ Paul: I removed the assumption $p_j - s_j \to \infty$ because we don't use it anymore. I will correct assumption numbering once we agree on content/presentation.

3 Oracle externally-informed penalties

We discuss here the properties of the externally-informed model selector $\hat{S}^{ei}$ (8) in the linear model (1). Our goal is to show that $\hat{S}^{ei}$ equals the data-generating truth S with probability 1, as n grows, and that this occurs under milder conditions or at a faster rate than when using a standard $\hat{S}^{sd}$ (2) that penalizes equally all blocks. In Section 3.1, we derive the properties of $\hat{S}^{ei}$. Those of $\hat{S}^{sd}$ follow by considering $b = 1$ blocks. In Section 3.2, we discuss the benefits of $\hat{S}^{ei}$ over $\hat{S}^{sd}$ in relation to the relevance of the external information, that is to what extent the blocks discriminate between truly active and inactive signals. In this section we assume that penalties are set by an oracle that knows the true pattern of sparsity and signal strength. In Section 3.3, we establish a generalized convergence result for $\hat{S}^{ei}$ under weak conditions on signal strength. This result plays an important role for the data-based procedures in Section 4.

3.1 Oracle properties

We first outline our proof strategy. For any model $M \in \mathcal{M}$, let $\tilde{\boldsymbol{\beta}}^{(M)} = (\mathbf{X}_M^\top \mathbf{X}_M)^{-1}\mathbf{X}_M^\top \mathbf{y} \in \mathbb{R}^{|M|}$ be the MLE under model M, and

$$C(M) \;:=\; \tfrac{1}{2}\|\mathbf{X}_M \tilde{\boldsymbol{\beta}}^{(M)}\|^2 - \sum_{j=1}^{b} \kappa_j |M_j| \quad \text{and} \quad NC(M) \;:=\; \frac{e^{C(M)}}{\sum_{M' \in \mathcal{M}} e^{C(M')}}. \tag{9}$$

Lemma 3.1. *$\hat{S}^{ei}$ satisfies $\hat{S}^{ei} \in \arg\max_{M \in \mathcal{M}} NC(M)$*

By Lemma 3.1, $\hat{S}^{ei}$ selects $M \in \mathcal{M}$ with the largest $NC(M)$. By the BIC approximation in (7), $C(M)$ and $NC(M)$ can be understood as, respectively, the unscaled and scaled pseudo-posterior probability for model M in a Bayesian variable selection framework. We refer to $C(M)$ and $NC(M)$ as the unnormalized and normalized scores for M, respectively. Lemma 3.2, reproduced from Rossell (2022), proves that the expected sum of the normalized scores for $M \neq S$ bounds the probability of an incorrect model selection.

Lemma 3.2. *(i)* $P(\hat{S}^{ei} \neq S) \ \leq \ 2\sum_{M\in\mathcal{M}\setminus\{S\}} \mathbb{E}\left(NC(M)\right)$.

(ii) For any $M, M' \in \mathcal{M}$*, such that* $M \neq M'$*,* $NC(M) \ \leq \ \left(1 + e^{C(M')-C(M)}\right)^{-1}$.

Using Lemma 3.2 (i) we show the variable selection consistency of $\hat{S}^{ei}$ by guaranteeing the vanishing, in expectation, of the sum of normalized scores across $M \neq S$. We thus show consistency in a strong sense, that is $NC(S) \xrightarrow{L_1} 1$. Taking $M' = S$ in Lemma 3.2 (ii), bounding $\mathbb{E}(NC(M))$ requires bounding the expectation of a simple function of $C(S) - C(M)$, which involves quadratic forms of least-squares estimators. To obtain such bounds we use concentration inequalities, carry out the integrals required by the expectations, and finally bound sums across $M \neq S$.

We now give two conditions that are sufficient to asymptotically recover S.

(A4) For each block j, there exists a sequence e_j such that $\lim_{n\to\infty} e_j > 0$ and for every sufficiently large n,

$$\kappa_j \ = \ \ln(p_j - s_j)(1 + e_j)$$

(A5) For each block j, there exists $g_j \to \infty$ such that for every sufficiently large n,

$$\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j} \ - \ \sqrt{\kappa_j} \ = \ \sqrt{\ln(s_j)} + g_j.$$

where $\rho(\mathbf{X}) = \min_{M\in\mathcal{M}:\, M\not\supseteq S} \lambda_{\min}\left(n^{-1}\mathbf{X}_{S\setminus M}^\top\left(I_n - \mathbf{X}_M\left(\mathbf{X}_M^\top\mathbf{X}_M\right)^{-1}\mathbf{X}_M^\top\right)\mathbf{X}_{S\setminus M}\right)$, and $\gamma := \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j) \in (\frac{1}{2}, 1)$.

♠ Paul: For simplicity. I changed A4 to a slightly stronger assumption so γ is always bounded away from 0 and we don't have to say in the later discussions that we focus on the penalties κ_j such that $\ln(p_j - s_j) = O(f_j)$. I could also change the definition of γ to alleviate notation and have everywhere just γ instead of $(1-\gamma)$.

Assumption A4 is a requirement on the strength of block penalties and A5 a betamin condition on signal strength. In A5, $\rho(\mathbf{X}) \geq 0$ relates to how distinguishable the other models $M \in \mathcal{M}$ are from S (Wainwright 2010). In our proof one could take a slightly milder assumptions, but Assumption A4 and A5 are presented here to facilitate interpretation and comparison to literature.

We next state our main theorems on the strong variable selection consistency of $\hat{S}^{ei}$ and the associated rates. In Theorem 3.4, $\delta \in (0,1)$ and $\gamma^* \in (1/2,1)$ should be understood as being arbitrarily close to 1 and 1/2 respectively. After discussing the theorems, we will discuss the tightness of Assumptions A4-A5 in relation to related necessary conditions for consistent variable selection.

Theorem 3.3. *Under Assumptions A1, A4 and, A5, it holds that*

$$\lim_{n\to\infty} \sup_{\beta^*\in\mathfrak{B}} \sum_{M\in\mathcal{M}\setminus\{S\}} \mathbb{E}\left(NC(M)\right) = 0 \qquad \textit{and} \qquad \lim_{n\to\infty} \inf_{\beta^*\in\mathfrak{B}} P(\hat{S}^{ei} = S) = 1.$$

♠ Paul: I added $\sup$ / $\inf$ over all the possible data generating β^* everywhere. Because our results are uniform of the set of possible $(s_1,\ldots,s_b)$-sparse β^* (but not on all possible S of size s). This is a usual way of writing results in the literature. (maybe too technical for JASA though)

Theorem 3.4. *Assume A1, A4, and A5. Then, there exists a constant $c>0$ such that for all sufficiently large n and any $\delta \in (0,1)$,*

$$\sup_{\beta^*\in\mathfrak{B}} P(\hat{S}^{ei} \neq S) \;\leq\; c\sum_{j=1}^{b} e^{-\frac{\delta}{2}\left[\kappa_j-\ln(p_j-s_j)\right]} + e^{-\frac{\delta}{2}\left[(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j}-\sqrt{\kappa_j})^2-\ln(s_j)\right]}. \tag{10}$$

Moreover, suppose that for some fixed $\gamma^* \in (1/2,1)$ and all $j=1,\ldots,b$, the κ_j are set at the oracle values such that

$$\sqrt{\kappa_j^*} \;=\; \frac{1}{2}\sqrt{\frac{(1-\gamma^*)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j} + \frac{1}{2}\sqrt{\frac{6}{(1-\gamma^*)n\rho(\mathbf{X})}}\frac{1}{\beta^*_{\min,j}}\Big(\ln(p_j-s_j)-\ln(s_j)\Big). \tag{11}$$

and that $\lim_{n\to\infty} g(\gamma^*)\sqrt{n\rho(\mathbf{X})/6}\ \beta^*_{\min,j}/(\sqrt{\ln(p_j-s_j)}+\sqrt{\ln(s_j)}) > 1$ where $g(\gamma^*) = \sqrt{(-1+2\gamma^*)(1-\gamma^*)}/(1+\sqrt{2-2\gamma^*})$. Then Assumptions A4 and A5 hold. Moreover, if Assumption A1 holds too, then for all sufficiently large n and any $\delta\in(0,1)$,

$$\sup_{\beta^*\in\mathfrak{B}} P(\hat{S}^{ei} \neq S) \;\leq\; 2c\sum_{j=1}^{b} e^{-\frac{\delta}{2}\left[\frac{(1-\gamma^*)n\rho(\mathbf{X})}{24}\beta^*_{\min,j}{}^2-\ln\max\{p_j-s_j,s_j\}\right]}. \tag{12}$$

♠ Paul: I now have a fixed γ^* in the definition of κ^*. This change removes some ambiguity because κ^* was defined for some γ that itself depended on κ^*, and importantly, it also eases the discussion of oracle rates. I had to require a more complex betamin condition though with this $g(\gamma^*)$. In (10) the first term of each summand decreases exponentially in κ_j, while the second term decreases exponentially in $(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j} - \sqrt{\kappa_j})^2$. The choice $\kappa_j = \kappa_j^*$ ensures that both terms are equal and hence approximates the values minimizing (10). We refer to these ideal κ_j^* as *oracle penalties* because they depend on quantities s_j and $\beta^*_{\min,j}$ that are unknown in practice. In (12), we give a simplified upper bound on the rate associated to the κ_j^* that is tightest when $n\rho(\mathbf{X})6^{-1}{\beta^*_{\min,j}}^2 = o(\ln(p_j/s_j - 1))$.

We observe that the results of Theorem 3.4, when applied to non-informed ℓ_0 selector $\hat{S}^{sd}$, nearly match the minimax rates for the orthonormal case, where $\mathbf{X}^\top\mathbf{X} = n\mathbb{I}_p$ and $\rho(\mathbf{X}) = 1$. Butucea et al. (2018) show that, in this case, the choice $\sqrt{\tilde{\kappa}} = (1/2)\sqrt{n/2}\beta^*_{\min} + (1/2)\sqrt{2/n}\ln(p/s-1)/\beta^*_{\min}$ is exactly minimax for the Hamming loss and asymptotically minimax for the probability of incorrect recovery, when $\beta_j^* \geq 0$ for all j. The minimax rate for the Hamming loss is then shown to be the sum of a term decreasing exponentially in $\tilde{\kappa} - \ln(p-s)$ and a term decreasing exponentially in $(\sqrt{n/2}\beta^*_{\min} - \sqrt{\tilde{\kappa}})^2 - \ln(s)$, similarly to our rate in (10).

Finally, we compare our sufficient Assumptions A4 and A5 to assumptions on κ_j and $\beta_{\min,j}$ that are necessary for consistent variable selection with $\hat{S}^{ei}$. Tight necessary assumptions follow from guaranteeing S is preferred over the models that differ from S by only one entry. In particular, we consider, for each j, the subset of models that over-fit by only one variable from block B_j,

$$\mathcal{O}_j \;=\; \{M \in \mathcal{M} \mid M = S \cup \{i\} \text{ where } i \in B_j \setminus S_j\},. \tag{13}$$

Our necessary conditions involve the next two quantities

$$\underline{\lambda}_j \;:=\; \lambda^2_{\min}(C^j) \quad \text{ and } \quad \bar{\lambda} \;:=\; \lambda_{\max}\Big(n^{-1}\mathbf{X}_S^\top\mathbf{X}_S\Big), \tag{14}$$

where, C^j is the matrix in $\mathbb{R}^{(p_j-s_j)\times(p_j-s_j)}$ such that for any $M^k, M^l \in \mathcal{O}_j$, $C^j_{k,l} = \operatorname{corr}\left(Z_{M^k}, Z_{M^l}\right)$ and $Z^2_{M^k} = \|\mathbf{X}_{M^k}\tilde{\boldsymbol{\beta}}^{(M^k)}\|^2 - \|\mathbf{X}_S\tilde{\boldsymbol{\beta}}^{(S)}\|^2$. That is, small $\underline{\lambda}_j$ indicates that truly inactive variables in block j are highly correlated with each other, whereas $\underline{\lambda}_j = 1$ that they are uncorrelated. Similarly, $\bar{\lambda}$ relates to the level of correlation between active variables across all blocks.

Proposition 3.5. *(i) If for some $j = 1, \ldots, b$, $\lim_{n\to\infty} \kappa_j/(\underline{\lambda}_j \ln(p_j - s_j)) < 1$, then $\lim_{n\to\infty} \sup_{\boldsymbol{\beta}^*\in\mathfrak{B}} P(\hat{S}^{ei} = S) = 0$.*

(ii) Assume A1. If for some $j = 1, \ldots, b$, $\kappa_j \to \infty$ and $\lim_{n\to\infty} \sqrt{n\bar{\lambda}}\beta^*_{\min,j} - \sqrt{2\kappa_j} < \infty$ then $\lim_{n\to\infty} \sup_{\boldsymbol{\beta}^*\in\mathfrak{B}} P(\hat{S}^{ei} = S) < 1$.

(iii) Assume A1. If for some $j \in \{1, \ldots, b\}$, $\lim_{n\to+\infty}(p_j - s_j)^{\underline{\lambda}_j} = +\infty$ and $\lim_{n\to\infty} \sqrt{n\bar{\lambda}}\beta^*_{\min,j} - \sqrt{2\underline{\lambda}_j \ln(p_j - s_j)} < \infty$, then $\lim_{n\to\infty} \sup_{\boldsymbol{\beta}^*\in\mathfrak{B}} P(\hat{S}^{ei} = S) < 1$.

Proposition 3.5 (i) shows that the block penalty κ_j cannot be too small compared to the log number of inactive parameters $p_j - s_j$ in the block. Proposition 3.5 (ii) further shows that the signal strength $\beta^*_{\min,j}$ cannot be too small with respect to the block penalties. Proposition 3.5 (iii) follows from combining (i) and (ii). The assumption $(p_j - s_j)^{\underline{\lambda}_j} \to \infty$ requires that an *effective* number of inactive variables in block B_j, in the sense that it accounts for their correlation through $\underline{\lambda}_j$, diverges, so that the regime in block B_j is truly high-dimensional and the block penalty κ_j is must diverge as required in (ii). Proposition 3.5 (iii) then shows the signal strength $\beta^*_{\min,j}$ cannot be too small with respect to the log of that *effective* number of inactive variables and yields the necessary betamin assumption for consistency

$$\lim_{n\to\infty} \sqrt{n\bar{\lambda}}\beta^*_{\min,j} - \sqrt{2\underline{\lambda}_j \ln(p_j - s_j)} = \infty \qquad \text{for all } j = 1, \ldots, b. \tag{15}$$

♠ Paul: David had asked about the assumption $\lim_{n\to+\infty}(p_j - s_j)^{\underline{\lambda}_j} = +\infty$ in (iii) in a later comment. It requires that an "effective" number of inactive variables goes to infinity. A2 is not enough. If $p_j - s_j \to \infty$ but correlation increases a lot, then you can end up with an "effective" number of inactives that is fixed or decreasing. I added an explanation but that may be too technical for JASA

It follows from Proposition 3.5 that Assumptions A4-A5 are fairly tight. Indeed, it holds that $\underline{\lambda}_j \in [0,1]$ and $\underline{\lambda}_j = 1$ corresponds to the worst case. In that worst case, by (i), a necessary assumption is that, for all j, $\kappa_j = \ln(p_j - s_j)(1 + e_j)$ such that $\lim_{n\to+\infty} e_j \geq 0$, closely matching A4. In that worst case, the necessary condition (15) also ressembles A5 where $\rho(\mathbf{X})$ is replaced by $\bar{\lambda}$ and up to a constant factor, a factor logarithmic in s_j and factor $1-\gamma$ which is bounded in $(0, 1/2)$. That is, Assumption A5 and (15) imply the same scalings of $(n, p_j, s_j, \beta^*_{\min,j})$ up to a logarithmic factor in s_j.

3.2 Benefits of externally-informed penalties

The benefits of the oracle-informed selector $\hat{S}^{ei}$ over the oracle standard ℓ_0 selector $\hat{S}^{sd}$, the class of (uninformed) methods with best selection properties (Gao & Aragam 2025), depends on whether and how different are the blocks in the number of active signals s_j and signal strengths $\beta^*_{\min,j}$. We discuss two types of benefits, softening the conditions for model selection consistency and improving the associated convergence rates.

♡ [David: This section was hard to follow. First we present (17)-(16) and argue that they give conditions under which the informed selector is consistent and the standard selector isn't. We then give an example in "For example,...". And then we present Corollary 3.6 without any explanation on whether it has anything to do with (17)-(16) or with the example that we just gave. The narrative feels disconnected. And then the interpretation of the corollary is complicated, silly me got lost which is a bad sign. See comments below but basically we need to re-write the section in a more accessible way.]♠

Paul: I rewrote it completely, I dropped the discussion of "ranges" of penalties that was natural in the sequence model for how the sufficient assumptions were formulated, but less so for regression. I think the ideas below are easier to grasp and more aligned with traditional literature on high-dimensional stats.

By Theorem 3.4, Assumptions A4–A5 give ranges of penalties that are sufficient for asymptotic support recovery with $\hat{S}^{ei}$ and $\hat{S}^{sd}$ (taking $b = 1$). For each of the two selectors, we set penalties to the lowest end of the respective ranges to understand the minimum scaling of n with respect to the $(p_j, s_j, \beta^*_{\min,j})$ for which consistency can be guaranteed. For the informed selector $\hat{S}^{ei}$ with block penalties set to $\kappa_j = \ln(p_j - s_j)(1 + e)$ for some arbitrarily small sequence e such that $\lim_{n\to\infty} e > 0$, a sufficient condition on n to have consistency is, for some arbitrarily small sequence g such that $\lim_{n\to\infty} g = \infty$ and up to constant factors and factors $(1 - \gamma_j)$ bounded in $(0, 1/2)$,

$$\sqrt{n} \geq \max_{j=1\dots,b} \left\{ \frac{\sqrt{\ln(p_j - s_j)(1 + e)} + \sqrt{\ln(s_j)}}{\sqrt{\rho(\mathbf{X})}\beta^*_{\min,j}} \right\} + g. \tag{16}$$

For the standard selector $\hat{S}^{sd}$ with single penalty set to $\kappa = \ln(p - s)(1 + e')$, for some arbitrarily small sequence e' such that $\lim_{n\to\infty} e' > 0$, a sufficient condition on n to have consistency is, for some arbitrarily small sequence g' such that $\lim_{n\to\infty} g' = \infty$ and up to a constant factor and a factor $(1 - \gamma)$ bounded in $(0, 1/2)$,

$$\sqrt{n} \geq \frac{\sqrt{\ln(p - s)(1 + e')} + \sqrt{\ln(s)}}{\sqrt{\rho(\mathbf{X})}\beta^*_{\min}} + g'. \tag{17}$$

Since for every $j = 1, \dots, b$, $p_j - s_j \leq p - s$, $s_j \leq s$ and $\beta^*_{\min,j} \geq \beta^*_{\min}$, the growth rate of n required in (16) is slower than that required in (17).

Variable selection consistency can then be guaranteed with the informed selector $\hat{S}^{ei}$ under milder conditions than with the standard selector $\hat{S}^{sd}$. How much milder depends on how much sparser the global $\boldsymbol{\beta}^*$ is compared to the block $\boldsymbol{\beta}^*_j$'s. Consider a two block example with discriminative external information that splits the global number of inactive variable $p - s = n^\alpha$ ($\alpha > 1$) into a first block where $p_1 - s_1 = \ln(n)$ and second block $p_2 - s_2 = n^\alpha - \ln(n)$. Further, suppose the number of active variables is the same across block, $s_1 = s_2 = O(1)$, such that the block proportions of active variables are $s_1/p_1 = O(1/\ln(n))$ and $s_2/p_2 = O(1/n^\alpha)$. For simplicity, asumme the design is orthonormal such that $\mathbf{X}^\top\mathbf{X} = n$

and $\rho(\mathbf{X}) = 1$. With $\hat{S}^{sd}$, recovery can be guaranteed only for signals of size at least $O(\sqrt{\ln(n)/n})$. With $\hat{S}^{ei}$, in the sparser block 2, signals must be of the same order of size to be recovered but in the less sparse block 1, signals as small as $O(\sqrt{\ln(\ln(n))/n})$ can be recovered. In Supplement XXX, we consider other scenarios of informativeness of the block partition and provide illustrations of the gains achievable.

Proposition 3.5 (iii) implies that, similarly, the growth rate of n necessary for consistency with $\hat{S}^{ei}$ is slower than the growth rate necessary for consistency with $\hat{S}^{sd}$. It follows that in some settings n may not meet the necessary growth rate required for consistency with $\hat{S}^{sd}$ but may meet the sufficient growth rate required for consistency with $\hat{S}^{ei}$. Corollary 3.6 next describes such settings.

Corollary 3.6. *Assume A1, A4, A5 and* $\lim_{n\to\infty}(p-s)^{\underline{\lambda}} = \infty$ *where* $\underline{\lambda}$ *as in* (14) *for the case* $b = 1$. *If* $\lim_{n\to\infty} \sqrt{n\overline{\lambda}}\beta^*_{\min} - \sqrt{2\underline{\lambda}\ln(p-s)} < \infty$ *where* $\underline{\lambda}$ *as in* (14), *then* $\lim_{n\to\infty} \sup_{\beta^*\in\mathfrak{B}} P(\hat{S}^{sd} = S) < 1$ *and* $\lim_{n\to\infty} \inf_{\beta^*\in\mathfrak{B}} P(\hat{S}^{ei} = S) = 1$.

Corollary 3.6 assumes a truly high-dimensional setting, where the global *effective* number of inactive variables $(p-s)^{\underline{\lambda}}$ diverges, in the fashion of Proposition 3.5 (iii). It shows then that $\hat{S}^{ei}$, by incorporating external information, may attain variable selection consistency when otherwise it would not be possible. Supplement XXX provides an illustration in Example 2.

We now discuss differences in the probability of correct selection, when both procedures are consistent. Assume that, for some $\gamma^* \in (1/2, 1)$, block penalties are set to their oracle values κ_j^* in (11) and similarly that the single penalty takes its oracle value κ^*. Let OR^{ei} be the oracle convergence rate for κ_j^* in (12) and OR^{sd} that for κ^*. Then, up to constant factors,

$$\frac{OR^{ei}}{OR^{sd}} = \sum_{j=1}^{b} e^{-\frac{\delta}{2}\left[\frac{n\rho(\mathbf{X})(1-\gamma^*)}{24}\left(\beta^{*\,2}_{\min,j} - \beta^{*\,2}_{\min}\right) + \ln\max\{p-s,s\} - \ln\max\{p_j-s_j,s_j\}\right]}. \tag{18}$$

where δ is arbitrarily close to 1. Since $\beta^*_{\min,j} \geq \beta^*_{\min}$ and $\max\{p-s,s\} \geq \max\{p_j-s_j,s_j\}$, equation (18) shows that for every n large enough $OR^{ei} = O(OR^{sd})$ for any block partition and if for some j, $\beta^*_{\min,j} < \beta^*_{\min}$, $OR^{ei} < OR^{sd}$, i.e. the oracle convergence rate in (12) for κ^*_j is faster than for κ^*. The magnitude of the gain depends on how informative the blocks are. For any sparse setting where $s_j < p_j - s_j$ for every j, assuming that $\beta^*_{\min,b} = \beta^*_{\min}$ without loss of generality and that $\gamma = \gamma'$ for simplicity, it holds that

$$\frac{OR^{ei}}{OR} = \frac{p_b - s_b}{p - s} + \sum_{j<b} \frac{p_j - s_j}{p - s} e^{-\frac{n\rho(\mathbf{X})(1-\gamma^*)}{24}\left({\beta^*_{\min,j}}^2 - {\beta^*_{\min}}^2\right)} \tag{19}$$

There are then two sources of improvement in convergence rates: in any block j, a smaller number of truly zero means, that is $p_j - s_j < p - s$, or a larger smallest signal, $\beta^*_{\min,j} > \beta^*_{\min}$. Depending on the sparsity of the block, one of the two effects prevails. In blocks where $\beta^*_{\min,j} \approx \beta^*_{\min}$, improvement comes mainly from having a smaller number of truly zero means. In those blocks, the oracle sets $\kappa^*_j < \kappa^*$, so that smaller signals are detected. If $\beta^*_{\min,j}$ is much greater than $\beta^*_{\min}$, the oracle sets $\kappa^*_j > \kappa^*$ and the probability of false positives is reduced. See Supplement XXX, for illustrations in different scenarios of sparsity regimes and informativeness of the blocks.

3.3 Generalized convergence

We now derive a convergence result for $\hat{S}^{ei}$ that generalizes Theorem 3.3 by forgoing the betamin Assumption A5 and is key to the proofs of Section 4. Rather than requiring large enough active signals, here we partition active signals into small, intermediate and large signals and show that, if one sets a sufficiently large penalty κ_j, all inactive and all small active signals are discarded. On the other hand, all large signals (relative to κ_j) are retained. We next formally define small, intermediate and large signals. For fixed κ_j, and for γ and

g_j as defined in Assumption A5, let

$$\begin{aligned} S_j^S(\kappa_j) &:= \left\{\beta_i^* \in S_j \,\middle|\, \sqrt{n\bar{\lambda}}|\beta_i^*| = o\left(\sqrt{\kappa_j}\right)\right\} \\ S_j^L(\kappa_j) &:= \left\{\beta_i^* \in S_j \,\middle|\, \sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}|\beta_i^*| - \sqrt{\kappa_j} = \sqrt{\ln(s_j)} + g_j\right\} \\ S_j^I(\kappa_j) &:= S_j \setminus \left(S_j^L(\kappa_j) \cup S_j^S(\kappa_j)\right). \end{aligned} \tag{20}$$

where $\bar{\lambda} = \lambda_{\max}\left(n^{-1}\mathbf{X}_S^\top \mathbf{X}_S\right)$. The subset $S_j^S(\kappa_j)$ gathers truly active signals in S_j that are small with respect to the penalty κ_j, $S_j^L(\kappa_j)$ those that are large in that they satisfy Assumption A5, and $S_j^I(\kappa_j)$ those that are neither large nor small. Moreover, let $\mathcal{T}(\kappa)$ be the set of models that contain all large signals in $S^L(\kappa) = \cup_{j=1}^b S_j^L(\kappa_j)$, and neither truly inactive signals in $V \setminus S$ nor small signals in $S^S(\kappa) = \cup_{j=1}^b S_j^S(\kappa_j)$. That is,

$$\mathcal{T}(\kappa) := \left\{M \in \mathcal{M} \,\middle|\, M = S^L(\kappa) \cup R,\ R \in \mathcal{P}\left(S^I(\kappa)\right)\right\}, \tag{21}$$

where for a set A, $\mathcal{P}(A)$ denotes the power set of A. We show that, if one sets sufficiently large penalties, that is, meeting Assumption A4, the normalized scores $NC(M)$ in (9) concentrate on $\mathcal{T}(\kappa)$. We also require the following assumption:

(A6) For each block j, $|S_j^I(\kappa_j)| = O(1)$ and $|S_j^S(\kappa_j)| = O(p_j - s_j)$.

Assumption A6 is mild and can be relaxed. The $|S_j^I(\kappa_j)|$ can grow moderately with $p_j - s_j$ in our proof, and one expects $|S_j^I(\kappa_j)|$ to be small because there is a relatively small gap between the range of β_i^*'s included in S_j^S and S_j^L. Further, if $|S_j^S(\kappa_j)|/(p_j - s_j) \to \infty$ the same result holds replacing, in A4, $p_j - s_j$ by $|S_j^S(\kappa_j)|$ for all j.

Theorem 3.7. *Assume A1, A4, and A6, for every $j = 1, \ldots, b$, then,*
$\lim_{n\to\infty} \sum_{M \in \mathcal{M}\setminus\mathcal{T}(\kappa)} \sup_{\beta^* \in \mathfrak{B}} \mathbb{E}\left(NC(M)\right) = 0$ *and* $\lim_{n\to\infty} \inf_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \in \mathcal{T}(\kappa)) = 1$.

Theorem 3.7 shows that, asymptotically, all inactive and small active signals are discarded, large signals are retained and intermediate signals in $S^I(\kappa)$ may or may not be retained.

4 Data-based externally-informed penalization

We propose next a data analysis method to set block penalties that does not require an oracle and discuss its properties. Our goal is to have a method that adapts to the relevance of the external-information. In Section 3.2, we saw that the relevance is driven by the varying sparsity levels and signal strengths across blocks. We define then an estimator of the proportion of truly active variables in each block. This estimator is based on the connection between informed ℓ_0 penalties and Bayesian variable selection (7) and by empirical Bayes techniques. Such connection also allows the use of Bayesian computational methods. Despite its Bayesian motivation, our method is fully data-dependent and does not require any prior distribution.

Recall that, by the BIC approximation in (7), $NC(M)$ is the pseudo-posterior probability for model M in a Bayesian framework where the κ_j's are a function of the block prior inclusion probability θ_j given after (8). Therein, θ_j is a prior guess for s_j/p_j, the proportion of active coefficients in block B_j. A standard empirical Bayes strategy to set $\hat{\theta}_j$ consists in maximizing the marginal likelihood of $\mathbf{y}$ given $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_b)$, i.e.

$$\hat{\boldsymbol{\theta}} := \arg\max_{\boldsymbol{\theta}} p(\mathbf{y} \mid \boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \int p(\mathbf{y} \mid \boldsymbol{\beta}, M) dP(\boldsymbol{\beta}, M \mid \boldsymbol{\theta}).$$

Lemma 4.1 shows that $\hat{\boldsymbol{\theta}}$ satisfies a fixed point equation, whereby the prior inclusion probability $\hat{\theta}_j$ for block j equals the average posterior inclusion probability across variables in that block given $\hat{\boldsymbol{\theta}}$.

Lemma 4.1. *Let $p(M \mid \boldsymbol{\theta})$ as in (6) and $P(\beta_i \neq 0 \mid \mathbf{y}, \boldsymbol{\theta}) = \sum_{M \in \mathcal{M}: i \in M} P(M \mid \mathbf{y}, \boldsymbol{\theta})$ be the posterior inclusion probability of variable j. Then $\hat{\boldsymbol{\theta}}$ satisfies*

$$\hat{\theta}_j = \frac{1}{p_j} \sum_{i \in B_j} P(\beta_i \neq 0 \mid \mathbf{y}, \hat{\boldsymbol{\theta}}).$$

Interpreting $NC(M)$ as the pseudo-posterior probability of model M motivates estimating the number of active variables s_j in block j by $\hat{s}_j = p_j \hat{\theta}_j$ where $P(M \mid \mathbf{y}, \boldsymbol{\theta})$ is replaced by $NC(M)$,

$$\hat{s}_j := \sum_{i \in B_j} \sum_{M \in \mathcal{M}: i \in M} NC(M). \tag{22}$$

We have the following asymptotic bounds on the expectation of $\hat{s}_j / p_j$.

Proposition 4.2. *Assume A1, A4, and A6. Then*

$$\frac{|S_j^L(\kappa_j)|}{p_j} \leq \lim_{n \to \infty} \mathbb{E}\left(\frac{\hat{s}_j}{p_j}\right) \leq \frac{s_j - |S_j^S(\kappa_j)|}{p_j} \quad \textit{for all } j = 1, \ldots, b \textit{ and } \boldsymbol{\beta}^* \in \mathfrak{B}.$$

As a consequence, if the betamin condition in Assumption A5 also holds, then $\hat{s}_j / p_j$ is an asymptotically unbiased estimator of s_j / p_j because $S_j = S_j^L(\kappa_j)$ and $S_j^S(\kappa_j) = \emptyset$ for all j. The proposition also shows that, when Assumption A5 is not met, $\hat{s}_j / p_j$ is asymptotically downward biased by at least the proportion of small signals $|S_j^S(\kappa_j)|/p_j$, but it is guaranteed to be larger than the proportion of large signals $|S_j^L(\kappa)/p_j$ (i.e., those satisfying A5).

We propose a two-step procedure that uses $\hat{s}_j$ in (22) to set adaptive block penalties that leverage the difference in sparsity across blocks. In the first step, we use a global penalty across blocks κ° and obtain the corresponding $\hat{s}_j$. In the second step, we use $\hat{s}_j$ to set block penalties $\kappa_j^A = \ln(p_j - \hat{s}_j) + \frac{1}{2}\ln(n)$. This choice of block penaltiesamounts to setting $\theta_j \approx (p_j - \hat{s}_j + 1)^{-1}$ in the BIC approximation (7). Such block penalties κ_j^A are sufficiently large for Theorem 3.3 to hold, under assumption that p grows at most polynomially in n,

(A6) There exists $\alpha > 0$ constant such that $p \leq n^\alpha$. ♠ Paul: This a new assumption, the price to pay for the new stricter assumption A4.

Algorithm 1 - Adaptive informed selector

1. Set $\kappa_j = \kappa^\circ = \ln(p) + \frac{1}{2}\ln(n)$ for $j = 1, \ldots, b$. Compute $\hat{s}_j / p_j$ in (22) for $j = 1, \ldots, b$.
2. Obtain $\hat{S}^{A,ei}$ solving (8) with $\kappa_j^A = \ln(p_j - \hat{s}_j) + \frac{1}{2}\ln(n)$.

We refer to $\hat{S}^{A,ei}$ as the adaptive informed selector. Step 1 approximates posterior model probabilities under equal prior inclusion probabilities $\theta_j = (p+1)^{-1}$ in (6) across blocks. Step 1 can be carried out with any Bayesian computational method that returns variable posterior inclusion probabilities. Step 2 can be done with any computational method to find the model with best informed ℓ_0 criterion. In our examples in Section 6 we use MCMC for both steps.

In Section 3, the selection properties were driven by $p_j - s_j$ and the gains in conditions for consistency by how the $p_j - s_j$ compared to $p - s$, whereas here they are driven by $p_j - |S_j^L(\kappa^\circ)|$ and it how compares to $p - |S^L(\kappa^\circ)|$. This occurs because recovery of small signals in Step 1 is not guaranteed. Consider the following betamin assumption:

(A9) For each block j, there exists $d_j \to \infty$ such that for all sufficiently large n,

$$\sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{2}}\beta^*_{\min,j} - \sqrt{\ln\left(p_j - |S_j^L(\kappa^\circ)|\right) + \frac{1}{2}\ln(n)} = \sqrt{\ln(s_j)} + d_j.$$

where $\xi = \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/(\ln(p_j - s_j) + \frac{1}{2}\ln(n)))$.

Under Assumptions A1, A6, and A9, we show that, with probability going to 1, κ_j^A and $\beta^*_{\min,j}$ satisfy Assumptions A4–A5 and that Theorem 3.3 applies. The next result on the consistency of $\hat{S}^{A,ei}$ then follows. In our proof, Assumption A6 can be relaxed by setting larger block penalties in Step 2.

Theorem 4.3. *Assume A1, A6, and A9, then* $\lim_{n\to\infty} \inf_{\beta^* \in \mathfrak{B}} P(\hat{S}^{A,ei} = S) = 1$.

Theorem 4.3 shows the consistency of $\hat{S}^{A,ei}$ under a betamin assumption that ressembles the betamin assumption required for the consistency of the oracle selector in Theorem 3.4 where $\ln(p_j - s_j)$ is replaced by $\ln(p_j - |S_j^L(\kappa^\circ)|) + \frac{1}{2}\ln(n)$.

We compare next $\hat{S}^{A,ei}$ to its non-informed counterpart $\hat{S}^{A,sd}$ that sets in Step 2 a single common penalty $\kappa^A = \ln(p - \hat{s}) + \frac{1}{2}\ln(n)$. By Theorem 4.3, $\hat{S}^{sd,A}$ is variable selection consistent under the betamin assumption

$$\sqrt{\frac{(1-\xi')n\rho(\mathbf{X})}{2}}\beta^*_{\min} - \sqrt{\ln\left(p - |S^L(\kappa^\circ)|\right) + \frac{\ln(n)}{2}} = \sqrt{\ln(s)} + d, \tag{23}$$

where $\xi' = \frac{1}{2}(1 + \ln(p-s)/(\ln(p-s) + \frac{1}{2}\ln(n)))$ and $d \to \infty$. Since $\ln(p - |S^L(\kappa^\circ)|) \geq \ln(p_j - |S^L_j(\kappa^\circ)|)$ and $\ln(s) \geq \ln(s_j)$, Assumption A9 for $\hat{S}^{A,ei}$ is milder than (23) for $\hat{S}^{A,sd}$, in particular in less sparse blocks where $p_j - |S^L_j(\kappa^\circ)|$ is smaller $p - |S^L(\kappa^\circ)|$. Although not shown here, necessary betamin assumptions are also milder for $\hat{S}^{A,ei}$ than for $\hat{S}^{A,sd}$.

5 Structural transfer learning

♡ [David: We need to be careful not to open a can of worms. (1) The algorithm boils down to a specific choice of blocks, I'm not sure that listing/naming it as a separate algorithm is a good idea. (2) One could think about other ways how to define the blocks, for example let B_l denote variables that were identified in l of the previous studies. Another example, it may be that $\hat{S}^{(1)}$ is more informative than $\hat{S}^{(2)}$, so one would want to somehow weight $\hat{S}^{(1)}$ more. In other words, there's already fancier methodology to regress prior inclusion probabilities on the $\hat{S}^{(k)}$, I would clarify that our main contribution is the theory for our simplified transfer learning algorithm.]

This section studies the properties of a structural transfer learning algorithm based on the adaptive informed selector $\hat{S}^{A,ei}$. We consider a transfer learning setting where one is given a set of selected variables $\hat{S}^{(k)}$ for each source dataset $k = 1, \ldots, K$, and one wants to leverage these sets to improve selection in the target dataset. That is, the similarity between source and target data generating processes is thought to lie in the sets of variables associated with the respective outcomes, unlike the similarity in respective parameters values considered in transfer learning for estimation (Li et al. 2021, Tian & Feng 2023, Abba et al. 2024).

A naive approach could be to force the inclusion in the target model of variables selected in any of the source datasets. Such an approach is prone to negative transfers as sets of truly active variables in the target and source datasets may in a good case be similar but not identical, and in a bad case completely unrelated. Instead, in the Tran-s-ell0 algorithm below, we treat $\hat{S}^{(k)}$ as external information and perform variable selection in the target data with the adaptive informed selector $\hat{S}^{A,ei}$. In particular, in a first step, for each variable $i = 1, \ldots, p$, we define an external information vector $z_i = I(i \in \cup_{k=1}^{K} \hat{S}^{(k)})$. That is, variables in the target dataset are split into a first block with variables that were selected in ≥ 1 source datasets, and a second block gathering the rest. In a second step, we use the adaptive informed selector of Section 4 with that partition to select variables in the target dataset.

Algorithm 2 - Tran-s-ell0

(i) Construct $B_1 = \cup_{k=1}^{K} \hat{S}^{(k)} \cap V$ and $B_2 = V \setminus B_1$.

(ii) Obtain $\hat{S}^{A,ei}$ by running Algorithm 1 on the target dataset.

We remark that, in Step (i), other strategies are possible to form blocks from the source $\hat{S}^{(k)}$. For example one could define $z_i = 1 + \sum_{k=1}^{K} I(i \in \hat{S}^{(k)})$, that is define $K + 1$ blocks by counting in how many source datasets a variable was selected. Our theory continues to apply to such alternatives. In practice however, a more advanced strategy is to replace z_i by a vector $\boldsymbol{z}_i \in \{0, 1\}^K$ where $z_{ik} = I(i \in \hat{S}^{(k)})$ and to regress prior inclusion probabilities on $\boldsymbol{z}_i$ (van de Wiel et al. 2019). Here we focus on the simpler block partitioning, because such settings require a more advanced theoretical treatment beyond our scope.

We derive next the variable selection properties Tran-s-ell0. These follow from Theorem 4.3 with the block numbers of inactive and active (large to small) signals induced by the partition defined from the $\hat{S}^{(k)}$,

$$
\begin{aligned}
&s_1 = |B_1 \cap S| = |\cup_{k=1}^{K} \hat{S}^{(k)} \cap S|,\\
&s_2 = |B_2 \cap S| = |S \setminus \cup_{k=1}^{K} \hat{S}^{(k)}|,\\
&p_1 - s_1 = |B_1 \setminus S| = |(\cup_{k=1}^{K} \hat{S}^{(k)} \cap V) \setminus S|,\\
&p_2 - s_2 = |B_2 \setminus S| = |V \setminus (\cup_{k=1}^{K} \hat{S}^{(k)} \cup S)|,\\
&p_1 - |S_1^L(\kappa^\circ)| = |B_1 \setminus S^L(\kappa^\circ)| = |(\cup_{k=1}^{K} \hat{S}^{(k)} \cap V) \setminus S^L(\kappa^\circ)|,\\
&p_2 - |S_2^L(\kappa^\circ)| = |B_2 \setminus S^L(\kappa^\circ)| = |V \setminus (\cup_{k=1}^{K} \hat{S}^{(k)} \cup S^L(\kappa^\circ))|,
\end{aligned}
\tag{24}
$$

where $S^L(\kappa^\circ)$ is the subset of large active signals in S as defined in (20) for κ° as in Step 1 of Algorithm 1.

Consider the betamin assumption

(A10) There exist $d_1, d_2 \to \infty$ such that for all sufficiently large n,

$$\sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{6}}\beta^*_{\min,1} - \sqrt{\ln\left(|(\cup_{k=1}^{K} \hat{S}^{(k)} \cap V) \setminus S^L(\kappa^\circ)|\right) + \frac{\ln(n)}{2}} = \sqrt{\ln\left(|\cup_k S^{(k)} \cap S|\right)} + d_1,$$

and,

$$\sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{6}}\beta^*_{\min,2} - \sqrt{\ln\left(|V \setminus (\cup_{k=1}^{K} \hat{S}^{(k)} \cup S^L(\kappa^\circ))|\right) + \frac{\ln(n)}{2}} = \sqrt{\ln\left(|S \setminus \cup_{k=1}^{K} \hat{S}^{(k)}|\right)} + d_2,$$

where $\xi = \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/(\ln(p_j - s_j) + 0.5\ln(n)))$. Assumption A10 is the betamin Assumption A9 where the block number of active and inactive parameters are as in (24) and is sufficient for Tran-s-ell0 to be selection consistent.

Corollary 5.1. *Assume A1, A6, and A10. Then in Step (ii) of Tran-s-ell0,* $\lim_{n\to\infty} \inf_{\beta^* \in \mathfrak{B}} P(\hat{S}^{A,ei} = S) = 1$.

Theorem 5.1 shows that Tran-s-ell0 is selection consistent under milder conditions than a standard, non-informed, ℓ_0 selector $\hat{S}^{A,sd}$, which, recall, is consistent under (23). The gains in sufficient conditions for consistency depend on the similarity between the true set of active variables in the target dataset and the sets of variables selected in the source datasets. Observe that in block B_1 the more similar S is to the $\hat{S}^{(k)}$, the smaller is $|(\cup_{k=1}^{K} S^{(k)} \cap V) \setminus S^{L}(\kappa^{\circ})|$ and the milder is A10 compared to (23). Tran-s-ell0 allows the recovery of smaller signals among the variables selected in the source datasets.

The adaptive nature of $\hat{S}^{A,ei}$ on which Tran-s-ell0 is based, renders our transfer learning algorithm robust to negative transfer. Indeed, in the last step, in block B_1 that gathers the transferred variables, the penalty is set to $\kappa_1^A = \ln(p_1 - \hat{s}_1) + \ln(n)/2$ where $p_1 - \hat{s}_1$ is an estimate of $|(\cup_{k=1}^{K} \hat{S}^{(k)} \cap V) \setminus S|$. That is, the larger the dissimilarity between the target set of active variables and the set of transferred variables, the larger the penalty incurred by the latter and the less likely they are to be selected.

Remarkably, our structural transfer learning algorithm achieves gains under comparatively fewer requirements than transfer learning algorithms for parameter estimation and prediction. Those latter algorithms tend to either assume the sets of variables are the same between source and target datasets e.g Abba et al. (2024) or make strong assumptions on the similarity of design matrices e.g Li et al. (2021), Tian & Feng (2023). This is required to avoid negative transfer because parameter estimates are highly sensitive to adjustment sets. Gains in estimation achievable through transfer learning were in fact shown to be inherently limited by the dissimilarity between source and target design matrices (Liu et al. 2025), an issue also known as the *covariate shift* problem. Our results on structural transfer learning require no such assumption: the variable sets in the source and target datasets may differ largely, even in size.

In addition, many (but not all) transfer learning algorithms for parameter estimation and prediction, e.g Jiang et al. (2016) and Li et al. (2021), require access to the source datasets. Tran-s-ell0 only requires sets of selected variables. It is then more widely applicable and enjoys stronger privacy and communication efficiency guarantees.

6 Numerical illustrations

We illustrate the performance of our adaptive informed selector $\hat{S}^{A,ei}$ (Algorithm 1) on simulated data under different scenarios of relevance of the external information (Section 6.1) and that of the Tran-s-ell0 (Algorithm 2) on simulated and empirical data (Section 6.2). In Step 1 of Algorithm 1, run on its own or within Algorithm 2, we use $\kappa^{\circ} = \frac{1}{2}\ln(n) + \ln(p)$ and the MCMC algorithm in function `bestIC` in R package `mombf`. Briefly, each visited model is scored with the normalized score (the pseudo-posterior probability given by the BIC approximation (5)). In Step 2 of Algorithm 1, we obtain $\hat{S}^{A,ei}$ by scoring all the models visited by the MCMC in Step 1. We obtain similarly, for comparison purposes, $\hat{S}^{A,sd}$, the uninformed counterparts of $\hat{S}^{A,ei}$ (with a single common penalty). We also compare $\hat{S}^{A,ei}$ and Tran-s-ell0 to the standard ℓ_0 penalty EBIC (Chen & Chen 2008), the LASSO (Tibshirani 1996) and SCAD (Fan & Li 2001). For the last two methods, the penalization parameter is chosen by cross-validation. The R code to reproduce our analyses is at https://github.com/paulrognonvael/varsel.extdata.

6.1 Adaptive informed selector

We simulate data from the linear model (1) with standard Gaussian errors ($\sigma^2 = 1$), sample size $n \in [20, 700]$, and variable values drawn from a multivariate Gaussian with zero means, unit variances, and pairwise correlations $\text{cor}(x_{ij}, x_{il}) = 0.5$ for $j \neq l$.

Table 1: Sparsity and block assumptions in Examples 1–3

Example	$p - s$	s	s_1/p_1	s_2/p_2	$\beta^*_{\min,1}$	$\beta^*_{\min,2} = \beta^*_{\min}$
1	$3n/2$	$3\ln(n)$	$O(n^{-1}\ln(n))$	$O(n^{-1/2}\ln(n))$	0.5	0.33
2	n	$3\ln(n)$	$O(n^{-1}\ln(n))$	$2/3$	0.5	0.33
3	n	$3\ln(n)$	$O(n^{-1}\ln(n))$	$O(n^{-1}\ln(n))$	0.5	0.5

We consider three different examples with varying assumptions on the sparsity and informativeness of blocks summarized in Table 1. In Examples 1 and 2, external information is discriminative as it singles out a block B_2 that is less sparse than block B_1. In Example 1, the blocks are moderately discriminative, in that $(s_1/p_1)/(s_2/p_2)$ is a power of n. In Example 2, they are highly discriminative in that $\boldsymbol{\beta}^*$ is sparse overall, but it is non-sparse within block 2 ($s_2 > p_2 - s_2$). Example 3 is a non-discriminative average random guess where each block has half of the inactive parameters. The nonzero regression parameters β_i^*, $i \in S$, are set uniformly spaced in $(1/2, 1)$ except for one parameter in each block, which take values $\beta_{\min,1}^*$ and $\beta_{\min,2}^*$ in the last two columns of Table 1.

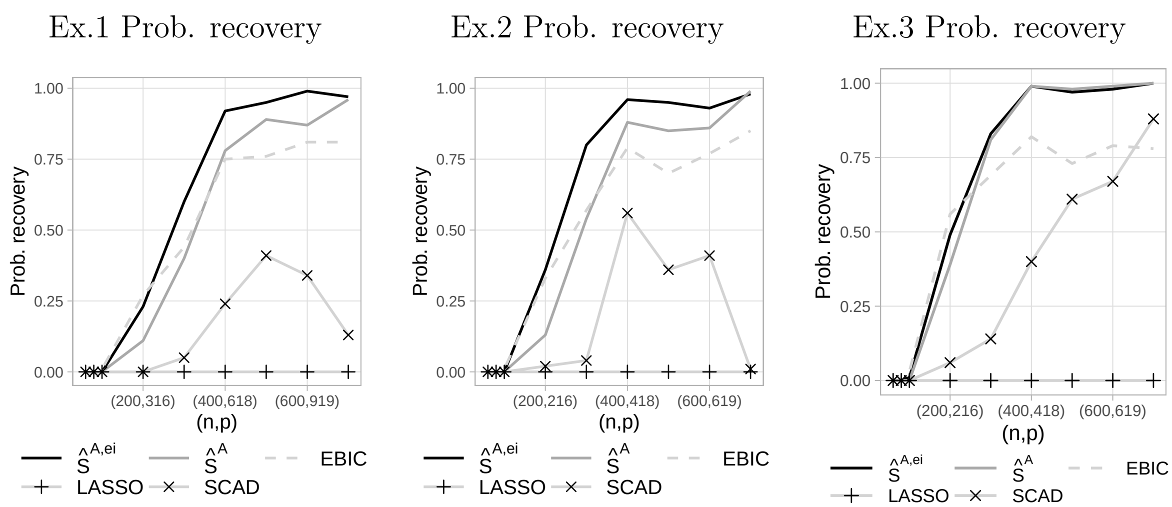

Figure 1: Probability of correct selection with $\hat{S}^{ei}$, $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD in Example 1 (left), 2 (middle), 3 (right).

For each sample size, we generate 100 datasets and report in Figure 1 the empirical probability of correct recovery for $\hat{S}^{A,ei}$, $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD. In Examples 1 and 2 where blocks are discriminative, $\hat{S}^{A,ei}$ outperforms $\hat{S}^{A,sd}$ and performs similarly to the EBIC for small n. For large n, $\hat{S}^{A,ei}$ outperforms the EBIC and performs similarly to $\hat{S}^{A,sd}$. In the non-discriminative block setting of Example 4, $\hat{S}^{A,ei}$ performs very similarly to $\hat{S}^{A,sd}$. LASSO and SCAD have consistently lower probability of recovery than the ℓ_0

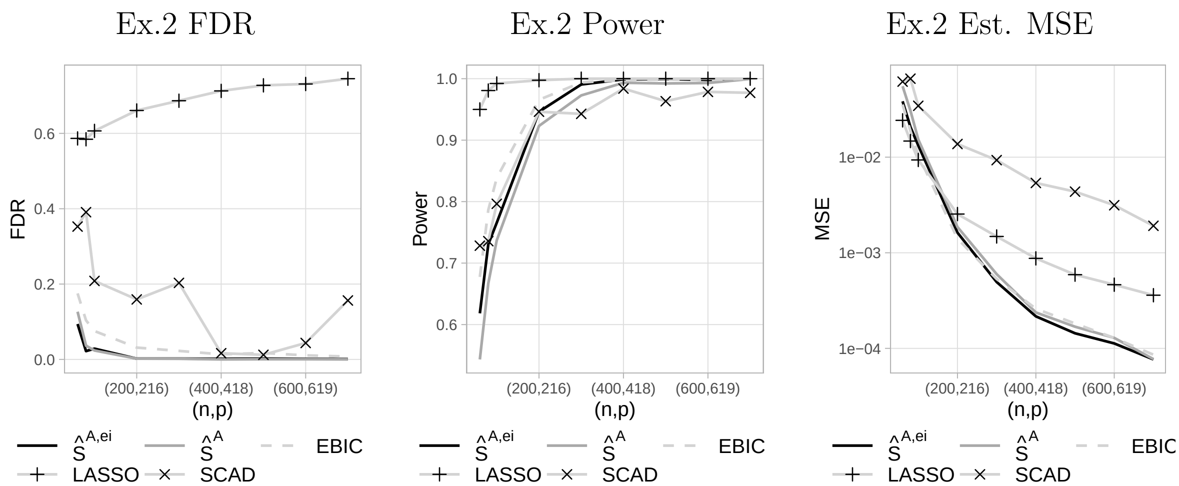

Figure 2: FDR (left), power (middle) and estimation MSE (right) with $\hat{S}^{A,ei}$, $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD in Example 2.

methods. The average false discovery rate (FDR) and power in Example 3 reported in Figure 2 shed light on those differences. Compared to the uninformed $\hat{S}^{A,sd}$, our informed $\hat{S}^{A,ei}$ has equally low FDR but higher power. LASSO and EBIC have higher power than $\hat{S}^{A,ei}$ but higher FDR. In terms of mean squared error, $\hat{S}^{A,ei}$, $\hat{S}^{A,sd}$ and EBIC perform similarly and better than SCAD and LASSO. Similar findings are observed in the other examples, and the corresponding plots can be found in Supplement XXX. We also report in Supplement XXX the average computation time to obtain $\hat{S}^{A,b}$ as a function of dimension p. In all three examples, the average time does not exceed 20 seconds.

6.2 Tran-s-ell0 algorithm

We apply next Tran-s-ell0 to simulated data. We consider target sample sizes n that range from 20 to 700 and, for each sample size, generate 100 sets of target and source datasets. In each target dataset, a data-generating $\boldsymbol{\beta}^*$ is drawn such that $p - s = 3n/2$, $s = 3\ln(n)$, and nonzero coefficients are uniformly spaced in $(1/3, 1)$. Variable values are drawn from a multivariate Gaussian with zero means, unit variances, and pairwise

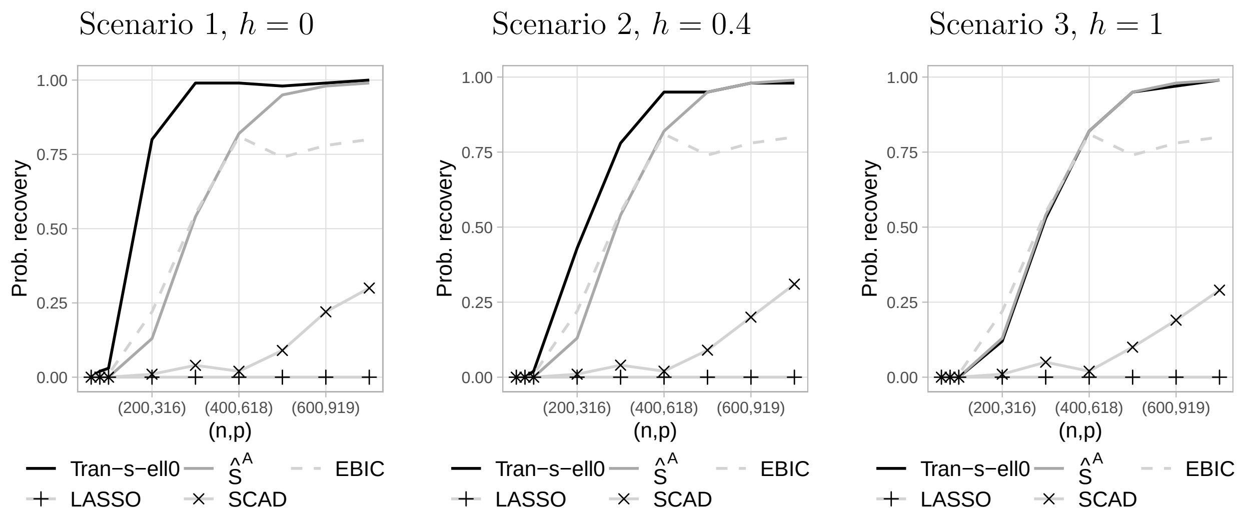

Figure 3: Probability of correct selection with Tran-s-ell0 algorithm, $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD in Example 1 (left), 2 (middle), 3 (right).

correlations $\text{cor}(x_{ij}, x_{il}) = 0.5$ for $j \neq l$. For each target dataset, we generate two source datasets with sample size 700 and data-generating parameters that are equal to the target $\boldsymbol{\beta}^*$ where a proportion h of the nonzero target coefficients is set to 0 and an equal number of the zero target coefficients is set uniformly spaced in $(1/3, 1)$. Source variable are drawn from the same multivariate Gaussian distribution as in the target data. We select variables in the source datasets with a standard ℓ_0 penalty $\kappa = \frac{1}{2}\ln(n) + \ln(p)$ and feed the selected sets to Tran-s-ell0. We consider three scenarios of similarity between source and target sets of active variables and report the average probability of recovery in Figure 3. In Scenario 1, the source and target sets of active variables are the same, that is $h = 0$. In Scenario 2, the source and target sets of active variables are similar but different with $h = 0.4$. Scenario 3 is an example of completely irrelevant transfer where $h = 1$. In Scenario 1 and 2, Tran-s-ell0 algorithm outperforms all the other selectors. In Scenario 3, $\hat{S}^{A,ei}$ performs similarly to $\hat{S}^{A,sd}$, better than LASSO and SCAD, slightly worse than EBIC for small n but signficantly better for large n. ♠ Paul: This is the simulation I initially did. But now I think I should EITHER only simulate sets of variables selected on the sources OR change the source designs to stress our point on the absence of requirement on the similarity of the designs.

Next, we apply Tran-s-ell0 to a biological data set. In molecular biology, it is common to use animal models (e.g. mice) to study genomic mechanisms such as gene expression. Findings from such studies often serve as a basis to study human diseases, for example, to assess whether genes that played key roles in the animal model are also important in humans. Specifically, Calon et al. (2012) compared mice where gene TGFB was silenced to mice where it was not, showed that TGFB plays a crucial role in colon cancer progression, and found a list of 172 genes that were associated with TGFB. We use Tran-s-ell0 to leverage this set of 172 genes in the selection of genes associated to TGFB in a colon cancer dataset of $n = 262$ human samples analyzed in Rossell & Telesca (2017). We regress TGFB expression on that of $p = 1000$ genes. All variables are standardized to zero mean and unit variance. In this dataset, the runtime of Tran-s-ell0 was 49 seconds.

Table 2 reports the genes selected with Tran-s-ell0, $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD. Compared to the uninformed selector $\hat{S}^{A,sd}$, Tran-s-ell0 focuses discoveries on the genes that were found to be associated to TGFB in mice: it selects more genes from the mouse list, and less genes from outside the list. LASSO and SCAD select a much larger number of genes than Tran-s-ell0 but these ℓ_1 methods typically exhibit relatively large FDR in correlated design (Bogdan et al. 2015), as illusatred in our simulations of Section 6.1. The extra discoveries of LASSO and SCAD should then be considered with caution. Finally, the EBIC selects the same genes from the mouse list as Tran-s-ell0 but also three more genes from outside the list. In our simulations, EBIC had a slighlty larger FDR than our informed selector but also a slightly larger power, for small sample size as here. Then, to assessthe relevance of the extra discoveries of EBIC and the quality of the models selected by EBIC and Tran-s-ell0, we report in Table 3 the associated cross-validated mean squared error (CV-MSE) in estimation and some pseudo-posterior inclusion probabilities, that is,

Table 2: Selected genes in each block

	In mouse list		Outside of mouse list	
Method	Number of discoveries	Genes	Number of discoveries	Genes
Tran-s-ell0	4	ESM1, GAS1, HIC1, CILP	0	-
$\hat{S}^{A,sd}$	2	IGFBP3, CILP	2	LGALS1, ARHGEF1
EBIC	4	ESM1, GAS1, HIC1, CILP	3	KCNJ5-AS1, ZBTB46, OXER1
LASSO	22	GAS1, HIC1, IGFBP3, CILP, DNAJB5, ...	55	KCNJ5-AS1, ZBTB46, OXER1, LGALS1, ...
SCAD	11	GAS1, HIC1, IGFBP3, CILP, DNAJB5, ...	5	POLQ, CRIP2, ZBTB46, VSIG4, LGALS1

$p(\beta_i) = \sum_{\mathcal{M}} NC(M)I(i \in M)$. We observe that, compared to the EBIC, Tran-s-ell0 leads to lower mean squared error and to stronger confidence in the genes selected in from the mouse list. We also remark that the model selected by Tran-s-ell0 remains the same when using in Step (ii) an informed selector different from $\hat{S}^{A,ei}$ that has larger power than EBIC in simulations. Those latter results are discussed in Supplement XXX.

Table 3: $\hat{S}^{A,sd}$ vs EBIC: cross-valitated MSE and pseudo-posterior probabilities

		Pseudo-posterior probabilities			
Method	CV-MSE	ESM1	GAS1	HIC1	CILP
EBIC	0.54	0.55	0.88	0.81	0.95
Tran-s-ell0	0.49	0.74	0.90	0.95	0.99

7 Discussion

We discussed how variable selection properties can be improved by leveraging external information on the likelihood of variables to be truly associated to the outcomes. We first analyzed the gains achievable by an oracle ideal selector, providing a benchmark for future concrete selection methods. We proposed then a data-based selector that recovers most of the benefits of the oracle selector. As a particular case, we introduced a flexible and robust structural transfer learning algorithm where the set of variables selected in source datasets is transferred to ease variable selection in a target dataset.

A question for future work is to understand the benefits of leveraging external information and the transfer of selected variables in estimation. Our simulations suggest that improvements can be achieved, but we have not discussed related theory and estimation procedures. This work also focused on external information in variable selection in linear regression. The benefits of incorporating external information and how to make them a reality in other structural learning problems like graphical model selection is another interesting research avenue.

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SUPPLEMENTARY MATERIAL

Tightness of conditions for variable selection consistency in linear regression: comparison of our sufficient and necessary conditions of Section 3 applied to standard

ℓ_0 penalties to results in literature.

Examples of benefits of $\hat{S}^{ei}$: Illustrations in four examples of sparsity and informativeness of the blocks of the benefits of $\hat{S}^{ei}$ in oracle selection properties.

Complementary material on numerical illustrations: Reports of false discovery rates, power, mean squared error and computation time in the numerical illustrations of the performance of $\hat{S}^{ei}$ and Tran-s-ell0.

Proofs

8 Limits on variable selection consistency in linear regression with standard (non-informed) selectors

Sufficient and necessary assumptions for variable selection consistency in (1) under correlated random designs for a (non-informed) oracle selector that knows s were derived by Wainwright (2010) and Wang et al. (2010). We compare those to our assumptions for the standard ℓ_0 selector $\hat{S}^{sd}$. Specifically, we compare the scaling of n implied by the assumptions under six regimes of interest of $(p, s, \beta^*_{\min})$. For simplicity, the effect of the conditioning of $\mathbf{X}$ is ignored, i.e $\rho(\mathbf{X})$, $\bar{\lambda}$ and $\underline{\lambda}$ are considered fixed and bounded away from 0. Wainwright (2010) and Wang et al. (2010) derive sufficient and necessary assumptions in a setting that slightly differs from the one considered here. They study a slightly weaker notion of consistency and necessary conditions are given for the harder task of recovering S when the latter is chosen uniformly at random among subsets of size s. Although their assumptions and ours are then not entirely comparable, the comparison gives an idea of the tightness of our results.

Theorem 3.3 shows variable selection consistency with $\hat{S}^{sd}$ under Assumptions A4 and A5. By (S.59) in the proof of Theorem 3.4, an assumption slightly less stringent than A5, and easier to analyse, is sufficient together with A4. That assumption is:

for each block j, there exists $g_j \to \infty$ such that for every sufficiently large n,

$$\frac{(1-\gamma)n\rho(\mathbf{X})}{6}{\beta^*_{\min,j}}^2 \;-\; \kappa_j \;=\; \ln(s_j) + g_j. \tag{S.25}$$

where $\gamma := \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j) \in (\frac{1}{2}, 1)$

Consider assumptions A5 and A4 for the standard ℓ_0 selector $\hat{S}^{sd}$ with single penalty κ. Their combination implies an assumption on the scaling of $(n, p, s, \beta^*_{\min})$. To simplify the analysis of that assumption, we assume $\kappa = (1+\varepsilon)\ln(p-s)$ for some fixed $\varepsilon > 0$. This choice guarantees that κ meets assumption A4 and that $1-\gamma$ is constant and bounded away from 0. A sufficient assumption on $(n, p, s, \beta^*_{\min})$ that follows from assumptions A5 and (S.25) is then that there exists $t \to \infty$, growing at an arbitrarily slow rate, such that:

$$n = \frac{12(1+\varepsilon)}{\varepsilon}\,\frac{(1+\varepsilon)\ln(p-s) + \ln(s)}{\rho(\mathbf{X}){\beta^*_{\min}}^2} \;+\; t \tag{S.26}$$

Theorem 1 in Wainwright (2010) shows a sufficient assumption on $(n, p, s, \beta^*_{\min})$ for the optimal selector that knows s to be variable selection consistent in random Gaussian designs is:

$$n > (c_1 + 2048)\max\left\{\log\binom{p-s}{s}, \frac{\log(p-s)}{\rho(\mathbf{X}){\beta^*_{\min}}^2}\right\} \tag{S.27}$$

for some $c_1 > 0$.

An immediate initial conclusion is that for any regime of $(n, p, s, \beta^*_{\min})$ such that $\ln(s) = O({\beta^*_{\min}}^2)$, (S.26) is less stringent than (S.27). It is also the case when ${\beta^*_{\min}}^2 = \Theta(1)$ and $s < p/2$. Table **??** gives, for six regimes of interest, the scalings of n implied by (S.26) and (S.27). In regimes 1, 2 and 5 the scalings implied by our sufficient assumptions match those implied by the sufficient assumption of Wainwright (2010). Our assumption is tighter in regimes 3 and 6, with the difference being particularly large in regime 3. In that regime, results by Wainwright (2010) imply a sample size growing linearly in the dimension, while our results shows a logarithmic growth in dimension is sufficient. In regime 4 where $s = o(p)$, our assumption is loser by a term of order $s\ln(s)$. Overall, our results refine the known limits on variable selection in the standard non-informed setting and show a well calibrated standard ℓ_0 penalty performs as well as an optimal selector in the vast majority of regimes.

In this thesis we establish the necessity of assumption (15) which, when applied to $\hat{S}^{sd}$ ($\hat{S}^{ei}$ with $b=1$), implies

$$\lim_{n\to\infty} \sqrt{n\bar{\lambda}}\beta^*_{\min} - \underline{\lambda}\sqrt{\ln(p-s)n} \;=\; \frac{\underline{\lambda}\sqrt{\ln(p-s)}}{\bar{\lambda}{\beta^*_{\min}}^2} + t' \tag{S.28}$$

where $t' \to \infty$ and $1/t = o({\beta^*_{\min}}^2)$. In other words, t must grow faster than ${\beta^*_{\min}}^2$ shrinks. For comparison, Theorem 1 in Wang et al. (2010) gives, for the *harder* problem of recovering S when S is chosen at random in $\mathcal{M}$, the necessary assumption

$$n > \max\left\{f_1(p,s,\lambda),\ldots,f_k(p,s,\lambda),s\right\} \tag{S.29}$$

where

$$f_m(p,s,\lambda) := \frac{\ln\binom{p-s+m}{m} - 1}{\frac{1}{2}\ln\left(1 + m\beta^2_{\min,j}\left(1 - \frac{m}{p-s+m}\right)\right)} \quad \text{for} \;\; m = 1,\ldots,s$$

Table **??** gives, for six regimes of interest, the scalings of n implied by (S.28) and (S.29).

Observe first that the scalings implied by our necessary assumption match those implied by our sufficient assumption in all regimes but 4 and 6. In regimes 4 and 6, where $s = o(p)$, our sufficient assumption is stricter by linear and logarithmic terms in the sparsity s. In regimes 1, 2 and 4, the scalings implied by our necessary assumption and that of Wang et al. (2010) match. In regime 5, our necessary assumption is tighter than that of Wang et al. (2010). In regimes 3 and 6, the necessary assumptions of Wang et al. (2010) are stricter than mine and also stricter than our sufficient assumption. This may be explained by the harder nature of the recovery problem considered by Wang et al. (2010).

9 Examples of benefits of $\hat{S}^{ei}$

We illustrate here the benefits of $\hat{S}^{ei}$ over $\hat{S}^{sd}$ with four concrete examples contemplating different scenarios of sparsity regimes and informativeness of the blocks, summarized in Table 4. We assume an orthonormal design such that $\rho(\mathbf{X}) = 1$ for simplicity and focus on a sparse setting where $\ln(s) = o(\ln(p - s))$, with two blocks ($b = 2$). The smallest non-zero $|\beta_i^*|$, $\beta^*_{\min}$ is assumed to be located in block 2, without loss of generality.

Table 4: Sparsity and block assumptions in Examples 1–4

Example	$p-s$	p_1-s_1	p_2-s_2	s_1	s_2
1	$3n/2$	$3n/2-\sqrt{n}$	$\sqrt{n}$	$3\ln(n)/2$	$3\ln(n)/2$
2	$e^{n/20}$	$e^{n/20}-n^2$	n^2	$3\ln(n)/2$	$3\ln(n)/2$
3	n	$n-\ln(n)$	$\ln(n)$	$3\ln(n)/2$	$3\ln(n)/2$
4	n	$n/2$	$n/2$	$3\ln(n)/2$	$3\ln(n)/2$

In Examples 1 to 3, external information is discriminative as it singles out a block B_1 that is sparser than block B_2. In Example 1 the blocks are moderately discriminative, in that $(p_1 - s_1)/(p_2 - s_2)$ is a power of n. In Example 2 they are highly discriminative, since $(p_1 - s_1)/(p_2 - s_2)$ is exponential in n and in Example 3 they are highly discriminative in that $\boldsymbol{\beta}^*$ is sparse overall, but it is non-sparse within block 2 ($s_2 > p_2 - s_2$). Example 4 is a non-discriminative random guess where each block has half of the inactive parameters.

Figures **??** and **??** plot the ranges of penalty values κ and (κ_1, κ_2) ensuring selection consistency with $\hat{S}^{sd}$ and $\hat{S}^{ei}$ given in (**??**) and (**??**) in the four examples, assuming $\beta^*_{\min,1} = 2/3$, $\beta^*_{\min,2} = \beta^*_{\min} = 1/10$, $f = g = \ln(ln(n))$ and $f_j = g_j = \ln(ln(n))$ for all j. In Example 2, asymptotic recovery with $\hat{S}^{sd}$ is not possible in the range of values of n considered while it is with $\hat{S}^{ei}$. In the other examples, asymptotic recovery with $\hat{S}^{sd}$ is possible, but it requires larger n than with $\hat{S}^{ei}$. Example 1 shows that the gain can be large even when blocks are moderately discriminative, and Example 3 when some blocks are non-sparse. The gain in terms of the value of n making recovery possible is close to null in Example 4 when external information is non-discriminative.

Figure **??** plots the ratio OR^{ei}/OR in (19) for Examples 1 to 3 where blocks are discriminative with γ set to 3/4 for simplicity. The smallest signals in each block are such that $\beta^*_{\min} = \beta^*_{\min,2} = 1/5$ and $\beta^*_{\min,1} = 1.3\beta^*_{\min}$ to guarantee the recovery is possible with both $\hat{S}^{sd}$ and $\hat{S}^{b}$. Figure **??** shows that in the three examples, the convergence is much faster with $\hat{S}^{b}$. The more discriminative the blocks are, the larger the gain. ♠ Paul: I have made simplifications assumptions on γ in the discussions of the oracle convergence rates but in reality κ^* depends on γ and γ depends on κ^*.

10 Proofs

Section 10.1 presents results auxiliary to the rest of the proofs, as well as the proofs of that auxiliary results. Section 10.2 the proofs of Section 3 and Section 10.3 the proofs of Sections 4 and 5.

10.1 Auxiliary results

In this section we collect some technical results auxiliary to the rest of the proofs.

10.1.1 Tail bounds

Lemma 10.1. *(i) If $\mathbf{y} \sim N_p(0, n^{-1}I_p)$, $p > 1$, and $a \geq \sqrt{2\ln(p)/n}$,*

$$P\left(\max_{i\in\{1,\dots,p\}} |y_i| > a\right) \leq \frac{e^{-\frac{n}{2}\left(a^2 - \frac{2\ln(p)}{n}\right)}}{\sqrt{\pi \ln(p)}}.$$

(ii) If $y \sim N(\mu, \sigma^2)$ and $a \leq |\mu|$,

$$P(|y| > a) = P\left(z < \frac{|\mu| - a}{\sigma}\right) + P\left(z < -\frac{a + |\mu|}{\sigma}\right)$$

Proof. Part (i) By the union bound and the identical distribution of the y_i's,

$$P\Big(\max_{i\in\{1,\dots,p\}}|y_i|>a\Big)\le \textstyle\sum_{i=1}^{p} P\left(|\sqrt{n}y_i|>\sqrt{n}a\right)=p\,P\left(|z|>\sqrt{n}a\right)$$

where $z\sim N(0,1)$. By symmetry and using the standard tail bound for standard normal, $P(z\ge\delta)\le(2\pi)^{-1/2}\delta^{-1}e^{-\delta^2/2}$ for $\delta\ge 0$, we obtain that

$$P\Big(\max_{i\in\{1,\dots,p\}}|y_i|>a\Big)\le p\,2\,P\left(z>\sqrt{n}a\right)\le \tfrac{1}{\sqrt{\pi}}\tfrac{\sqrt{2}}{\sqrt{n}a}pe^{-\frac{n}{2}a^2}.$$

Since $a\ge\sqrt{2\ln(p)/n}$, we have that $\sqrt{2}/\sqrt{n}a\le 1/\sqrt{\ln(p)}$. Taking $a^2=a^2-\frac{2\ln(p)}{n}+\frac{2\ln(p)}{n}$, we get

$$P\Big(\max_{i\in\{1,\dots,p\}}|y_i|>a\Big)\le \tfrac{1}{\sqrt{\pi\ln(p)}}e^{-\frac{n}{2}\left(a^2-\frac{2\ln(p)}{n}\right)}.$$

Part (ii) We have

$$P(|y|>a)=P(y>a)+P(y<-a)=P(z>\frac{a-\mu}{\sigma})+P(z<-\frac{a+\mu}{\sigma})$$

where $z\sim N(0,1)$. If $\mu>0$, $P(z>\frac{a-\mu}{\sigma})=P(z>\frac{a-|\mu|}{\sigma})=P(z<\frac{|\mu|-a}{\sigma})$ by symmetry of the standard Gaussian, and $P(z<-\frac{a+\mu}{\sigma})=P(z<-\frac{a+|\mu|}{\sigma})$, then $P(|y_i|>a)=P(z<\frac{|\mu|-a}{\sigma})+P(z<-\frac{a+|\mu|}{\sigma})$. If $\mu\le 0$, then $P(z>\frac{a-\mu}{\sigma})=P(z>\frac{a+|\mu|}{\sigma})=P(z<-\frac{a+|\mu|}{\sigma})$, by symmetry of the standard Gaussian, and $P(z<-\frac{a+\mu}{\sigma})=P(z<\frac{|\mu|-a}{\sigma})$. Then for any μ, $P(|y_i|>a)=P(z<\frac{|\mu|-a}{\sigma})+P(z<-\frac{a+|\mu|}{\sigma})$. ∎

Lemma 10.2. *Let $W\sim\chi^2_\nu(\mu)$ with $\mu\ge 0$, then for any $w>\mu+\nu$*

$$P(W>w)\le e^{-\left(\frac{w+\mu}{2}-\sqrt{2w(2\mu+\nu)-2\mu\nu-\nu^2}\right)}.$$

Moreover, assume w, ν and μ are functions of n such that w is increasing, $\nu=o(w)$, and $\mu=o(w)$. Then, for any $\phi\in(0,1)$ and n large enough

$$P(W>w)\le e^{-\phi\frac{w}{2}}.$$

Proof. By (n.d.), Lemma 8.1 we have that for any $x > 0$

$$P(W > (\nu + \mu) + 2\sqrt{(\nu + 2\mu)x} + 2x) \le e^{-x}.$$

The function $f : x \mapsto (\nu + \mu) + 2\sqrt{(\nu + 2\mu)x} + 2x$ is one-to-one between $\mathbb{R}^+$ and $(\nu + \mu, \infty)$. Hence, we have that for any $w > \mu + \nu$,

$$P(W > w) \le e^{-f^{-1}(w)} \;=\; e^{-\left(\frac{w+\mu}{2} - \sqrt{2w(2\mu+\nu) - 2\mu\nu - \nu^2}\right)}.$$

Observe that

$$\frac{w + \mu}{2} - \sqrt{2w(2\mu + \nu) - 2\mu\nu - \nu^2} = \frac{w}{2}\left(1 + \frac{\mu}{w} - \sqrt{\frac{8(2\mu + \nu)}{w}\left(1 - \frac{2\nu\mu - \nu^2}{2(2w\mu + w\nu)}\right)}\right).$$

Since $\nu = o(w)$ and $\mu = o(w)$ by assumption, we have $\frac{w+\mu}{2} - \sqrt{2w(2\mu + \nu) - 2\mu\nu - \nu^2} = \frac{w}{2}(1 + o(1))$. Therefore, for any $\phi \in (0, 1)$ and every n large enough.

$$P(W > w) \le e^{-\phi\frac{w}{2}}.$$

□

Lemma 10.3. *Let $W \sim \chi^2_\nu(\mu)$ with $\mu > 0$. For any $w < \mu$,*

$$P(W < w) \le \frac{e^{-\frac{1}{2}(\sqrt{\mu} - \sqrt{w})^2}}{(\mu/w)^{\nu/4}}.$$

Proof. The result follows directly from Rossell (2022), Lemma S2. □

Lemma 10.4. *Let $W \sim \chi^2_\eta(\mu)$ with $\mu \ge 0$. Assume that h, c, η and μ are functions of n such that h is positive and increasing, c is positive, $\eta = o(c \log(h))$, and $\mu = o(c \log(h))$. Let $\bar{u}, \underline{u}$ in $(0, 1)$ such that $1 > \bar{u} > \underline{u} \ge \left(1 + h^\psi e^{-(\eta+\mu)/c}\right)^{-1}$ where $\psi \in (0, 1)$, then for every n large enough, we have*

$$\int_{\underline{u}}^{\bar{u}} P\left(W > c\log\left(\frac{h}{1/u - 1}\right)\right) du \le \frac{1}{h^\psi}\left(\bar{u} - \underline{u} + \log\left(\frac{\bar{u}}{\underline{u}}\right)\right).$$

Proof. For any $u \in [\underline{u}, \bar{u}]$, we have $c\log\left(\frac{h}{1/u-1}\right) \geq c\log\left(\frac{h}{1/\underline{u}-1}\right)$. Since $\underline{u} \geq (1 + h^{\psi}e^{-(\eta+\mu)/c})^{-1}$ by assumption we also have that, for any $u \in (\underline{u}, \bar{u})$,

$$c\log\left(\frac{h}{1/u-1}\right) \geq c(1-\psi)\log(h) + \eta + \mu.$$

It follows,

$$\frac{\eta}{c\log\left(\frac{h}{1/u-1}\right)} \leq \frac{\eta}{c(1-\psi)\log(h)+\eta+\mu} \quad \text{and} \quad \frac{\mu}{c\log\left(\frac{h}{1/u-1}\right)} \leq \frac{\mu}{c(1-\psi)\log(h)+\eta+\mu}.$$

By assumption $\eta = o(c\log(h))$ and $\mu = o(c\log(h))$, then for any $u \in (\underline{u}, \bar{u})$, $\eta = o\left(c\log\left(\frac{h}{1/u-1}\right)\right)$ and $\mu = o\left(c\log\left(\frac{h}{1/u-1}\right)\right)$. By Lemma 10.2, for any $\psi \in (0,1)$ and every n large enough,

$$\int_{\underline{u}}^{\bar{u}} P\left(W > c\log\left(\frac{h}{1/u-1}\right)\right) du < \frac{1}{h^{\psi}} \int_{\underline{u}}^{\bar{u}} (1/u-1)^{\psi} du. \tag{S.30}$$

Applying the change of variables $v = 1/u - 1$ to the integral on the right-hand side above gives

$$\int_{\underline{u}}^{\bar{u}} (1/u-1)^{\psi} du = \int_{1/\bar{u}-1}^{1/\underline{u}-1} \frac{v^{\psi}}{(v+1)^2} dv. \tag{S.31}$$

Rewrite $v^{\psi} = (v-1+1)^{\psi}$. Since $\bar{u} < 1$, we have that for any $v > 1/\bar{u}-1$, $v-1 > -1$. Note that for any $x \geq -1$ and $r \in [0,1]$ $(1+x)^r \leq 1 + rx$. Then, for any $v > 1/\bar{u}-1$, $v^{\psi} = (v-1+1)^{\psi} \leq 1 + \psi(v-1) \leq 1 + \psi(v+1)$. Applying this last inequality to the right-hand side in (S.31) gives

$$\int_{\underline{u}}^{\bar{u}} (1/u-1)^{\psi} du < \int_{1/\bar{u}-1}^{1/\underline{u}-1} \frac{1}{(v+1)^2} + \frac{\psi}{v+1} dv = \bar{u} - \underline{u} + \psi\log\left(\frac{\bar{u}}{\underline{u}}\right). \tag{S.32}$$

The result follows inputting the bound from (S.32) in (S.30) and using that $\psi < 1$ and $\log(\bar{u}/\underline{u}) \geq 0$ ($\underline{u} \leq \bar{u}$). □

Lemma 10.5. *Let $W \sim \chi^2_{\eta}(\mu)$ with $\mu \geq 0$. Assume that h, c, η and μ are functions of n such that h is positive and increasing, c is positive, $\eta = o(c\log(h))$, and $\mu = o(c\log(h))$. Then for any $\alpha \in (0,1)$ and every large enough n, we have*

$$\int_0^1 P\left(W > c\log\left(\frac{h}{1/u-1}\right)\right) du = o\left(h^{-\alpha}\right).$$

Proof. Since a probability is bounded by 1, for any $a \in (0,1)$,

$$\int_0^1 P\left(W > c\log\left(\frac{h}{1/u-1}\right)\right) du \;\leq\; 2a + \int_a^{1-a} P\left(W > c\log\left(\frac{h}{1/u-1}\right)\right) du. \qquad \text{(S.33)}$$

Take $a = \left(1 + h^{\psi} e^{-(\eta+\mu)/c}\right)^{-1}$ for some $\psi \in (\alpha, 1)$. By Lemma 10.4 with $\underline{u} = a$ and $\bar{u} = 1-a$, we have that

$$\int_0^1 P\left(W > c\log\left(\frac{h}{1/u-1}\right)\right) du \;\leq\; \frac{2}{1 + h^{\psi} e^{-(\eta+\mu)/c}} + \frac{1 - 2a + \log(h^{\psi} e^{-(\eta+\mu)/c})}{h^{\psi}}.$$

Since $1 + h^{\psi} e^{-(\eta+\mu)/c} > h^{\psi} e^{-(\eta+\mu)/c}$ and $1 - 2a - \frac{\eta+\mu}{c} \leq \log(h^{\psi})$ for every n large enough, we have

$$\int_0^1 P\left(W > 2\log\left(\frac{h}{1/u-1}\right)\right) du \;\leq\; h^{-\psi}\left(2e^{(\eta+\mu)/c} + 2\log(h^{\psi})\right)$$

We have that

$$\frac{h^{-\psi} 2e^{(\eta+\mu)/c}}{h^{-\alpha}} = e^{-(\psi-\alpha)\log(h) + \log(2) + (\eta+\mu)/c} \;=\; e^{-(\psi-\alpha)\log(h)\left(1 - \frac{\eta+\mu}{c(\psi-\alpha)\log(h)} - \frac{\log(2)}{(\psi-\alpha)\log(h)}\right)} \qquad \text{(S.34)}$$

and similarly that

$$\frac{h^{-\psi} 2\log(h^{\psi})}{h^{-\alpha}} \;=\; e^{-(\psi-\alpha)\log(h)\left(1 - \frac{\log(\psi\log(h))}{(\psi-\alpha)\log(h)} - \frac{\log(2)}{(\psi-\alpha)\log(h)}\right)}. \qquad \text{(S.35)}$$

Since $\alpha < \psi$ as stated above, and by assumption h is increasing, $\eta = o(c\log(h))$ and $\mu = o(c\log(h))$, both expressions in (S.34) and (S.35) vanish as n grows. Hence,

$$\int_0^1 P\left(W > 2\log\left(\frac{h}{1/u-1}\right)\right) du \;=\; o(h^{-\alpha}).$$

□

10.1.2 Bounds related to model normalized scores

Let $NC(M)$ be the normalized score for model M defined in (9), $\mathcal{M}$ the set of models under consideration. Lemma 10.6 generalizes Lemma 3.2 and shows that the probability of not selecting a set of models is bounded above by the expected sum of the normalized scores of the models outside the set. Lemma 10.7 shows the difference in squared errors for two nested models is chi-squared distributed. Lemma 10.8 and Lemma 10.9 give bounds on the noncentrality parameter of that chi-squared distribution. Lemma 10.10 provides a bound on the expected normalized score of any $M \in \mathcal{M}$ based on the pairwise comparison $C(T) - C(M)$ (*cf* (9)) for some $T \subseteq S$. It essentially shows that, if the block penalties diverge and the signals in $M \setminus T$ are small, $NC(M)$ is small. Prior to stating Lemma 10.10 we outline the proof strategy.

Lemma 10.6. *For $\hat{S}^{ei}$ as in (8) and any k models $M^1, \ldots, M^k$*

$$P(\hat{S}^{ei} \notin \{M^1, \ldots, M^k\}) \;\le\; (k+1) \sum_{M \in \mathcal{M} \setminus \{M^1, \ldots, M^k\}} \mathbb{E}\left(NC(M)\right).$$

Proof. Suppose that $NC(M^1) + \ldots + NC(M^k) > \frac{k}{k+1}$ then for any $M \notin \{M^1, \ldots, M^k\}$, we have that

$$NC(M) = 1 - \sum_{M' \neq M} NC(M') < 1 - \sum_{M' \in \{M^1, \ldots, M^k\}} NC(M') < \frac{1}{k+1}.$$

In addition, if $NC(M^1) + \ldots + NC(M^k) > \frac{k}{k+1}$ then necessarily $\max_{l=1,\ldots,k} NC(M_l) > \frac{1}{k+1} > NC(M)$ for any $M \notin \{M^1, \ldots, M^k\}$, and therefore $\hat{S}^{ei} \in \{M^1, \ldots, M^k\}$. Consequently,

$$P(\hat{S}^{ei} \notin \{M^1, \ldots, M^k\}) \;\le\; P\Big(NC(M^1) + \ldots + NC(M^k) \le \frac{k}{k+1}\Big).$$

Moreover, we have

$$P\Big(NC(M^1) + \ldots + NC(M^k) \le \frac{k}{k+1}\Big) \;=\; P\Big(\sum_{M \in \mathcal{M} \setminus \{M^1, \ldots, M^k\}} NC(M) \ge \frac{1}{k+1}\Big).$$

The result follows from the Markov's inequality applied to the right-hand side above. $\square$

Lemma 10.7. *Let M, Q be any two nested models such that $M \subseteq Q$. Then*

$$L_{QM} \;=\; \|\mathbf{X}_Q \tilde{\boldsymbol{\beta}}^{(Q)}\|^2 - \|\mathbf{X}_M \tilde{\boldsymbol{\beta}}^{(M)}\|^2 \;\sim\; \chi^2_{|Q \setminus M|}\left(\mu_{QM}\right),$$

where $\chi_k^2(\mu)$ denotes, when $\mu > 0$, the noncentral chi-squared distribution with k degrees of freedom and noncentrality parameter μ and, when $\mu = 0$, the chi-squared distribution with k degrees of freedom χ_k^2. The parameter μ_{QM} is given by

$$\mu_{QM} \;:=\; \|\Big(I_n - P_M\Big)\mathbf{X}_{Q\setminus M}\boldsymbol{\beta}^*_{Q\setminus M}\|^2 \tag{S.36}$$

where $P_M = \mathbf{X}_M\left(\mathbf{X}_M^\top\mathbf{X}_M\right)^{-1}\mathbf{X}_M^\top$.

Proof. This result follows directly from Lemma S7 in Rossell (2022) taking $\phi^* = 1$. $\square$

Lemma 10.8. *For any $T \subseteq S$ and $M \in \mathcal{M}$ such that $T \not\subseteq M$, let $Q_T = M \cup T$. The non-centrality parameter defined in (S.36) satisfies:*

$$\mu_{Q_T M} \;\geq\; n\,\rho(\mathbf{X})\sum_{j=1}^{b}|T_j \setminus M_j|\min_{i\in T_j\setminus M_j}\beta_i^{*2}. \tag{S.37}$$

Proof. The non-centrality parameter $\mu_{Q_T M}$, as defined in (S.36), satisfies

$$\begin{aligned}
\mu_{Q_T M} &= \|\Big(I_n - P_M\Big)\mathbf{X}_{Q_T\setminus M}\boldsymbol{\beta}^*_{Q_T\setminus M}\|^2 \\
&= \|\Big(I_n - P_M\Big)\mathbf{X}_{T\setminus M}\boldsymbol{\beta}^*_{T\setminus M}\|^2 \\
&= n{\beta^*_{T\setminus M}}^\top\left(\tfrac{1}{n}\mathbf{X}_{T\setminus M}^\top(I_n - P_M)\mathbf{X}_{T\setminus M}\right)\beta^*_{T\setminus M} \\
&\geq n\lambda_{\min}\left(\tfrac{1}{n}\mathbf{X}_{T\setminus M}^\top(I_n - P_M)\mathbf{X}_{T\setminus M}\right)\|\beta^*_{T\setminus M}\|^2,
\end{aligned}$$

where the second equality follows from observing that $Q_T\setminus M = T\setminus M$. Since T is a subset of S, by reordering columns, $X_{S\setminus M} = [X_{T\setminus M}, X_{S\setminus(M\cup T)}]$ and therefore $\frac{1}{n}\mathbf{X}_{T\setminus M}^\top(I_n - P_M)\mathbf{X}_{T\setminus M}$ is a principal submatrix of $\frac{1}{n}\mathbf{X}_{S\setminus M}^\top(I_n - P_M)\mathbf{X}_{S\setminus M}$. Hence, Cauchy's interlacing theorem gives that $\lambda_{\min}\left(\frac{1}{n}\mathbf{X}_{T\setminus M}^\top(I_n - P_M)\mathbf{X}_{T\setminus M}\right) \geq \lambda_{\min}\left(\frac{1}{n}\mathbf{X}_{S\setminus M}^\top(I_n - P_M)\mathbf{X}_{S\setminus M}\right)$. Finally, by definition of $\rho(\mathbf{X})$ in Assumption (A5) we have that $\lambda_{\min}\left(\frac{1}{n}\mathbf{X}_{S\setminus M}^\top(I_n - P_M)\mathbf{X}_{S\setminus M}\right) \geq \rho(\mathbf{X})$, and further noting that $\|\beta^*_{T\setminus M}\|^2 \geq \sum_{j=1}^{b}|T_j \setminus M_j|\min_{i\in T_j\setminus M_j}\beta_i^{*2}$ gives the desired result. $\square$

Lemma 10.9. *For any $T \subseteq S$ and $M \in \mathcal{M}$, let $Q_T = M \cup T$, and $\mu_{Q_T T}$ as defined in (S.36), then,*

$$\mu_{Q_T T} \leq n\,\bar{\lambda} \sum_{j=1}^{b} |(S_j \cap M_j) \setminus T_j| \max_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2} \quad \textit{where} \quad \bar{\lambda} := \lambda_{\max}\Big(\frac{1}{n}\mathbf{X}_S^\top \mathbf{X}_S\Big) \tag{S.38}$$

and

$$\mu_{Q_T T} \geq n\,\lambda_{\min}\Big(\frac{1}{n}\mathbf{X}_S^\top \mathbf{X}_S\Big) \sum_{j=1}^{b} |(S_j \cap M_j) \setminus T_j| \min_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2} \tag{S.39}$$

Proof. Using the definition of $\mu_{Q_T T}$ in (S.36), we have that

$$\begin{aligned} \mu_{Q_T T} &= {\boldsymbol{\beta}^*_{Q_T \setminus T}}^\top \mathbf{X}_{Q_T \setminus T}^\top \Big(I_n - P_T\Big) \mathbf{X}_{Q_T \setminus T} \boldsymbol{\beta}^*_{Q_T \setminus T} \\ &= {\boldsymbol{\beta}^*_{M \setminus T}}^\top \mathbf{X}_{M \setminus T}^\top \Big(I_n - P_T\Big) \mathbf{X}_{M \setminus T} \boldsymbol{\beta}^*_{M \setminus T} \\ &= {\boldsymbol{\beta}^*_{(S \cap M) \setminus T}}^\top \mathbf{X}_{(S \cap M) \setminus T}^\top \Big(I_n - P_T\Big) \mathbf{X}_{(S \cap M) \setminus T} \boldsymbol{\beta}^*_{(S \cap M) \setminus T} \end{aligned} \tag{S.40}$$

where the second equality follows from $Q_T \setminus T = M \setminus T$ and the third equality from $\boldsymbol{\beta}^*_{M \setminus S} = 0$. We start by showing the upper bound in (S.38). Denote for any square matrix A, its largest eigenvalue $\lambda_{\max}(A)$. By (S.40), we have that

$$\mu_{Q_T T} \leq n\lambda_{\max}\Big(\frac{1}{n}\mathbf{X}_{(S \cap M) \setminus T}^\top \Big(I_n - P_T\Big) \mathbf{X}_{(S \cap M) \setminus T}\Big) \|\boldsymbol{\beta}^*_{(S \cap M) \setminus T}\|^2.$$

Let $B := \frac{1}{n}\mathbf{X}_{(S \cap M) \setminus T}^\top \Big(I_n - P_T\Big) \mathbf{X}_{(S \cap M) \setminus T}$, $C := \frac{1}{n}\mathbf{X}_{(S \cap M) \cup T}^\top \mathbf{X}_{(S \cap M) \cup T}$ and $D := \frac{1}{n}\mathbf{X}_T^\top \mathbf{X}_T$. D is a principal submatrix of C and B is the Schur complement of D of C. The inverse B^{-1} is then a principal submatrix of C^{-1}, and by Cauchy's interlacing theorem we have that $\lambda_{\min}(B^{-1}) \geq \lambda_{\min}(C^{-1})$ and then $\lambda_{\max}(B) \leq \lambda_{\max}(C)$. Since $T \subseteq S$ by assumption, we also have that $(S \cap M) \cup T \subseteq S$, then by interlacing again $\lambda_{\max}(C) \leq \lambda_{\max}\Big(\frac{1}{n}\mathbf{X}_S^\top \mathbf{X}_S\Big) = \bar{\lambda}$. The upper bound in (S.38) follows from the latter inequality and also observing that $\|\boldsymbol{\beta}^*_{(S \cap M) \setminus T}\|_2^2 \leq \sum_{j=1}^{b} |(S_j \cap M_j) \setminus T_j| \max_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2}$.

We now derive the lower bound in (S.39). By (S.40), we have that

$$\mu_{Q_T T} \geq n\lambda_{\min}\Big(\frac{1}{n}\mathbf{X}_{(S \cap M) \setminus T}^\top \Big(I_n - P_T\Big) \mathbf{X}_{(S \cap M) \setminus T}\Big) \|\boldsymbol{\beta}^*_{(S \cap M) \setminus T}\|_2^2.$$

Recall that B^{-1} is a principal submatrix of C^{-1}, hence by interlacing $\lambda_{\max}(B^{-1}) \leq \lambda_{\max}(C^{-1})$ and $\lambda_{\min}(B) \geq \lambda_{\min}(C)$. Since $T \subseteq S$ by assumption, we have that $(S \cap M) \cup T \subseteq S$, and hence $\lambda_{\min}(C) \geq \lambda_{\min}(\frac{1}{n}\mathbf{X}_S^\top \mathbf{X}_S)$. The bound in (S.39) is obtained by using the latter inequality and noting that also $\|\boldsymbol{\beta}^*_{(S\cap M)\setminus T}\|_2^2 \geq \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j| \min_{i\in(S_j\cap M_j)\setminus T_j} \beta_i^{*2}$. $\square$

Lemma 10.10 next gives a bound on the expected normalized score of any $M \in \mathcal{M}$. We outline now the strategy to prove it. Recall that by Lemma 3.2, for any M, M', the normalized score $NC(M)$ is bounded by a simple function of $C(M') - C(M)$. In the next lemma we take $M' = T$ where $T \subseteq S$. By the definition in (9), $C(T) - C(M) = \frac{1}{2}L_{TM} + \Delta_{MS}$, where for any two models $M, T \subseteq V$, we denote

$$\Delta_{MT} := \sum_{j=1}^b \kappa_j(|M_j| - |T_j|) \quad \text{and} \quad L_{TM} := \|\mathbf{X}_T\tilde{\boldsymbol{\beta}}^{(T)}\|^2 - \|\mathbf{X}_M\tilde{\boldsymbol{\beta}}^{(M)}\|^2. \tag{S.41}$$

To lower bound $C(T) - C(M)$, we use that L_{TM} can be expressed in terms of chi-squared variables. The idea is to take the union model $Q_T = T \cup M$, and to note that $L_{TM} = L_{Q_TM} - L_{Q_TT}$. Lemma 10.7 is used to bound L_{Q_TM}, L_{Q_TT}, and hence also L_{TM}.

If $M \supset T$, then $Q_T = M$ and $-L_{TM} = L_{Q_TT}$ and by Lemma 10.7, $-L_{TM} \sim \chi^2_{|Q_T\setminus T|}(\mu_{Q_TT})$, which has expectation $|Q_T \setminus T| + \mu_{Q_TT}$. It also holds that $\Delta_{MT} > 0$. Using properties of chi-squared distribution, we show that if one sets large enough κ_j (and thus Δ_{MT}), then $C(T) - C(M)$ is also large and $NC(M)$ vanishes. If $M \subset T$, then $Q_T = T$, $L_{TM} = L_{Q_TM}$, and by Lemma 10.7 $L_{TM} \sim \chi^2_{|Q_T\setminus M|}(\mu_{Q_TM})$, which has expectation $|Q_T \setminus M| + \mu_{Q_TM}$. If the noncentrality parameter μ_{Q_TM} is large enough, then L_{TM} and $C(T) - C(M)$ are also large, and, by the properties of the noncentral chi-squared distribution, $NC(M)$ vanishes. By Lemma 10.8, large μ_{Q_TM} can be achieved by setting a condition on signal strength and $\rho(\mathbf{X})$.

If $M \neq T$ is such that $M \not\supset T$ and $M \not\subset T$, simultaneously large enough κ_j and μ_{Q_TM} guarantee that $C(T) - C(M)$ is also large, and that $NC(M)$ vanishes.

Lemma 10.10. *For any $T \subseteq S$ and $M \in \mathcal{M} \setminus \{T\}$, denote $Q_T = M \cup T$ and $A_T := \gamma\Delta_{MT} + \frac{1-\gamma}{6}\mu_{Q_TM}$ (*cf (S.41) and (S.36)*). Suppose that, for some $\gamma \in (1/2, 1]$, it holds that $A_T > 0$, $|M \setminus T| = o(A_T)$, and $\mu_{Q_TT} = o(A_T)$. For any $\psi \in (0,1)$ and every n large enough,*

$$\mathbb{E}\left(NC(M)\right) \;\; \leq \;\; e^{-\psi A_T}.$$

Proof. Since $M \in \mathcal{M} \setminus \{T\}$, by Lemma 3.2 (ii), $NC(M) \leq (1 + e^{C(T)-C(M)})^{-1} \in [0,1]$. The first step of the proof is to use that for any random variable $Z \geq 0$ we have $\mathbb{E}(Z) = \int_0^\infty P(Z > u)du$, so that

$$\begin{aligned}
\mathbb{E}\left(NC(M)\right) \;\; &\leq \;\; \int_0^1 P\left((1 + e^{C(T)-C(M)})^{-1} \geq u\right) \mathrm{d}u \\
&= \;\; \int_0^1 P\left(C(T) - C(M) \leq \ln\left(\tfrac{1}{u} - 1\right)\right) \mathrm{d}u \\
&= \;\; \int_0^1 P\Big(-\tfrac{1}{2}L_{TM} \geq \Delta_{MT} - \ln\left(\tfrac{1}{u} - 1\right)\Big)\mathrm{d}u.
\end{aligned}$$

The second step of the proof is to use the union bound to upper bound the probability in the integrand above. Let $Q_T = M \cup T$ and recall that $L_{TM} = L_{Q_TM} - L_{Q_TT}$. For any γ

$$-\tfrac{1}{2}L_{TM} - (\Delta_{MT} - \ln\left(\tfrac{1}{u} - 1\right)) \;\; = \;\; \left(\tfrac{1}{2}L_{Q_TT} - \ln\left(\frac{e^{\gamma\Delta_{MT}}}{\frac{1}{u} - 1}\right)\right) - \left(\tfrac{1}{2}L_{Q_TM} + (1-\gamma)\Delta_{MT}\right).$$

Observe that for any random variables U, V, and any $\epsilon, \gamma' \geq 0$, the event $\{U - V \geq 0\}$ implies $\{U \geq \gamma'\epsilon\} \cup \{V < \gamma'\epsilon\}$. Let $U = \frac{1}{2}L_{Q_TT} - \ln\left(\frac{e^{\gamma\Delta_{MT}}}{1/u-1}\right)$ and $V = \frac{1}{2}L_{Q_TM} + (1-\gamma)\Delta_{MT}$. Take $\epsilon = \mu_{Q_TM}$ and $\gamma' = \frac{1}{6}(1-\gamma)$, and observe that $A_T := \gamma\Delta_{MT} + \frac{1-\gamma}{6}\mu_{Q_TM} = \gamma\Delta_{MT} + \gamma'\mu_{Q_TM}$. We then have

$$\begin{aligned}
\{U \geq \gamma'\epsilon\} \;\; &= \;\; \{\tfrac{1}{2}L_{Q_TT} \geq \ln\left(\frac{e^{\gamma\Delta_{MT}}}{\frac{1}{u} - 1}\right) + \gamma'\mu_{Q_TM}\} \;\; = \;\; \{\tfrac{1}{2}L_{Q_TT} \geq \ln\left(\frac{e^{A_T}}{\frac{1}{u} - 1}\right)\} \\
\{V < \gamma'\epsilon\} \;\; &= \;\; \{\tfrac{1}{2}L_{Q_TM} < -(1-\gamma)\Delta_{MT} + \gamma'\mu_{Q_TM}\} \;\; = \;\; \{\tfrac{1}{2}L_{Q_TM} < \gamma'(\mu_{Q_TM} - 6\Delta_{MT})\}.
\end{aligned}$$

By the union bound we have that

$$\mathbb{E}\left(NC(M)\right) \leq \int_0^1 P\Big(\tfrac{1}{2}L_{Q_TT} \geq \ln\left(\frac{e^{A_T}}{\frac{1}{u} - 1}\right)\Big)\mathrm{d}u + P\Big(\tfrac{1}{2}L_{Q_TM} < \gamma'(\mu_{Q_TM} - 6\Delta_{MT})\Big). \quad \text{(S.42)}$$

The third and final step of the proof is to upper bound each of the terms in the right-hand side of (S.42). The intuition is that both T and M are nested within Q_T, and therefore $L_{Q_T T}$ and $L_{Q_T M}$ follow chi-squared distributions. We first bound the first term. If $M \subset T$ then $Q_T = T$, $L_{Q_T T} = 0$, and this term is zero. Suppose now that $M \not\subset T$. Then, by Lemma 10.7, $L_{Q_T T} \sim \chi^2_{|Q_T \setminus T|}(\mu_{Q_T T})$ with $|Q_T \setminus T| = |M \setminus T|$. By assumption, $A_T > 0$, $|M \setminus T| = o(\ln(e^{A_T}))$ and $\mu_{Q_T T} = o(\ln(e^{A_T}))$, then by Lemma 10.5, for $\alpha \in (\psi, 1)$, and every n large enough,

$$\int_0^1 P\Big(L_{Q_T T} > 2\ln\Big(\frac{e^{A_T}}{1/u-1}\Big)\Big)\mathrm{d}u \;<\; e^{-\alpha A_T}. \tag{S.43}$$

We now bound the second term in (S.42). If $M \supset T$, then $Q_T = M$, $L_{Q_T M} = 0$, $\mu_{Q_T M} = 0$, and this term is zero. Alternatively, if $M \not\supset T$ then, by Lemma 10.7, $L_{Q_T M} \sim \chi^2_{|Q_T \setminus M|}(\mu_{Q_T M})$ with $|Q_T \setminus M| = |T \setminus M|$. Clearly, when $\mu_{Q_T M} \le 6\Delta_{MT}$ this term is also zero, so suppose that $\mu_{Q_T M} > 6\Delta_{MT}$. We have

$$P\left(L_{Q_T M} < 2\gamma'(\mu_{Q_T M} - 6\Delta_{MT})\right) \;\le\; P\left(L_{Q_T M} < 2\gamma'\mu_{Q_T M}\right)$$

and, by Lemma 10.3 and using that $\gamma' \in (0, \frac{1}{12})$, we obtain that

$$P\left(L_{Q_T M} < 2\gamma'\mu_{Q_T M}\right) \;\le\; (2\gamma')^{\frac{|T\setminus M|}{4}} e^{-\frac{1}{2}(1-\sqrt{2\gamma'})^2\mu_{Q_T M}} \;\le\; \left(\tfrac{1}{6}\right)^{\frac{|T\setminus M|}{4}} e^{-\frac{1}{2}(1-\sqrt{2\gamma'})^2\mu_{Q_T M}}.$$

Since $\mu_{Q_T M} > 6\Delta_{MT}$, we get

$$A_T \;=\; \gamma'\mu_{Q_T M} + \gamma\Delta_{MT} \;<\; \frac{1-\gamma}{6}\mu_{Q_T M} + \frac{\gamma}{6}\mu_{Q_T M} \;=\; \frac{1}{6}\mu_{Q_T M} \;\le\; \frac{1}{2}(1-\sqrt{2\gamma'})^2\mu_{Q_T M},$$

where the last inequality follows from the fact that $\gamma' \in (0, \frac{1}{12})$. We then get

$$P\left(L_{Q_T M} < 2\gamma'(\mu_{Q_T M} - 6\Delta_{MT})\right) \;\le\; \left(\frac{1}{4}\right)^{\frac{|T\setminus M|}{6}} e^{-A_T} \le e^{-A_T} \le e^{-\alpha A_T} \tag{S.44}$$

Summing the bounds in (S.43) and (S.44) gives that for every n large enough $\mathbb{E}\left(NC(M)\right) < 2e^{-\alpha A_T}$. Since $\psi < \alpha$, for every n large enough, we have that $\mathbb{E}\left(NC(M)\right) < e^{-\psi A_T}$ as claimed. $\square$

10.2 Proofs of Section 3

10.2.1 Proof of Lemma 3.1

First note that by discarding constant terms

$$\arg\max_{M\in\mathcal{M}}\left\{\max_{\beta\in\mathcal{L}_M}\ell(\mathbf{y};\boldsymbol{\beta})-\sum_{j=1}^{b}\kappa_j|M_j|\right\}=\arg\min_{M\in\mathcal{M}}\left\{\min_{\beta\in\mathcal{L}_M}\left\{\tfrac{1}{2}\|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\|^2\right\}+\sum_{j=1}^{b}\kappa_j|M_j|\right\}.$$

We also have that

$$\min_{\beta\in\mathcal{L}_M}\tfrac{1}{2}\|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\|^2=\tfrac{1}{2}\|\mathbf{y}-\mathbf{X}_M\tilde{\boldsymbol{\beta}}^{(M)}\|^2=\tfrac{1}{2}\|\mathbf{y}\|^2-\tfrac{1}{2}\|\mathbf{X}_M\tilde{\boldsymbol{\beta}}^{(M)}\|^2,$$

where in the last equality we used that $\tilde{\boldsymbol{\beta}}^{(M)}=(\mathbf{X}_M^T\mathbf{X}_M)^{-1}\mathbf{X}_M^T\mathbf{y}$. The maximization in (8) can be then replaced with the maximization of $C(M)=\frac{1}{2}\|\mathbf{X}_M\tilde{\boldsymbol{\beta}}^{(M)}\|^2-\sum_{j=1}^{b}\kappa_j|M_j|$ which is equivalent to maximizing $NC(M)$.

10.2.2 Proof of Lemma 3.2

Part (i) follows directly from Lemma 10.6 by taking $\{M^1,\ldots,M^k\}=\{S\}$. Part (ii) follows from

$$NC(M)=\Big(1+\sum_{N\neq M}e^{C(N)-C(M)}\Big)^{-1}<\Big(1+e^{C(M')-C(M)}\Big)^{-1}.$$

10.2.3 Proof of Theorem 3.3

The proof strategy is to first use Lemma 10.10 with $T=S$ to show that for every $M\neq S$, $\mathbb{E}(NC(M))\leq e^{-\psi A_S}$ for every large enough n and any $\psi\in(0,1)$, where $A_S=\gamma\Delta_{MS}+\frac{1-\gamma}{6}\mu_{Q_SM}$ (*cf* (S.41) and (S.36)), $\gamma\in(1/2,1)$ is defined in Assumption A5 and $Q_S=M\cup S$. Assumption A4 and the fact that $M\setminus S\subseteq S^C$ ensure the assumptions of Lemma 10.10 are met. The second step is to obtain a lower bound for A_S, which gives a new upper bound for $\mathbb{E}(NC(M))$. The final step is to use these bounds to get an upper-bound on $\sum_{M\in\mathcal{M}\setminus\{S\}}\mathbb{E}\left(NC(M)\right)$ that asymptotically vanishes under Assumptions A4 and A5. We then use Lemma 3.2 to conclude on the vanishing of $P(\hat{S}^{ei}\neq S)$.

First, to show that $\mathbb{E}(NC(M)) \le e^{-\psi A_S}$ for any $M \in \mathcal{M} \setminus \{S\}$, we show that A_S satisfies the conditions of Lemma 10.10, taking $T = S$. That is, we wish to show that, $A_S > 0$, $|M \setminus S| = o(A_S)$, and $\mu_{Q_S S} = o(A_S)$. Observe that Δ_{MS}, defined in (S.41), can be rewritten as $\Delta_{MS} = \sum_{j=1}^{b}(|M_j \setminus S_j| - |S_j \setminus M_j|)\kappa_j$. By Lemma 10.8 with $T = S$, for every $n \in \mathbb{N}$ we have

$$\begin{aligned} A_S &= \gamma\Delta_{MS} + \frac{1-\gamma}{6}\mu_{Q_S M} \\ &\ge \gamma\sum_{j=1}^{b}|M_j \setminus S_j|\kappa_j + \sum_{j=1}^{b}|S_j \setminus M_j|\left(\tfrac{1-\gamma}{6}n\rho(\mathbf{X}){\beta^*_{\min,j}}^2 - \gamma\kappa_j\right) \end{aligned} \tag{S.45}$$

Since $M \ne S$, $|M \setminus S| \ne 0$ or $|S \setminus M| \ne 0$, then by Assumptions A4 and A5, for every n large enough, $A_S > 0$. We immediately have $\mu_{Q_S S} = o(A_S)$ because $\beta^*_{Q_S \setminus S} = \beta^*_{M \setminus S} = 0$ (any parameter outside the true support S is by definition 0) and hence $\mu_{Q_S S} = 0$. If $|M \setminus S| = 0$, $|M \setminus S| = o(A_S)$ also immediately. Consider now the case $|M \setminus S| \ne 0$. By Assumption A5, the last term in (S.45) is nonnegative, and hence

$$\frac{|M \setminus S|}{A_S} = \frac{|M \setminus S|}{\gamma\Delta_{MS} + \frac{1-\gamma}{6}\mu_{Q_S M}} \le \left[\gamma\sum_{j=1}^{b}\frac{|M_j \setminus S_j|}{|M \setminus S|}\kappa_j\right]^{-1} \le \left[\gamma \min_{j=1,\ldots,b}\kappa_j\right]^{-1}$$

where the last inequality follows from $\sum_{j=1}^{b}\frac{|M_j \setminus S_j|}{|M \setminus S|} = 1$. By Assumption A4 we have that $\min_j \kappa_j \to \infty$ as $n \to \infty$, and hence $|M \setminus S| = o(A_S)$. Thus, by Lemma 10.10, for any $\psi \in (0,1)$ and all n large enough, $\mathbb{E}(NC(M)) \le e^{-\psi A_S}$.

For the second step of the proof, let A_S^* be the lower bound for A_S given in (S.45). That is

$$A_S^* := \gamma\sum_{j=1}^{b}|M_j \setminus S_j|\kappa_j + \sum_{j=1}^{b}|S_j \setminus M_j|\left(\tfrac{1-\gamma}{6}n\rho(\mathbf{X}){\beta^*_{\min,j}}^2 - \gamma\kappa_j\right).$$

By (S.45), we have, for all n large enough,

$$\mathbb{E}(NC(M)) \le e^{-\psi A_S^*}. \tag{S.46}$$

Assumption A5 implies there exist $g'_j \to \infty$ such that

$$\frac{(1-\gamma)n\rho(\mathbf{X})}{6}{\beta^*_{\min,j}}^2 - \kappa_j = \ln(s_j) + g'_j. \tag{S.47}$$

Let $\delta \in (0,1)$ and denote $\bar{m}_j = \max\left\{\frac{2\ln(p_j-s_j)}{f_j}, \frac{2\ln(s_j)}{g'_j}\right\}$, where f_j is given in Assumption A4. Take $\psi = \max_{j=1,\dots,b} \frac{\xi+\delta+\bar{m}_j}{1+\bar{m}_j}$ for some $\xi \in (0, 1-\delta)$ then $\psi \in (0,1)$ and we have, for every $j = 1, \dots, b$,

$$\psi \;>\; \frac{\delta + \frac{2\ln(p_j-s_j)}{f_j}}{1+\frac{2\ln(p_j-s_j)}{f_j}} \;=\; \frac{\delta f_j/2 + \ln(p_j - s_j)}{f_j/2 + \ln(p_j - s_j)} \tag{S.48}$$

$$\psi \;>\; \frac{\delta + \frac{2\ln(s_j)}{g'_j}}{1+\frac{2\ln(s_j)}{g'_j}} \;=\; \frac{\delta g'_j/2 + \ln(s_j)}{g'_j/2 + \ln(s_j)} \;\geq\; \frac{\delta g'_j/2 + \ln(s_j)}{g'_j + \ln(s_j)}. \tag{S.49}$$

In Assumption A5, γ is defined as $\gamma := \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j)$, we then have

$$\gamma\kappa_j \;\geq\; \frac{1}{2}\Big(1 + \frac{\ln(p_j - s_j)}{\kappa_j}\Big)\kappa_j \;=\; \ln(p_j - s_j) + \frac{1}{2}(\kappa_j - \ln(p_j - s_j)) \;=\; \ln(p_j - s_j) + \frac{1}{2}f_j.$$

Hence, by (S.48), we have

$$\psi\gamma\kappa_j \;\geq\; \psi\Big(\ln(p_j - s_j) + \frac{1}{2}f_j\Big) \;\geq\; \ln(p_j - s_j) + \delta\frac{1}{2}f_j. \tag{S.50}$$

Further,

$$\psi\left(\tfrac{1-\gamma}{6}n\rho(\mathbf{X})\beta^{*\,2}_{\min,j} - \gamma\kappa_j\right) \;\geq\; \psi\left(\ln(s_j) + g'_j\right) \;\geq\; \ln(s_j) + \delta\frac{1}{2}g'_j. \tag{S.51}$$

where the first inequality follows from (S.47) and the second inequality from (S.49).

In (S.46), $\psi A^*_S = \sum_{j=1}^b |M_j \setminus S_j|\psi\gamma\kappa_j + \sum_{j=1}^b |S_j \setminus M_j|\psi\left(\frac{1-\gamma}{6}n\rho(\mathbf{X})\beta^{*\,2}_{\min,j} - \gamma\kappa_j\right)$. Then by (S.50) and (S.51), we get

$$\mathbb{E}(NC(M)) \;\leq\; \exp\left\{-\sum_{j=1}^{b}|M_j \setminus S_j|(\ln(p_j - s_j) + \delta\tfrac{f_j}{2}) - \sum_{j=1}^{b}|S_j \setminus M_j|(\ln(s_j) + \delta\tfrac{g'_j}{2})\right\}. \tag{S.52}$$

For the final step of the proof, denote $\mathcal{S} = \sum_{M\in\mathcal{M}\setminus\{S\}} \mathbb{E}\left(NC(M)\right)$ for convenience. By (S.52) we have

$$\mathcal{S} \leq \sum_{M\in\mathcal{M}\setminus\{S\}} e^{-\sum_{j=1}^b |M_j\setminus S_j|\left(\ln(p_j-s_j)+\delta\frac{f_j}{2}\right) - \sum_{j=1}^b |S_j\setminus M_j|\left(\ln(s_j)+\delta\frac{g'_j}{2}\right)}.$$

Observe that if $|M_j \setminus S_j| = 0$ and $|S_j \setminus M_j| = 0$ for all j, then $M = S$ and the summand in the right-hand side above is 1. Then by adding and subtracting 1 we get

$$\mathcal{S} \leq \sum_{M\in\mathcal{M}} e^{-\sum_{j=1}^b |M_j\setminus S_j|\left(\ln(p_j-s_j)+\delta\frac{f_j}{2}\right) - \sum_{j=1}^b |S_j\setminus M_j|\left(\ln(s_j)+\delta\frac{g'_j}{2}\right)} - 1.$$

We can split the sum in the right-hand side above into sums over the models that have the same number of inactive variables and missing the same number of truly active variables in every block. That is, the models M such that for all j, $|M_j \setminus S_j| = u_j$ and $|S_j \setminus M_j| = w_j$ with $u_j \in \{0, \ldots, p_j - s_j\}$ and $w_j \in \{0, \ldots, s_j\}$. Denote those sums

$$S_{\mathbf{w}}^{\mathbf{u}} = \sum_{M \in \mathcal{M}: \forall j |M_j \setminus S_j| = u_j, |S_j \setminus M_j| = w_j} e^{-\sum_{j=1}^{b} u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2}\right) - \sum_{j=1}^{b} w_j \left(\ln(s_j) + \delta \frac{g_j'}{2}\right)}.$$

We get

$$\mathcal{S} \leq -1 + \sum_{w_1=0}^{s_1} \cdots \sum_{w_b=0}^{s_b} \sum_{u_1=0}^{p_1 - s_1} \cdots \sum_{u_b=0}^{p_b - s_b} S_{\mathbf{w}}^{\mathbf{u}}. \tag{S.53}$$

The number of models having, for all j, u_j inactive parameters and missing w_j out of the s_j active parameters is $\prod_{j=1}^{b} \binom{p_j - s_j}{u_j} \binom{s_j}{w_j}$. We thus have

$$\begin{aligned} S_{\mathbf{w}}^{\mathbf{u}} &= \left(\prod_{j=1}^{b} \binom{p_j - s_j}{u_j} \binom{s_j}{w_j} \right) e^{-\sum_{j=1}^{b} u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2}\right) - \sum_{j=1}^{b} w_j \left(\ln(s_j) + \delta \frac{g_j'}{2}\right)} \\ &= \prod_{j=1}^{b} \binom{p_j - s_j}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2}\right)} \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g_j'}{2}\right)}. \end{aligned}$$

Inputting the expression above in (S.53) gives

$$\begin{aligned} \mathcal{S} &\leq -1 + \sum_{w_1=0}^{s_1} \cdots \sum_{w_b=0}^{s_b} \sum_{u_1=0}^{p_1 - s_1} \cdots \sum_{u_b=0}^{p_b - s_b} \prod_{j=1}^{b} \binom{p_j - s_j}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2}\right)} \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g_j'}{2}\right)} \\ &\leq -1 + \prod_{j=1}^{b} \left(1 + \sum_{u_j=1}^{p_j - s_j} \binom{p_j - s_j}{u_j} e^{-u_j (\ln(p_j - s_j) + \delta \frac{f_j}{2})} \right) \left(1 + \sum_{w_j=1}^{s_j} \binom{s_j}{w_j} e^{-w_j (\ln(s_j) + \delta \frac{g_j'}{2})} \right) \end{aligned}$$

where the second inequality follows from first factorizing over terms in u_j and w_j and then taking the term in 0 out of every sum. A standard bound on binomial coefficient for $1 \leq k \leq n$ is

$$\binom{n}{k} \leq \left(\frac{n\, e}{k} \right)^k \leq (n\, e)^k = e^{k(\ln(n) + 1)}. \tag{S.54}$$

Then

$$\mathcal{S} \quad \leq \quad -1 + \prod_{j=1}^{b} \left(1 + \sum_{u_j=1}^{p_j - s_j} e^{-u_j \left(\delta \frac{f_j}{2} - 1\right)} \right) \left(1 + \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta \frac{g_j'}{2} - 1\right)} \right). \tag{S.55}$$

Denote

$$d_j \;=\; e^{1 - \delta \frac{f_j}{2}}, \qquad h_j \;=\; e^{1 - \delta \frac{g_j'}{2}}$$

where both expressions go to zero as n increases since $f_j \to \infty$ and $g'_j \to \infty$. For every j, by the properties of geometric sums, we have

$$1 + \sum_{u_j=1}^{p_j-s_j} e^{-u_j\left(\delta\frac{f_j}{2}-1\right)} = \frac{1-d_j^{p_j-s_j+1}}{1-d_j}$$
$$1 + \sum_{w_j=1}^{s_j} e^{-w_j\left(\delta\frac{g'_j}{2}-1\right)} = \frac{1-h_j^{s_j+1}}{1-h_j}.$$

Since both expressions converge to 1 as n grows, and b is fixed we get that

$$\lim_{n\to\infty} \mathcal{S} = \lim_{n\to\infty} \sum_{M\in\mathcal{M}\setminus\{S\}} \mathbb{E}\left(NC(M)\right) = 0.$$

By Lemma 3.2, $P(\hat{S}^{ei} \neq S) \leq 2\,\mathcal{S}$ and then $\lim_{n\to\infty} P(\hat{S}^{ei} = S) = 1$.

10.2.4 Proof of Theorem 3.4

First, we prove the upper bound on $P(\hat{S}^{ei} \neq S)$ in (10). We assume here A1, A4, and A5. Under the same assumptions, in the proof of Theorem 3.3, the following bound was shown in (S.55):

$$\sum_{M\in\mathcal{M}\setminus\{S\}} \mathbb{E}\left(NC(M)\right) \;\leq\; -1 + \prod_{j=1}^{b}\left(1 + \sum_{u_j=1}^{p_j-s_j} e^{-u_j\left(\delta\frac{f_j}{2}-1\right)}\right)\left(1 + \sum_{w_j=1}^{s_j} e^{-w_j\left(\delta\frac{g'_j}{2}-1\right)}\right). \tag{S.56}$$

Denote $S(u_j) = \sum_{u_j=1}^{p_j-s_j} e^{-u_j\left(\delta\frac{f_j}{2}-1\right)}$, $S(w_j) = \sum_{w_j=1}^{s_j} e^{-w_j\left(\delta\frac{g'_j}{2}-1\right)}$, $d_j = e^{1-\delta\frac{f_j}{2}}$, and $h_j = e^{1-\delta\frac{g'_j}{2}}$. For every j, we have, by the properties of geometric sums:

$$S(u_j) = d_j\,\frac{1-d_j^{p_j-s_j}}{1-d_j}$$
$$S(w_j) = h_j\,\frac{1-h_j^{s_j}}{1-h_j}.$$

Developing the product in the right-hand side in (S.56) and reordering the resulting terms gives

$$\sum_{M\in\mathcal{M}\setminus\{S\}} \mathbb{E}\left(NC(M)\right) \;\leq\; -1+1+\sum_{j=1}^{b}\Big[S(u_j)+S(w_j)\Big] + \mathcal{R}$$

where all the terms in $\mathcal{R}$ are product of two or more of the sums $S(u_1), \ldots, S(u_b), S(w_1), \ldots, S(w_b)$. Given that $\delta > 0$, $f_j \to \infty$ and $g'_j \to \infty$ by assumption, and hence $d_j \to 0$ and $h_j \to 0$, the $S(u_j)$ and $S(w_j)$ are smaller than 1 for all sufficiently large n for all j. Then each of the $2^{2b} - 2b - 1$ terms in $\mathcal{R}$ is bounded above by $\sum_{j=1}^{b} \left[S(u_j) + S(w_j)\right]$ and we get, for every n large enough,

$$\sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}\left(NC(M)\right) \leq (2^{2b} - 2b) \sum_{j=1}^{b} \left[d_j \frac{1 - d_j^{p_j - s_j}}{1 - d_j} + h_j \frac{1 - h_j^{s_j}}{1 - h_j}\right]. \tag{S.57}$$

Denote

$$r = \max_{j=1,\ldots,b} \left\{\frac{1 - d_j^{p_j - s_j}}{1 - d_j}, \frac{1 - h_j^{s_j}}{1 - h_j}\right\}. \tag{S.58}$$

By Lemma 3.2, (S.57) and (S.58), we then obtain:

$$P(\hat{S}^{ei} \neq S) \; \leq \; 2 \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}\left(NC(M)\right) \; \leq \; 2(2^{2b} - 2b) r \, e \sum_{j=1}^{b} e^{-\delta \frac{f_j}{2}} + e^{-\delta \frac{g'_j}{2}}$$

By the definition of f_j in Assumption A4, that of g'_j in (S.47), and the fact that $2e < 6$, we get, for every n large enough, that

$$P(\hat{S}^{ei} \neq S) \; \leq \; (2^{2b} - 2b) 6r \sum_{j=1}^{b} e^{-\frac{\delta}{2}\left[\kappa_j - \ln(p_j - s_j)\right]} + e^{-\frac{\delta}{2}\left[\frac{(1-\gamma) n \rho(\mathbf{X})}{6} {\beta^*_{\min,j}}^2 - \kappa_j - \ln(s_j)\right]}. \tag{S.59}$$

We have $\frac{(1-\gamma) n \rho(\mathbf{X})}{6} {\beta^*_{\min,j}}^2 - \kappa_j > \left(\sqrt{\frac{(1-\gamma) n \rho(\mathbf{X})}{6}} \beta^*_{\min,j} - \sqrt{\kappa_j}\right)^2$ and then,

$$P(\hat{S}^{ei} \neq S) \; \leq \; (2^{2b} - 2b) 6r \sum_{j=1}^{b} e^{-\frac{\delta}{2}\left[\kappa_j - \ln(p_j - s_j)\right]} + e^{-\frac{\delta}{2}\left[(\sqrt{\frac{(1-\gamma) n \rho(\mathbf{X})}{6}} \beta^*_{\min,j} - \sqrt{\kappa_j})^2 - \ln(s_j)\right]}.$$

We have $d_j \to 0$ and $h_j \to 0$, then for every n large enough $\frac{1 - d_j^{p_j - s_j}}{1 - d_j} \to 1$ and $\frac{1 - h_j^{s_j}}{1 - h_j} \to 1$ for all j, and $r \to 1$. It follows that for any $c > (2^{2b} - 2b)6$ for n large enough the bound in (10) holds.

Second, we prove that, if for all $j = 1, \ldots, b$, it holds that

$$\lim_{n \to \infty} \frac{g(\gamma^*) \sqrt{n \rho(\mathbf{X})/6} \, \beta^*_{\min,j}}{\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)}} \geq 1 \tag{S.60}$$

with $g(\gamma^*) = \sqrt{(-1+2\gamma^*)(1-\gamma^*)}/(1+\sqrt{2-2\gamma^*})$ then κ_j^* defined in (11) satisfies Assumptions A4 and A5. Under (S.60), for every j, there exists a sequence x_j such that $\lim_{n\to\infty} x_j \geq 1$ and $g(\gamma^*)\sqrt{\frac{n}{6}}\beta^*_{\min,j} = x_j\Big(\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)}\Big)$ and $v_j = x_j\sqrt{(1-\gamma^*)}g(\gamma^*)^{-1}$ such that $\sqrt{\frac{(1-\gamma^*)n}{6}}\beta^*_{\min,j} = v_j\Big(\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)}\Big)$. Denote $a = \sqrt{\ln(p_j - s_j)}$ and $b = \sqrt{\ln(s_j)}$. We have

$$\sqrt{\kappa_j^*} \;=\; \frac{v_j}{2}(a+b) + \frac{(a^2-b^2)}{2v_j(a+b)} \;=\; \frac{v_j}{2}(a+b) + \frac{a-b}{2v_j} \;=\; \frac{v_j^2+1}{2v_j}a + \frac{v_j^2-1}{2v_j}b.$$

For n large enough, $b \geq 0$ and $v_j > 1$, since $\sqrt{(1-\gamma^*)}g(\gamma^*)^{-1} > 1$ for $\gamma^* \in (1/2, 1)$ and $\lim_{n\to\infty} x_j \geq 1$. It follows that $b(v_j^2 - 1)/2v_j \geq 0$ for n large enough and

$$\sqrt{\kappa_j^*} = \frac{v_j^2+1}{2v_j}a + \frac{v_j^2-1}{2v_j}b \geq \Big(\frac{v_j^2+1}{2v_j} + 1 - 1\Big)a = \Big(1 + \frac{(v_j-1)^2}{2v_j}\Big)\sqrt{\ln(p_j - s_j)} \quad \text{(S.61)}$$

which implies Assumption A4 since $\lim_{n\to\infty} v_j > 1$. We also have

$$\frac{\sqrt{(1-\gamma^*)n\rho(\mathbf{X})}}{6}\beta^*_{\min,j} - \sqrt{\kappa^*} = \frac{v_j^2-1}{2v_j}a + \frac{v_j^2+1}{2v_j}b \geq \Big(1 + \frac{(v_j-1)^2}{2v_j}\Big)\sqrt{\ln(s_j)}. \quad \text{(S.62)}$$

Let $\gamma = \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j^*)$ as defined in Assumption A5. By (S.61), $\gamma = \frac{1}{2}(1 + \max_j \Big(1 + \frac{(v_j-1)^2}{2v_j}\Big)^{-2}$. Since $\lim_{n\to\infty} x_j > 1$, then for sufficiently large n, $v_j > \sqrt{(1-\gamma^*)}g(\gamma^*)^{-1}$. Simple algebra shows the latter inequality implies $\frac{1}{2}(1 + \Big(1 + \frac{(v_j-1)^2}{2v_j}\Big)^{-2} < \gamma^*$ and $1 - \gamma > 1 - \gamma^*$. Then, by (S.62),

$$\frac{\sqrt{(1-\gamma)n\rho(\mathbf{X})}}{6}\beta^*_{\min,j} - \sqrt{\kappa^*} \geq \Big(1 + \frac{(v_j-1)^2}{2v_j}\Big)\sqrt{\ln(s_j)}.$$

for every sufficiently large n. Since $\lim_{n\to\infty} x_j \geq 1$ and $\sqrt{(1-\gamma^*)}g(\gamma^*)^{-1} > 1$ for $\gamma^* \in (1/2, 1)$, $\lim_{n\to\infty} v_j > 1$ and Assumption A5 holds.

Finally, we prove (12). Since Assumptions A4 and A5 hold for the κ_j^*, and because we assume A1, by the first part of the theorem, there exists $c > 0$ such that, for every n large enough,

$$P(\hat{S}^{ei} \neq S) \;\leq\; c \sum_{j=1}^{b} \; e^{-\frac{\delta}{2}\left[\kappa_j^* - \ln(p_j - s_j)\right]} + e^{-\frac{\delta}{2}\left[(\sqrt{\frac{(1-\gamma^*)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j} - \sqrt{\kappa_j^*})^2 - \ln(s_j)\right]}.$$

Simple algebra shows that the κ_j^* satisfy $\kappa_j^* - \ln(p_j - s_j) = \left(\sqrt{\frac{(1-\gamma^*)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j} - \sqrt{\kappa_j^*}\right)^2 - \ln(s_j)$ for all j. We then have

$$P(\hat{S}^{ei} \neq S) \ \leq \ 2c \sum_{j=1}^{b} \ e^{-\frac{\delta}{2}\left[\kappa_j^* - \ln(p_j - s_j)\right]}. \tag{S.63}$$

Denote

$d = \frac{1}{2}\sqrt{\frac{6}{(1-\gamma^*)n\rho(\mathbf{X})}}\frac{1}{\beta^*_{\min,j}}\Big(\ln(p_j - s_j) - \ln(s_j)\Big)$ such that $\sqrt{\kappa_j^*} = \frac{1}{2}\sqrt{\frac{(1-\gamma^*)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j} + d$. Then

$$e^{-\frac{\delta}{2}\left(\kappa_j^* - \ln(p_j - s_j)\right)} = e^{-\frac{\delta}{2}\left[\frac{(1-\gamma^*)n\rho(\mathbf{X})}{24}{\beta^*_{\min,j}}^2 + d^2 - \frac{1}{2}(\ln(p_j - s_j) + \ln(s_j))\right]}.$$

By considering separately the two possible maxima in $\ln\max\{p_j - s_j, s_j\}$, we get that

$$\frac{e^{-\frac{\delta}{2}\left(\kappa_j^* - \ln(p_j - s_j)\right)}}{e^{-\frac{\delta}{2}\left[\frac{(1-\gamma^*)n\rho(\mathbf{X})}{24}{\beta^*_{\min,j}}^2 - \ln\max\{p_j - s_j, s_j\}\right]}} = e^{-\frac{\delta}{2}\left[d^2 + \frac{1}{2}|\ln(p_j - s_j) - \ln(s_j)|\right]} < 1. \tag{S.64}$$

It follows that, by (S.63) and (S.64),

$$P(\hat{S}^{ei} \neq S) \ \leq \ 2c \sum_{j=1}^{b} \ e^{-\frac{\delta}{2}\left[\frac{(1-\gamma^*)n\rho(\mathbf{X})}{24}{\beta^*_{\min,j}}^2 - \ln\max\{p_j - s_j, s_j\}\right]}.$$

which proves (12).

10.2.5 Proof of Proposition 3.5

Consistent recovery of S implies that for for any subset of models $\mathcal{N}$, $\max_{M \in \mathcal{N}} NC(M)/NC(S) < 1$ with probability going to 1 as n grows, where NC is the normalized criterion defined in (9). In Part (i), for every $j = 1, \dots, b$, we obtain a necessary assumption on the κ_j for selection consistency by considering $\mathcal{N} = \mathcal{O}_j$ as defined in (13). In Part (ii), we obtain a necessary assumption on the $\beta^*_{\min,j}$ by considering $\mathcal{N} = \mathcal{U}_j$ where

$$\mathcal{U}_j \ = \ \{M \in \mathcal{M} \mid M = S \setminus \{i\} \text{ where } \beta_i = \beta_{\min,j}\}.$$

That is $\mathcal{U}_j$ the set of models that under-fit by only one variable i with smallest $\beta_i = \beta^*_{\min,j}$. In Part (iii), we use (i) and (ii) to get, for every $j = 1, \dots, b$, a necessary assumption on the scaling of $(n, p_j, s_j, \beta^*_{\min,j})$.

Part (i). The event $\hat{S}^{ei} = S$ (correct recovery of S) requires the event $\max_{M \in \mathcal{O}_j} \frac{NC(M)}{NC(S)} < 1$. We have

$$P(\hat{S}^{ei} = S) \leq P\Big(\max_{M \in \mathcal{O}_j} \frac{NC(M)}{NC(S)} < 1 \Big) = P\left(\max_{M \in \mathcal{O}_j} e^{C(M)-C(S)} < 1 \right).$$

Using that $C(S) - C(M) = L_{SM}/2 + \Delta_{MS}$ for L_{SM} and Δ_{MS} defined in (S.41), we obtain

$$P(\hat{S}^{ei} = S) \leq P\left(\max_{M \in \mathcal{O}_j} e^{-(\frac{1}{2}L_{SM} + \Delta_{MS})} < 1 \right) = P\Big(\max_{M \in \mathcal{O}_j} L_{MS} < 2\kappa_j \Big). \tag{S.65}$$

Taking square root on both sides of the inequality in the right-most probability in (S.65), we have

$$P(\hat{S}^{ei} = S) \leq P\Big(\max_{M \in \mathcal{O}_j} \sqrt{L_{MS}} < \sqrt{2\kappa_j} \Big). \tag{S.66}$$

By Lemma 10.7, for every $M \in \mathcal{O}_j$, $L_{MS} \in \chi^2_1$ and there exists $Z_M \sim N(0,1)$ such that $\sqrt{L_{MS}} = |Z_M|$. Since for every $M \in \mathcal{O}_j$, $|Z_M| \geq Z_M$, we have

$$P(\hat{S}^{ei} = S) \leq 1 - P\Big(\max_{M \in \mathcal{O}_j} Z_M \geq \sqrt{2\kappa_j} \Big). \tag{S.67}$$

The set $\mathcal{O}_j$ has cardinality $p_j - s_j$, then by Theorem 3.4 in Hartigan (2014) and our Assumption A2, for any $\varepsilon > 0$,

$$P\Big(\max_{M \in \mathcal{O}_j} Z_M \geq \sqrt{2\underline{\lambda}_j \ln(p_j - s_j)} (1 - \varepsilon) \Big) \to 1,$$

where $\underline{\lambda}_j$ is as defined in (14). If $\lim_{n \to \infty} \frac{\kappa_j}{\underline{\lambda}_j \ln(p_j - s_j)} < 1$, then $\lim_{n \to \infty} P\Big(\max_{M \in \mathcal{O}_j} Z_M \geq \sqrt{2\kappa_j} \Big) = 1$. Hence, by (S.67) we have that $\lim_{n \to \infty} P(\hat{S}^{ei} = S) = 0$, as claimed.

Part (ii). Consider any model $M \in \mathcal{U}_j$, that is under-fitting S by a variable in block j associated to $\beta^*_{\min,j}$. The event $\hat{S}^{ei} = S$ (correct recovery of S) requires the event $\frac{NC(M)}{NC(S)} < 1$ where NC is the normalized criterion in (9). That is S is preferred to model M.

$$P(\hat{S}^{ei} = S) \leq P\Big(\frac{NC(M)}{NC(S)} < 1 \Big) = P\left(e^{C(M)-C(S)} < 1 \right)$$

Using that $C(S) - C(M) = L_{SM}/2 + \Delta_{MS}$ for L_{SM} and Δ_{MS} defined in (S.41), and that $\Delta_{MS} = -\kappa_j$ we obtain

$$P(\hat{S}^{ei} = S) \leq P\left(e^{-(\frac{1}{2}L_{SM}+\Delta_{MS})} < 1\right) = P\Big(L_{SM} > 2\kappa_j\Big). \tag{S.68}$$

Taking square root on both sides of the inequality in the right-most probability in (S.68), we get

$$P(\hat{S}^{ei} = S) \leq P\Big(\sqrt{L_{SM}} > \sqrt{2\kappa_j}\Big). \tag{S.69}$$

By Lemma 10.7, $L_{SM} \in \chi_1^2(\mu_{SM})$ and there exists then $Z_M \sim N(\sqrt{\mu_{SM}}, 1)$ such that $\sqrt{L_{SM}} = |Z_M|$. We have

$$P(\hat{S}^{ei} = S) \leq P\Big(|Z_M| > \sqrt{2\kappa_j}\Big). \tag{S.70}$$

Consider first the case where $2\kappa_j > \mu_{SM}$. Using that $|Z_M| = |Z_M - \sqrt{\mu_{SM}} + \sqrt{\mu_{SM}}| \leq |Z_M - \sqrt{\mu_{SM}}| + \sqrt{\mu_{SM}}$, we get that

$$P\Big(|Z_M| > \sqrt{2\kappa_j}\Big) \leq P\Big(|Z_M - \sqrt{\mu_{SM}}| > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}\Big) = 2P\Big(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}\Big)$$

where $z \sim N(0,1)$ and the equality follows from the symmetry of the standard Gaussian distribution. Since $2\kappa_j > \mu_{SM}$, $P\Big(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}\Big) < 1/2$, $P\Big(|Z_M| > \sqrt{2\kappa_j}\Big) < 1$ and $P(\hat{S}^{ei} = S) < 1$ too by (S.70).

Consider next the case where $2\kappa_j \leq \mu_{SM}$. By Lemma 10.1 (ii), we have

$$P\Big(|Z_M| > \sqrt{2\kappa_j}\Big) = P\left(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}\right) + P\left(z < -\sqrt{2\kappa_j} - \sqrt{\mu_{SM}}\right)$$

where $z \sim N(0,1)$. Since $\kappa_j \to +\infty$, $\sqrt{2\kappa_j} + \sqrt{\mu_{SM}} \to \infty$ and we have that $P(z < -\sqrt{2\kappa_j} - \sqrt{\mu_{SM}}) \to 0$. Further, by Lemma 10.9 with $T = S$, $\sqrt{\mu_{SM}} \leq \sqrt{n\bar{\lambda}}\beta^*_{\min,j}$. By assumption we have $\lim_{n\to\infty} \sqrt{n\bar{\lambda}}\beta^*_{\min,j} - \sqrt{2\kappa_j} < \infty$. It follows that $\lim_{n\to\infty} \sqrt{\mu_{SM}} - \sqrt{2\kappa_j} < \infty$ and we have that $\lim\limits_{n\to\infty} P\left(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}\right) < 1$. Hence we get $\lim\limits_{n\to\infty} P\left(\hat{S}^{ei} = S\right) < 1$, as claimed.

Part (iii). First, we re-write

$$\sqrt{n\bar{\lambda}}\beta^*_{\min,j} - \sqrt{2\underline{\lambda}_j \ln(p_j - s_j)} \;=\; \sqrt{n\bar{\lambda}}\beta^*_{\min,j} - \sqrt{2\kappa_j} + \sqrt{2\underline{\lambda}_j \ln(p_j - s_j)}\Big(\sqrt{\tfrac{\kappa_j}{\underline{\lambda}_j \ln(p_j - s_j)}} - 1\Big).$$

By assumption $\lim_{n\to\infty}\sqrt{n\bar{\lambda}}\beta^*_{\min,j}-\sqrt{2\underline{\lambda}_j\ln(p_j-s_j)}<\infty$ and there exists $e\in\mathbb{R}^+$ such that

$$\lim_{n\to\infty}\sqrt{n\bar{\lambda}}\beta^*_{\min,j}-\sqrt{2\kappa_j}+\sqrt{2\underline{\lambda}_j\ln(p_j-s_j)}\left(\sqrt{\frac{\kappa_j}{\underline{\lambda}_j\ln(p_j-s_j)}}-1\right)\le e \tag{S.71}$$

Consider the case $\lim_{n\to\infty}\sqrt{\frac{\kappa_j}{\underline{\lambda}_j\ln(p_j-s_j)}}-1<0$. Then by Part (i), we have $\lim_{n\to\infty}P(\hat{S}^{ei}=S)<1$. Now consider the case $\lim_{n\to\infty}\sqrt{\frac{\kappa_j}{\underline{\lambda}_j\ln(p_j-s_j)}}-1\ge 0$. Then, $\kappa_j\to\infty$ since by assumption $\lim_{n\to\infty}\underline{\lambda}_j\ln(p_j-s_j)>0$. Moreover, condition (S.71) implies that $\lim_{n\to\infty}\sqrt{n\bar{\lambda}}\beta^*_{\min,j}-\sqrt{2\kappa_j}\le e$. By Part (ii), we have that $\lim_{n\to\infty}P(\hat{S}^{ei}=S)<1$.

10.2.6 Proof of Corollary 3.6

Since $\hat{S}^{sd}$ is $\hat{S}^{ei}$ with $b=1$, the assumptions of Proposition 3.5 (iii) for $\hat{S}^{sd}$ are met and $\lim_{n\to\infty}P(\hat{S}^{sd}=S)<1$. Since Assumptions A4 and A5 hold, by Theorem 3.3 we have that $\lim_{n\to\infty}P(\hat{S}^{ei}=S)=1$.

10.2.7 Proof of Theorem 3.7

The proof strategy is the same as that of Theorem 3.3, with suitable adjustments. The first step is to use Lemma 10.10 to bound $\mathbb{E}(NC(M))$ for every $M\notin\mathcal{T}(\kappa)$. The main difference is that in the proof of Theorem 3.3 we took $T=S$ in Lemma 10.10, whereas now we take a model $T=T^M\in\mathcal{T}(\kappa)$ that depends on M. Intuitively, T^M contains large truly non-zero parameters that are missed by M, and hence T^M should be chosen over M asymptotically. More precisely, we choose $T^M\in\mathcal{T}(\kappa)$ such that $T^M\setminus M\subseteq S^L(\kappa)$ and the elements in $M\setminus T^M$ are either inactive or in $S^S(\kappa)$. The latter condition and Assumption A4 ensure that the assumptions of Lemma 10.10 are met. We then get a bound $\mathbb{E}(NC(M))\le e^{-\psi A_{T^M}}$ for every large enough n and any $\psi\in(0,1)$, where $A_{T^M}=\gamma\Delta_{MT^M}+\frac{1-\gamma}{6}\mu_{Q_{T^M}M}$ (*cf* (S.41) and (S.36)), $\gamma\in(1/2,1)$ is defined in (20) and $Q_{T^M}=M\cup T^M$. The second step is to obtain a lower bound for A_{T^M}, which gives an upper bound for $\mathbb{E}(NC(M))$ (distinct to that obtained in the proof of Theorem 3.3). The final step is to get an upper-bound

$\sum_{M\in\mathcal{M}\setminus\mathcal{T}(\kappa)} \mathbb{E}\left(NC(M)\right)$ that vanishes (as n grows) under the assumptions of Theorem 3.7. Then Lemma 10.6 immediately implies that $P(\hat{S}^{ei} \notin \mathcal{T}(\kappa))$ also vanishes.

For the first step of the proof, recall that the set $\mathcal{T}(\kappa)$ contains all the models that are the union of $S^L(\kappa)$ (large signals) and some subset of $S^I(\kappa)$ (intermediate signals). For any $M \notin \mathcal{T}(\kappa)$, take the unique $T^M \in \mathcal{T}(\kappa)$ such that $T^M = \left[M \cap S^I(\kappa)\right] \cup S^L(\kappa)$, which implies $M \cap S^I(\kappa) = T^M \cap S^I(\kappa)$. That is, T^M contains all the large signals plus the intermediate signals in M. To show that $\mathbb{E}(NC(M)) \leq e^{-\psi A_{T^M}}$ we show that A_{T^M} satisfies the conditions of Lemma 10.10, taking $T = T^M$. That is, we wish to show that three conditions hold: $A_{T^M} > 0$, $|M \setminus T^M| = o(A_{T^M})$, and $\mu_{Q_{T^M}T^M} = o(A_{T^M})$.

Observe that Δ_{MT^M}, defined in (S.41), can be rewritten as $\Delta_{MT^M} = \sum_{j=1}^b (|M_j \setminus T_j^M| - |T_j^M \setminus M_j|)\kappa_j$ and then:

$$A_{T^M} = \sum_{j=1}^b |M_j \setminus T_j^M|\gamma\kappa_j + \tfrac{1-\gamma}{6}\mu_{Q_{T^M}M} - \sum_{j=1}^b |T_j^M \setminus M_j|\gamma\kappa_j$$

Since $\gamma < 1$, $(1-\gamma)/6 > 0$, by Lemma 10.8, for every $n \in \mathbb{N}$ we have

$$A_{T^M} \;\geq\; \sum_{j=1}^b |M_j \setminus T_j^M|\gamma\kappa_j + \sum_{j=1}^b |T_j^M \setminus M_j|\Big(\tfrac{1-\gamma}{6} n\rho(\mathbf{X}) \min_{i\in T_j^M\setminus M_j} \beta_i^{*2} - \gamma\kappa_j\Big). \tag{S.72}$$

We have that $T^M \subseteq \left(S^I(\kappa) \cup S^L(\kappa)\right)$ since $T^M \in \mathcal{T}(\kappa)$ and that $T^M \cap S^I(\kappa) = M \cap S^I(\kappa)$. It follows that $T^M \setminus M \subseteq S^L(\kappa)$. By definition of $S^L(\kappa)$, the rightmost component in (S.72) is nonnegative and, if $|T^M \setminus M| \neq 0$, it is positive, for every n large enough. By Assumption A4, component $\gamma\sum_{j=1}^b |M_j \setminus T_j^M|\kappa_j$ is nonnegative, and, if $|M \setminus T^M| \neq 0$, positive. Now, $M \neq T^M$ implies that necessarily $|M \setminus T^M| \neq 0$ or $|T^M \setminus M| \neq 0$ and then, for every n large enough, $A_{T^M} > 0$, establishing the first condition required by Lemma 10.10. Regarding its second condition, if $|M \setminus T^M| = 0$, we immediately have $|M \setminus T^M| = o(A_{T^M})$. If $|M \setminus T^M| \neq 0$, since the rightmost component in (S.72) is nonnegative, we have

$$\frac{|M \setminus T^M|}{A_{T^M}} \leq \left[\gamma \sum_{j=1}^b \frac{|M_j \setminus T_j^M|}{|M \setminus T^M|}\kappa_j\right]^{-1} \leq \left[\gamma \min_{j=1,\dots,b} \kappa_j\right]^{-1}$$

where the last inequality follows from $\sum_{j=1}^{b} \frac{|M_j \setminus T_j^M|}{|M \setminus T^M|} = 1$. By Assumption A4, $\min_{j=1,\dots,b} \kappa_j \to \infty$ as $n \to \infty$ and hence $|M \setminus T^M| = o(A_{T^M})$ when $|M \setminus T^M| \neq 0$ too.

Finally, consider the third condition in Lemma 10.10. In the proof of Theorem 3.3, $\mu_{Q_S S} = o(A_S)$ is immediate because $Q_S \setminus S = M \setminus S \subseteq S^C$, i.e. since all parameters in $M \setminus S$ are truly zero we have that $\mu_{Q_S S} = 0$. Here $M \setminus T^M$ is not necessarily a subset of S^C, hence $\mu_{Q_T^M T^M} \geq 0$. Note that, since $T^M \in \mathcal{T}(\kappa)$, we have that $S^L(\kappa) \subseteq T^M$. Moreover we have $M \cap S^I(\kappa) = T^M \cap S^I(\kappa)$. It follows that $M \setminus T^M \subseteq (S^I(\kappa) \cup S^L(\kappa))^C$, that is the elements of $M \setminus T^M$ are either inactive or belong to $S^S(\kappa)$. If $M \cap S^S(\kappa) = \emptyset$ then $M \setminus T^M \subseteq S^C$ and we immediately get $\mu_{Q_{T^M} T^M} = o(A_{T^M})$ because $\boldsymbol{\beta}^*_{M \setminus T^M} = \mu_{Q_{T^M} T^M} = 0$. Assume now that $M \cap S^S(\kappa) \neq \emptyset$. Using that the rightmost component in (S.72) is nonnegative and Lemma 10.9, we have

$$\frac{\mu_{Q_{T^M} T^M}}{A_{T^M}} \leq \frac{\mu_{Q_{T^M} T^M}}{\gamma \sum_{j=1}^{b} |M_j \setminus T_j^M| \kappa_j} \leq \frac{n\, \bar{\lambda} \sum_{j=1}^{b} |(S_j \cap M_j) \setminus T_j^M| \max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2}}{\gamma \sum_{j=1}^{b} |M_j \setminus T_j^M| \kappa_j}.$$

Observe that for all $j = 1, \dots, b$, $M_j \setminus T_j^M \supseteq (S_j \cap M_j) \setminus T_j^M$ and then $|M_j \setminus T_j^M| \geq |(S_j \cap M_j) \setminus T_j^M|$. We then get

$$\frac{\mu_{Q_{T^M} T^M}}{A_{T^M}} \leq \frac{n\, \bar{\lambda} \sum_{j=1}^{b} |(S_j \cap M_j) \setminus T_j^M| \max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2}}{\gamma \sum_{j=1}^{b} |(S_j \cap M_j) \setminus T_j^M| \kappa_j}.$$

Moreover, since $M \setminus T^M \subseteq (S^I(\kappa) \cup S^L(\kappa))^C$ as discussed earlier, we have, for all $j = 1, \dots, b$, $(S_j \cap M_j) \setminus T_j^M \subseteq S_j^S(\kappa_j)$. It follows $\max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2} \leq \max_{i \in S_j^S} \beta_i^{*2}$ and

$$\frac{\mu_{Q_{T^M} T^M}}{A_{T^M}} \leq \frac{n\, \bar{\lambda} \sum_{j=1}^{b} |(S_j \cap M_j) \setminus T_j^M| \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2}}{\gamma \sum_{j=1}^{b} |(S_j \cap M_j) \setminus T_j^M| \kappa_j}. \tag{S.73}$$

Let $\bar{r} := \sum_{j=1}^{b} n \bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2} / (\gamma \kappa_j)$. We show next that $\bar{r}$ is an upper bound on $\mu_{Q_{T^M} T^M} / A_{T^M}$. By restricting the sum in $\bar{r}$ to the j such that $|(S_j \cap M_j) \setminus T_j^M| \neq 0$ and multiplying the numerator and denominator of the summand by $|(S_j \cap M_j) \setminus T_j^M|$, we get

$$\bar{r} \;\; \geq \sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^{b} \frac{|(S_j \cap M_j) \setminus T_j^M| n \bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2}}{|(S_j \cap M_j) \setminus T_j^M| \gamma \kappa_j} \tag{S.74}$$

Note that for any collections $(\alpha_j, \delta_j) \in \mathbb{R} \times \mathbb{R} \setminus \{0\}$, $j = 1, \ldots, b$, we have

$$\sum_{j=1}^{b} \frac{\alpha_j}{\delta_j} = \sum_{j=1}^{b} \frac{\alpha_j \frac{1}{\delta_j}(\delta_j + \sum_{l\neq j} \delta_l)}{\sum_{j=1}^{b} \delta_j} = \frac{\sum_{j=1}^{b} \alpha_j (1 + \sum_{l\neq j} \frac{\delta_l}{\delta_j})}{\sum_{j=1}^{b} \delta_j} \tag{S.75}$$

Using (S.75) in the right-hand side of (S.74), we get

$$\begin{aligned}
\bar{r} &\geq \frac{\sum_{j=1, |(S_j \cap M_j)\setminus T_j^M| \neq 0}^{b} |(S_j \cap M_j) \setminus T_j^M| n\bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2} \left(1 + \sum_{l\neq j} \frac{|(S_l \cap M_l)\setminus T_l^M|\gamma\kappa_l}{|(S_j \cap M_j)\setminus T_j^M|\gamma\kappa_j}\right)}{\sum_{j=1, |(S_j \cap M_j)\setminus T_j^M| \neq 0}^{b} |(S_j \cap M_j) \setminus T_j^M| \gamma\kappa_j} \\
&\geq \frac{\sum_{j=1}^{b} |(S_j \cap M_j) \setminus T_j^M| n\bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2}}{\sum_{j=1}^{b} \left|(S \cap M_j) \setminus T_j^M\right| \gamma\kappa_j}
\end{aligned}$$

where last inequality follows from $\left(1 + \sum_{l\neq j} \frac{|(S_l \cap M_l)\setminus T_l^M|\gamma\kappa_l}{|(S_j \cap M_j)\setminus T_j^M|\gamma\kappa_j}\right) \geq 1$ for all j and from the identity $\sum_{j=1, |(S_j \cap M_j)\setminus T_j^M| \neq 0}^{b} |(S_j \cap M_j) \setminus T_j^M| \gamma\kappa_j = \sum_{j=1}^{b} \left|(S \cap M_j) \setminus T_j^M\right| \gamma\kappa_j$. Then, by (S.73),

$$\frac{\mu_{Q_{T^M} T^M}}{A_{T^M}} \;\leq\; \bar{r} \;=\; \sum_{j=1}^{b} \frac{n\bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2}}{\gamma\kappa_j}.$$

By definition of $S_j^S(\kappa_j)$, $n\bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2} = o(\kappa_j)$ for all j and $\mu_{Q_{T^M} T^M} = o(A_{T^M})$ when $M \cap S^S(\kappa) \neq \emptyset$ too. We can now apply Lemma 10.10 and get that for every $M \in \mathcal{M} \setminus \mathcal{T}(\kappa)$, for any $\psi \in (0, 1)$ and every n large enough, $\mathbb{E}(NC(M)) \;\leq\; e^{-\psi A_{T^M}}$.

The second step of the proof is to lower-bound $A_{T^M} = \gamma\Delta_{MT^M} + \frac{1-\gamma}{6}\mu_{Q_{T^M} M}$. Let

$$A^*_{T^M} \;:=\; \gamma \sum_{j=1}^{b} |M_j \setminus T_j^M| \kappa_j + \sum_{j=1}^{b} |T_j^M \setminus M_j| \left(\tfrac{1-\gamma}{6} n\rho(\mathbf{X}) \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2} - \gamma\kappa_j\right)$$

Recall that, for every j, $T_j^M \setminus M_j \subseteq S_j^L(\kappa_j)$. We have then $\min_{i \in T_j^M \setminus M_j} \beta_i^{*2} \geq \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2}$, and by (S.72), $A_{T^M} \geq A^*_{T^M}$. It follows that for any $\psi \in (0, 1)$ and for all n large enough,

$$\mathbb{E}(NC(M)) \;\leq\; e^{-\psi A^*_{T^M}}. \tag{S.76}$$

To conclude the second part of the proof we lower-bound $\psi A^*_{T^M} = \sum_{j=1}^{b} |M_j \setminus T_j^M| \psi\gamma\kappa_j + \sum_{j=1}^{b} |T_j^M \setminus M_j| \psi \left(\frac{1-\gamma}{6} n\rho(\mathbf{X}) \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2} - \gamma\kappa_j\right)$. To do this, we obtain a lower bound for $\psi\gamma\kappa_j$ and for $\psi\left(\frac{1-\gamma}{6} n\rho(\mathbf{X}) \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2} - \gamma\kappa_j\right)$.

The definition of $S_j^L(\kappa_j)$ implies there exists some $g'_j \to \infty$ such that

$$\frac{(1-\gamma)n\rho(\mathbf{X})}{6} \min_{i\in S_j^L(\kappa_j)} \beta_i^{*2} \; - \; \kappa_j \; = \; \ln(s_j) + g'_j. \tag{S.77}$$

Let $\delta \in (0,1)$ and denote $\bar{m}_j = \max\left\{\frac{2\ln(p_j-s_j)}{f_j}, \frac{2\ln(s_j)}{g'_j}\right\}$, where f_j is given in Assumption A4. Take $\psi = \max_{j=1,\dots,b} \frac{\xi+\delta+\bar{m}_j}{1+\bar{m}_j}$ for some $\xi \in (0, 1-\delta)$ then $\psi \in (0,1)$ and we have, for every $j = 1, \dots, b$,

$$\psi > \frac{\delta + \frac{2\ln(p_j-s_j)}{f_j}}{1 + \frac{2\ln(p_j-s_j)}{f_j}} = \frac{\delta f_j/2 + \ln(p_j - s_j)}{f_j/2 + \ln(p_j - s_j)} \tag{S.78}$$

$$\psi > \frac{\delta + \frac{2\ln(s_j)}{g'_j}}{1 + \frac{2\ln(s_j)}{g'_j}} = \frac{\delta g'_j/2 + \ln(s_j)}{g'_j/2 + \ln(s_j)} \geq \frac{\delta g'_j/2 + \ln(s_j)}{g'_j + \ln(s_j)}. \tag{S.79}$$

Recall that Assumptions A4 defines $f_j = \kappa_j - \ln(p_j - s_j)$ and $\gamma = \frac{1}{2}(1 + \max_j \frac{\ln(p_j-s_j)}{\kappa_j})$ respectively. Hence,

$$\gamma\kappa_j \;\geq\; \frac{1}{2}\Big(1 + \frac{\ln(p_j - s_j)}{\kappa_j}\Big)\kappa_j \;=\; \ln(p_j - s_j) + \frac{1}{2}(\kappa_j - \ln(p_j - s_j)) \;=\; \ln(p_j - s_j) + \frac{1}{2}f_j.$$

Hence, by (S.78), we have

$$\psi\gamma\kappa_j \;\geq\; \psi\Big(\ln(p_j - s_j) + \frac{1}{2}f_j\Big) \;\geq\; \ln(p_j - s_j) + \delta\frac{1}{2}f_j. \tag{S.80}$$

Further,

$$\psi\left(\tfrac{1-\gamma}{6}n\rho(\mathbf{X}) \min_{i\in S_j^L(\kappa_j)} \beta_i^{*2} - \gamma\kappa_j\right) \;\geq\; \psi\left(\ln(s_j) + g'_j\right) \;\geq\; \ln(s_j) + \delta\frac{1}{2}g'_j. \tag{S.81}$$

where the first inequality follows from (S.77) and the second inequality from (S.79). In (S.76), $\psi A^*_{T^M} = \sum_{j=1}^b |M_j \setminus T_j^M|\psi\gamma\kappa_j + \sum_{j=1}^b |T_j^M \setminus M_j|\psi\left(\frac{1-\gamma}{6}n\rho(\mathbf{X})\min_{i\in S_j^L(\kappa_j)} \beta_i^{*2} - \gamma\kappa_j\right)$. Then by (S.80) and (S.81), we get

$$\mathbb{E}(NC(M)) \;\leq\; \exp\left\{-\sum_{j=1}^b |M_j \setminus T_j^M|(\ln(p_j - s_j) + \delta\tfrac{f_j}{2}) - \sum_{j=1}^b |T_j^M \setminus M_j|(\ln(s_j) + \delta\tfrac{g'_j}{2})\right\}. \tag{S.82}$$

For the final step of the proof, denote $\mathcal{S} = \sum_{M\in\mathcal{M}\setminus\mathcal{T}(\kappa)} \mathbb{E}\left(NC(M)\right)$ for convenience. By (S.82) we have

$$\mathcal{S} \leq \sum_{M\in\mathcal{M}\setminus\mathcal{T}(\kappa)} e^{-\sum_{j=1}^b |M_j\setminus T_j^M|\left(\ln(p_j-s_j)+\delta\frac{f_j}{2}\right)-\sum_{j=1}^b |T_j^M\setminus M_j|\left(\ln(s_j)+\delta\frac{g'_j}{2}\right)}.$$

We split the sum in the right-hand side above into sums over models M such that $T^M = T$ for some common $T \in \mathcal{T}(\kappa)$. Denote for any $T \in \mathcal{T}$, $\mathcal{M}(T) := \{M \in \mathcal{M} \setminus \mathcal{T}(\kappa) \,|T^M = T\}$, then

$$\mathcal{S} \leq \sum_{T \in \mathcal{T}(\kappa)} \sum_{M \in \mathcal{M}(T)} e^{-\sum_{j=1}^b |M_j \setminus T_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |T_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g_j'}{2} \right)}. \tag{S.83}$$

The right hand-side of (S.83) is composed of a double sum over $T \in \mathcal{T}(\kappa)$ and over $M \in \mathcal{M}(T)$. Consider the sum over $M \in \mathcal{M}(T)$, add T to it, and denote it

$$\mathcal{S}(T) = \sum_{M \in \mathcal{M}(T) \cup T} e^{-\sum_{j=1}^b |M_j \setminus T_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |T_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g_j'}{2} \right)}. \tag{S.84}$$

In the summand in the right-hand side of (S.84), the case $M = T$ correspond to $|M_j \setminus T_j^M| = |T_j^M \setminus M_j| = 0$ for all j and the summand is then 1. By (S.83), we then get that

$$\mathcal{S} \leq \sum_{T \in \mathcal{T}(\kappa)} \Big(\mathcal{S}(T) - 1 \Big). \tag{S.85}$$

For each $T = T^M$, we further split $\mathcal{S}(T)$ into sums over subsets of models M that have u_j more parameters than T^M in block j, and are missing w_j parameters from T^M in block j. Specifically, consider models M such that, for all j, $|M_j \setminus T_j^M| = u_j$ and $|T_j^M \setminus M_j| = w_j$ with $u_j \in \{0, \ldots, p_j - s_j, \ldots, p_j - s_j + |S_j^S(\kappa_j)|\}$ and $w_j \in \{0, \ldots, |S_j^L(\kappa_j)|\}$. Denote by

$$S_{\mathbf{w}}^{\mathbf{u}}(T) = \sum_{M \in \mathcal{M}(T) \cup T : \forall j |M_j \setminus S_j| = u_j, |S_j \setminus M_j| = w_j} e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g_j'}{2} \right)}.$$

We get

$$\mathcal{S}(T) = \sum_{w_1 = 0}^{|S_1^L(\kappa)|} \cdots \sum_{w_b = 0}^{|S_b^L(\kappa)|} \sum_{u_1 = 0}^{p_1 - s_1 + |S_1^S|} \cdots \sum_{u_b = 0}^{p_b - s_b + |S_b^S|} S_{\mathbf{w}}^{\mathbf{u}}(T). \tag{S.86}$$

The number of models missing, for all j, w_j out of the $|S_j^L(\kappa)|$ large active parameters and having u_j inactive or small active parameters from B_j is $\prod_{j=1}^b \binom{p_j - s_j + |S_j^S(\kappa)|}{u_j} \binom{|S_j^L(\kappa)|}{w_j}$. We thus have

$$\begin{aligned} S_{\mathbf{w}}^{\mathbf{u}}(T) &= \left(\prod_{j=1}^b \binom{p_j - s_j + |S_j^S(\kappa)|}{u_j} \binom{|S_j^L(\kappa)|}{w_j} \right) e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g_j'}{2} \right)} \\ &= \prod_{j=1}^b \binom{p_j - s_j + |S_j^S(\kappa)|}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right)} \binom{|S_j^L(\kappa)|}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g_j'}{2} \right)}. \end{aligned}$$

Inputting the expression above in (S.86) and factorizing over terms in u_j and w_j gives

$$\mathcal{S}(T) \leq \prod_{j=1}^{b}\left(\sum_{u_j=0}^{p_j-s_j+|S_j^S(\kappa)|}\binom{p_j-s_j+|S_j^S(\kappa)|}{u_j}e^{-u_j(\ln(p_j-s_j)+\delta\frac{f_j}{2})}\right)$$
$$\cdot\left(\sum_{w_j=0}^{|S_j^L(\kappa)|}\binom{|S_j^L(\kappa)|}{w_j}e^{-w_j(\ln(s_j)+\delta\frac{g_j'}{2})}\right).$$

By the bound in (S.54) and taking the terms in $u_j=0$ and $w_j=0$ out of the sums above, we have

$$\mathcal{S}(T) \leq \prod_{j=1}^{b}\left(1+\sum_{u_j=1}^{p_j-s_j+|S_j^S(\kappa_j)|}e^{-u_j\left(\delta\frac{f_j}{2}-\ln\left(1+\frac{|S_j^S(\kappa_j)|}{p_j-s_j}\right)-1\right)}\right) \tag{S.87}$$
$$\cdot\left(1+\sum_{w_j=1}^{|S_j^L(\kappa_j)|}e^{-w_j\left(\delta\frac{g_j'}{2}+\ln\left(\frac{s_j}{|S_j^L(\kappa_j)|}\right)-1\right)}\right).$$

Denote

$$d_j \;=\; e^{1+\ln\left(1+\frac{|S_j^S(\kappa_j)|}{p_j-s_j}\right)-\delta\frac{f_j}{2}}, \qquad h_j \;=\; e^{1-\ln\left(\frac{s_j}{|S_j^L(\kappa_j)|}\right)-\delta\frac{g_j'}{2}}.$$

By assumption $|S_j^S(\kappa_j)| = O(p_j-s_j)$, and by definition of $|S_j^L(\kappa_j)|$ we have $s_j \geq |S_j^L(\kappa_j)|$. By Assumptions A4 and the definition of $|S_j^L(\kappa_j)$, we also have $f_j \to \infty$ and $g_j' \to \infty$, and then $\lim_{n\to\infty} = d_j = \lim_{n\to\infty} h_j = 0$. Using the properties of geometric series, for every j we have

$$1+\sum_{u_j=1}^{p_j-s_j+|S_j^S(\kappa_j)|}e^{-u_j\left(\delta\frac{f_j}{2}-\ln\left(1+\frac{|S_j^S(\kappa_j)|}{p_j-s_j}\right)-1\right)} = \frac{1-d_j^{p_j-s_j+|S_j^S(\kappa_j)|+1}}{1-d_j}$$
$$1+\sum_{w_j=1}^{|S_j^L(\kappa_j)|}e^{-w_j\left(\delta\frac{g_j'}{2}+\ln\left(\frac{s_j}{|S_j^L(\kappa_j)|}\right)-1\right)} = \frac{1-h_j^{|S_j^L(\kappa_j)|+1}}{1-h_j},$$

where both expressions converge to 1 as n grows. By (S.85) and (S.87):

$$\mathcal{S} \leq \sum_{T\in\mathcal{T}}\left(\prod_{j=1}^{b}\Big(\frac{1-d_j^{p_j-s_j+|S_j^S(\kappa_j)|+1}}{1-d_j}\Big)\Big(\frac{1-h_j^{|S_j^L(\kappa_j)|+1}}{1-h_j}\Big)-1\right).$$

Each of the summand vanishes as $n\to\infty$. Moreover, by assumption $|S^I(\kappa)| = O(1)$ and then $|\mathcal{T}(\kappa)| = 2^{|S^I(\kappa)|} = O(1)$. We thus have $\lim_{n\to\infty}\mathcal{S} = \lim_{n\to\infty}\sum_{M\in\mathcal{M}\setminus\mathcal{T}(\kappa)}\mathbb{E}\left(NC(M)\right) = 0$.

Further, by Lemma 10.6, $P(\hat{S}^{ei} \notin \mathcal{T}(\kappa)) \;\leq\; (|\mathcal{T}(\kappa)|+1)\mathcal{S} \;=\; (2^{|S^I(\kappa)|}+1)\mathcal{S}$. Since $\mathcal{S}$ vanishes and $|S^I(\kappa)| = O(1)$, $\lim_{n\to\infty} P(\hat{S}^{ei} \in \mathcal{T}(\kappa)) = 1$ as claimed.

10.3 Proofs of Section 4

10.3.1 Proof of Proposition 4.2

Denote

$$A_j := \frac{1}{p_j} \sum_{i \in B_j} \sum_{M \in \mathcal{T}(\kappa) | i \in M} NC(M) \quad \text{and} \quad C_j := \frac{1}{p_j} \sum_{i \in B_j} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa) | i \in M} NC(M).$$

For every $j = 1, \ldots, b$, we have the decomposition

$$\frac{\hat{s}_j}{p_j} = \frac{\sum_{i \in B_j} \sum_{M \in \mathcal{M} | i \in M} NC(M)}{p_j} = A_j + C_j. \tag{S.88}$$

To show the lower bound on $\hat{s}_j / p_j$, we decompose A_j

$$\begin{aligned} A_j &= \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j} \frac{I(i \in M_j)}{p_j} \\ &= \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in S_j^L(\kappa_j)} \frac{I(i \in M_j)}{p_j} + \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j \setminus S_j^L(\kappa_j)} \frac{I(i \in M_j)}{p_j} \\ &= \frac{|S_j^L(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j \setminus S_j^L(\kappa_j)} \frac{I(i \in M_j)}{p_j}. \end{aligned} \tag{S.89}$$

where the last equality follows from $I(i \in M_j) = 1$ for all $i \in S^L(\kappa)$ when $M \in \mathcal{T}(\kappa)$. The rightmost term above and C_j are nonnegative, then by the linearity of the expectation

$$\mathbb{E}\left(\frac{\widehat{s}_j}{p_j}\right) \geq \frac{|S_j^L(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} \mathbb{E}(NC(M)).$$

By Theorem 3.7, $\lim_{n \to \infty} \sum_{M \in \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 1$. It follows that $\lim_{n \to \infty} \mathbb{E}\left(\frac{\widehat{s}_j}{p_j}\right) \geq \frac{|S_j^L(\kappa_j)|}{p_j}$ for every $j = 1, \ldots, b$.

We now prove the upper bound. Recall that $\mathcal{T}(\kappa)$ by definition includes models that have no small signals, i.e. all parameters are in $S^L(\kappa) \cup S^I(\kappa)$. That is, for all $M \in \mathcal{T}(\kappa)$, we have that $I(i \in M_j) = 0$ for all $i \in B_j \setminus (S_j^L(\kappa_j) \cup S_j^I(\kappa_j))$. Hence, A_j in (S.89) satisfies

$$\begin{aligned} A_j &= \frac{|S_j^L(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in S_j^I(\kappa_j)} \frac{I(i \in M_j)}{p_j} \\ &\leq \frac{|S_j^L(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + \frac{|S_j^I(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) \end{aligned}$$

where the inequality follows from $\sum_{i\in S_j^I(\kappa_j)} I(i\in M_j) \leq |S_j^I(\kappa_j)|$ for all M. By (S.88), we then have

$$\frac{\hat{s}_j}{p_j} \leq \frac{|S_j^L(\kappa_j)| + |S_j^I(\kappa_j)|}{p_j} \sum_{M\in\mathcal{T}(\kappa)} NC(M) + C_j. \tag{S.90}$$

Moreover, for every $j = 1,\ldots,b$, C_j satisfies

$$C_j = \sum_{M\in\mathcal{M}\setminus\mathcal{T}(\kappa)} NC(M) \sum_{i\in B_j} \frac{I(i\in M_j)}{p_j} \leq \sum_{M\in\mathcal{M}\setminus\mathcal{T}(\kappa)} NC(M) \tag{S.91}$$

where the inequality follows from $\sum_{i\in B_j} \frac{I(i\in M_j)}{p_j} \leq 1$ for all M. Taking expectations in (S.90) and (S.91) gives

$$\mathbb{E}\left(\frac{\widehat{s}_j}{p_j}\right) \leq \frac{|S_j^L(\kappa_j)| + |S_j^I(\kappa_j)|}{p_j} \sum_{M\in\mathcal{T}(\kappa)} \mathbb{E}(NC(M)) + \sum_{M\in\mathcal{M}\setminus\mathcal{T}(\kappa)} \mathbb{E}(NC(M)).$$

By Theorem 3.7, we have on one hand $\lim_{n\to\infty} \sum_{M\in\mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 1$ and, on the other hand, $\lim_{n\to\infty} \sum_{M\in\mathcal{M}\setminus\mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 0$. It follows that $\lim_{n\to\infty} \mathbb{E}\left(\frac{\widehat{s}_j}{p_j}\right) \leq \frac{|S_j^L(\kappa_j)|+|S_j^I(\kappa_j)|}{p_j} = \frac{s_j - |S_j^S(\kappa_j)|}{p_j}$ for every $j = 1,\ldots,b$, which proves the upper bound.

10.3.2 Proof of Lemma 4.1

Denote the marginal likelihood of $\mathbf{y}$ given $\boldsymbol{\theta}$ by $H(\boldsymbol{\theta}) = p(\mathbf{y} \mid \boldsymbol{\theta})$. For every $j = 1,\ldots,b$, the partial derivative of its logarithm with respect to θ_j is

$$\frac{\partial \ln H(\boldsymbol{\theta})}{\partial\theta_j} = \frac{\partial H(\boldsymbol{\theta})}{\partial\theta_j} H(\boldsymbol{\theta})^{-1}. \tag{S.92}$$

Observe that:

$$H(\boldsymbol{\theta}) = \sum_{M\in\mathcal{M}} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta}) \quad \text{and} \quad \frac{\partial H(\boldsymbol{\theta})}{\partial\theta_j} = \sum_{M\in\mathcal{M}} p(\mathbf{y} \mid M, \boldsymbol{\theta}) \frac{\partial p(M \mid \boldsymbol{\theta})}{\partial\theta_j} \tag{S.93}$$

Recall that each model is defined as $M = (m_1,\ldots,m_p)$ where $m_i = I(\beta_i \neq 0)$ indicates whether variable j is included under M, and that our choice of model prior in (6) is

$$p(M \mid \boldsymbol{\theta}) = \prod_{j=1}^{b} \left(\theta_j\right)^{\sum_{i\in B_j} m_i} \left(1-\theta_j\right)^{p_j - \sum_{i\in B_j} m_i}.$$

Hence, simple algebra shows that for every $M \in \mathcal{M}$

$$\frac{\partial p(M \mid \boldsymbol{\theta})}{\partial \theta_j} = p(M \mid \boldsymbol{\theta}) \left(\frac{\sum_{i \in B_j} m_i}{\theta_j} - \frac{p_j - \sum_{i \in B_j} m_i}{1 - \theta_j} \right). \tag{S.94}$$

Replacing (S.94) into (S.93), and using that for any function f

$$\sum_{M \in \mathcal{M}} \sum_{i \in B_j} m_i f(M) = \sum_{i \in B_j} \sum_{M \in \mathcal{M}: m_i = 1} f(M),$$

we get

$$\begin{aligned} \frac{\partial H(\boldsymbol{\theta})}{\partial \theta_j} &= \frac{1}{\theta_j} \sum_{i \in B_j} \sum_{M \in \mathcal{M}: m_i = 1} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta}) \\ &- \frac{1}{1 - \theta_j} \left[p_j H(\boldsymbol{\theta}) - \sum_{i \in B_j} \sum_{M \in \mathcal{M}: m_i = 1} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta}). \right] \end{aligned} \tag{S.95}$$

Note that

$$\frac{\sum_{M \in \mathcal{M}: m_i = 1} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta})}{H(\boldsymbol{\theta})} = \frac{\sum_{M \in \mathcal{M}: m_i = 1} p(\mathbf{y}, M \mid \boldsymbol{\theta})}{p(\mathbf{y} \mid \boldsymbol{\theta})} = P(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta})$$

By (S.92), we then get the following expression of the partial derivative, for every $j = 1, \ldots, b$,

$$\frac{\partial \ln H(\boldsymbol{\theta})}{\partial \theta_j} = \frac{1}{\theta_j} \sum_{i \in B_j} P(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}) - \frac{1}{1 - \theta_j} \left[p_j - \sum_{i \in B_j} P(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}) \right]$$

Setting the partial derivatives to 0 and solving for θ_j gives the desired result.

10.3.3 Proof of Theorem 4.3

The proof strategy is to show Assumptions A1, A6 and A7 hold to apply Theorem 3.3. Recall that Theorem 4.3 makes Assumptions A1 and A9, hence it suffices to show that A6-A7 hold. We first derive a convenient decomposition of the penalties κ_j^A. The second step is to show that κ_j^A satisfies Assumption A4 with probability going to 1 as n grows. The third step is to show that Assumption A9 implies Assumption A5 for the κ_j^A with probability going to 1. The consistency of $\hat{S}^{A,ei}$ then follows from Theorem 3.3.

Denote for any $M \in \mathcal{M}$, $NC^{\circ}(M)$, the normalized criterion value for model M under Step 1 penalty κ°. For this choice of penalty and every $j = 1, \ldots, b$, we have

$$\begin{aligned}\frac{\widehat{s}_j}{p_j} &= \frac{1}{p_j}\sum_{i\in B_j}\sum_{M\in\mathcal{M}} NC^{\circ}(M)I(i\in M)\\ &= \frac{1}{p_j}\sum_{M\in\mathcal{M}} NC^{\circ}(M)\sum_{i\in B_j} I(i\in M)\\ &= \frac{s_j}{p_j}NC^{\circ}(S) + \sum_{M\in\mathcal{M}|M\neq S}\frac{|M_j|}{p_j}NC^{\circ}(M).\end{aligned}$$

Using that $NC^{\circ}(S) = 1 - \sum_{M\in\mathcal{M}|M\neq S} NC^{\circ}(M)$, we get

$$\frac{\hat{s}_j}{p_j} = \frac{s_j}{p_j} + \sum_{M\in\mathcal{M}|M\neq S}\frac{|M_j| - s_j}{p_j}NC^{\circ}(M).$$

Consider the decomposition of the sum in the right-hand side above between the sum over models M that contain more parameters than S in block j and the sum over those that contain less parameters than S in block b. Denote

$$O_j^{\circ} := \sum_{M\in\mathcal{M}||M_j|>s_j}\frac{|M_j| - s_j}{p_j}NC^{\circ}(M) \quad \text{and} \quad U_j^{\circ} := \sum_{M\in\mathcal{M}||M_j|<s_j}\frac{s_j - |M_j|}{p_j}NC^{\circ}(M).$$

We have

$$\frac{\hat{s}_j}{p_j} = \frac{s_j}{p_j} \ + \ O_j^{\circ} - \ U_j^{\circ}. \tag{S.96}$$

Observe that we have the following decomposition of Step 2 penalties

$$\kappa_j^A = \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln\left(\frac{p_j - \hat{s}_j}{p_j - s_j}\right).$$

By (S.96), it follows that

$$\kappa_j^A \;=\; \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln\left(1 - \frac{p_j(O_j^{\circ} - U_j^{\circ})}{p_j - s_j}\right), \tag{S.97}$$

completing the first step of the proof.

We continue with the second step of the proof: showing that the κ_j^A's satisfy Assumption A4 with probability going to 1. Recall that Assumption A4 states that there exists $f_j \to \infty$ (as $n \to \infty$) such that for every sufficiently large n,

$$\kappa_j \;=\; \ln(p_j - s_j) + f_j.$$

Since U_j° is nonnegative, a lower bound on κ_j^A is

$$\kappa_j^A \geq \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln\left(1 - \frac{p_j O_j^\circ}{p_j - s_j}\right). \tag{S.98}$$

Plugging in the definition of O_j°, we have that

$$\frac{p_j O_j^\circ}{p_j - s_j} = \sum_{M \in \mathcal{M} | |M_j| > s_j} \frac{|M_j| - s_j}{p_j - s_j} NC^\circ(M) \leq \sum_{M \in \mathcal{M} | |M_j| > s_j} NC^\circ(M)$$

where the inequality follows from $(|M_j| - s_j)/(p_j - s_j) \leq 1$ for all M. Note that if M is such that $|M_j| > s_j$, then $M \notin \mathcal{T}(\kappa^\circ)$ (this follows immediately from the definition of $\mathcal{T}(\kappa)$ in (21)) and therefore $\sum_{M \in \mathcal{M} | |M_j| > s_j} NC^\circ(M) \leq \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ)} NC^\circ(M)$. Moreover, κ° satisfies Assumption A4 and the assumptions of Theorem 3.7 are met for κ°. Then, by Theorem 3.7, $\lim_{n\to\infty} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ)} \mathbb{E}(NC^\circ(M) =) \lim_{n\to\infty} \sum_{M \in \mathcal{M} | |M_j| > s_j} \mathbb{E}(NC^\circ(M)) = 0$. It follows that $\frac{p_j O_j^\circ}{p_j - s_j}$ vanishes in probability and so does $\ln\left(1 - \frac{p_j O_j^\circ}{p_j - s_j}\right)$.

With probability going to 1, we then have that

$$\kappa_j^A \geq \ln(p_j - s_j) + \ln(\sqrt{n}) \tag{S.99}$$

and hence that the κ_j^A's satisfy Assumption A4.

For the third part of the proof, we now show that Assumption A9 implies Assumption A5 for the κ_j^A. Assumption A5 for the κ_j^A states that for each block j there exists $g_j \to \infty$ such that for large enough n,

$$\sqrt{\frac{(1-\gamma) n \rho(\mathbf{X})}{6}} \beta^*_{\min,j} - \sqrt{\kappa_j^A} = \sqrt{\ln(s_j)} + g_j.$$

where γ takes value

$$\gamma = \frac{1}{2}\left(1 + \max_j \frac{\ln(p_j - s_j)}{\kappa_j^A}\right). \tag{S.100}$$

To show that Assumption A9 implies Assumption A5 for the κ_j^A with probability going to

1, it suffices to show that the following two inequalities

$$\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j} \geq \sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j}, \quad \text{and} \tag{S.101}$$

$$-\sqrt{\kappa_j^A} \geq -\sqrt{\ln\left(p - |S^L(\kappa^\circ)|\right) + \frac{1}{2}\ln(n)} \tag{S.102}$$

hold with probability going to 1 for γ as in (S.100) and $\xi = \frac{1}{2}\Big(1 + \max_j \frac{\ln(p_j - s_j)}{\ln(p_j-s_j)+0.5\ln(n)}\Big)$ (defined in Assumption A9). We first show (S.101) holds with probability going to 1 and then that (S.102) does too.

By (S.99) we have that with probability going to 1, for any j

$$\frac{\ln(p_j - s_j)}{\kappa_j^A} \leq \frac{\ln(p_j - s_j)}{\ln(p_j - s_j) + \ln(n)/2}.$$

It follows that

$$\gamma = \frac{1}{2}\Big(1 + \max_{j=1,\dots,b} \frac{\ln(p_j - s_j)}{\kappa_j^A}\Big) \leq \frac{1}{2}\Big(1 + \max_j \frac{\ln(p_j - s_j)}{\ln(p_j - s_j) + \ln(n)/2}\Big) = \xi$$

and (S.101) holds with probability going to 1.

We now upper bound κ_j^A to show (S.102) holds with probability going to 1. By (S.97), we can write

$$\kappa_j^A = \kappa_j^{EB} + \ln(\widehat{s}_j) = \ln(p_j - s_j) + \ln\left(\sqrt{n}\right) + \ln\left(1 - \frac{p_j(O_j^\circ - U_j^\circ)}{p_j - s_j}\right).$$

Since $O_j^\circ \geq 0$, we obtain that

$$\kappa_j^A \leq \ln\left(p_j - s_j\right) + \ln(\sqrt{n}) + \ln\left(1 + \frac{p_j}{p_j - s_j}U_j^\circ\right).$$

We split the sum in U_j° between models in $\mathcal{T}(\kappa^\circ)$ and those not in $\mathcal{T}(\kappa^\circ)$.

$$U_j^\circ = \sum_{M \in \mathcal{T}(\kappa^\circ) || M_j| < s_j} \frac{s_j - |M_j|}{p_j} NC^\circ(M) + \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ) || M_j| < s_j} \frac{s_j - |M_j|}{p_j} NC^\circ(M).$$

If $M \in \mathcal{T}(\kappa^\circ)$, then by definition $|M_j| \geq |S_j^L(\kappa^\circ)|$ and thus $s_j - |M_j| \leq s_j - |S_j^L(\kappa^\circ)|$. A bound on $s_j - |M_j|$ for $M \notin \mathcal{T}(\kappa^\circ)$ is simply $s_j - |M_j| \leq s_j$. It follows that

$$U_j^\circ \leq \frac{s_j - |S_j^L(\kappa^\circ)|}{p_j} \sum_{M \in \mathcal{T}(\kappa^\circ) || M_j| < s_j} NC^\circ(M) + \frac{s_j}{p_j} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ) || M_j| < s_j} NC^\circ(M)$$

By Theorem 3.7, $\sum_{M\in\mathcal{T}(\kappa^\circ)||M_j|<s_j} NC^\circ(M)$ and $\sum_{M\in\mathcal{M}\setminus\mathcal{T}(\kappa^\circ)||M_j|<s_j} NC^\circ(M)$ converge in probability to 1 and 0 respectively. We then get that, with probability going to 1,

$$\frac{p_j}{p_j - s_j} U_j^\circ \leq \frac{s_j - |S_j^L(\kappa^\circ)|}{p_j - s_j} \tag{S.103}$$

By (S.103), with probability going to 1,

$$\begin{aligned}
\kappa_j^A &\leq \ln\left(p_j - s_j\right) + \ln(\sqrt{n}) + \ln\left(1 + \frac{s_j - |S_j^L(\kappa^\circ)|}{p_j - s_j}\right) \\
&= \ln\left(p_j - |S_j^L(\kappa^\circ)|\right) + \frac{1}{2}\ln(n). \qquad \text{(S.104)}
\end{aligned}$$

which shows (S.102) holds with probability going to 1 and that Assumption A9 implies Assumption A5 holds for the κ_j^A with probability going to 1.

Since Assumptions A4 and A5 hold with probability going to 1, by Theorem 3.3, $\lim_{n\to\infty} P(\hat{S}^{A,ei} = S) = 1$, as claimed.

10.3.4 Proof of Theorem ??

The proof strategy is to show that Assumptions A1, A6, and A9 hold to apply Theorem 4.3 where the block number of active and inactive parameters are given in (24). Recall that Theorem **??** assumes A1 and A6. Hence, it suffices to show that Assumption A9 holds.

By Assumption A10, with probability going to 1,

$$\begin{aligned}
&s_1 \to |\cup_{k=1}^K (S^{(k)} \cap S)| \\
&s_2 \to |\cap_{k=1}^K (S \setminus S^{(k)})| \\
&p_1 - |S_1^L(\kappa^\circ)| \to |\cup_{k=1}^K (S^{(k)} \setminus S^L(\kappa^\circ))|, \\
&p_2 - |S_2^L(\kappa^\circ)| \to \left|\cap_{k=1}^K \left(S^{(k)} \cup S^L(\kappa^\circ)\right)^C\right|.
\end{aligned}$$

Then, with probability going to 1, Assumption A11 implies Assumption A9 holds. By Theorem 4.3, we have $\lim_{n\to\infty} P(\hat{S}^{A,ei} = S) = 1$ in step 2 of Tran-s-ell0.

10.4 Proofs of Section ??

10.4.1 Proof of Proposition ??

First note that by discarding constant term

$$\arg\max_{M\in\mathcal{M}}\left\{\max_{\beta\in\mathcal{L}_M}\ell(\mathbf{y};\boldsymbol{\beta})-\sum_{j=1}^{b}\kappa_j|M_j|\right\}=\arg\min_{M\in\mathcal{M}}\left\{\min_{\beta\in\mathcal{L}_M}\left\{\tfrac{1}{2}\|\mathbf{y}-\sqrt{n}\boldsymbol{\beta}\|^2\right\}+\sum_{j=1}^{b}\kappa_j|M_j|\right\}$$

We also have $\min_{\beta\in\mathcal{L}_M}\frac{1}{2}\|\mathbf{y}-\sqrt{n}\boldsymbol{\beta}\|^2=\frac{1}{2}\|\mathbf{y}-\sqrt{n}\tilde{\boldsymbol{\beta}}_M\|^2=\frac{1}{2}\|\mathbf{y}\|^2-\frac{1}{2}\|\sqrt{n}\tilde{\boldsymbol{\beta}}_M\|^2$. The maximization in (8) can be then replaced with:

$$\arg\max_{M\in\mathcal{M}}\left\{\tfrac{1}{2}\|\sqrt{n}\tilde{\boldsymbol{\beta}}_M\|^2-\sum_{j=1}^{b}\kappa_j|M_j|\right\},\qquad M=M_1\cup\ldots\cup M_b. \tag{S.105}$$

Under the assumptions of (**??**), we can write

$$\tfrac{1}{2}\|\sqrt{n}\tilde{\boldsymbol{\beta}}_M\|^2-\sum_{j=1}^{b}\kappa_j|M_j| \;=\; \sum_{j=1}^{b}\sum_{i\in M_j}(\tfrac{n}{2}\tilde{\beta}_i^2-\kappa_j).$$

Then (S.105) can be maximized with respect to each M_j by including $i\in\hat{S}_j$ whenever $n\tilde{\beta}_i^2\geq 2\kappa_j$.

10.4.2 Proof of Lemma ??

Denote by $\hat{\boldsymbol{\beta}}$ the minimizer of the problem (**??**). The MLE under the full model is $\tilde{\boldsymbol{\beta}}=\frac{1}{\sqrt{n}}\mathbf{y}$ and the optimized function in (**??**) can be then rewritten as

$$\tfrac{1}{2}\mathbf{y}^\top\mathbf{y}+\tfrac{n}{2}\sum_{j=1}^{b}\sum_{i\in B_j}\left(\beta_i^2-2\tilde{\beta}_i\beta_i+2\tfrac{\lambda_j}{n}|\beta_i|\right)$$

which we optimize with respect to each β_i separately. For any $i\in B_j$ we have $\hat{\beta}_i=0$ if and only if $|\tilde{\beta}_i|\leq\lambda_j/n$. Similarly for (**??**), denoting $\hat{\boldsymbol{\beta}}$ the minimizer of the problem, for any $i\in B_j$ we have $\hat{\beta}_i=0$ if and only if $|\tilde{\beta}_i|\leq\lambda_j/(|\beta_i^\circ|n)$. If $\boldsymbol{\beta}^\circ=\tilde{\boldsymbol{\beta}}$, then for any $i\in B_j$ $\hat{\beta}_i=0$ if and only if $|\tilde{\beta}_i|\leq\sqrt{\lambda_j/n}$. If $\boldsymbol{\beta}^\circ$ is a LASSO estimate with penalization λ°, then for any $i\in B_j$ $\hat{\beta}_i=0$ if and only if $\tilde{\beta}_i^2+\mathrm{sign}(\lambda^\circ/n-\tilde{\beta}_i)\lambda^\circ/(n\tilde{\beta}_i)-\lambda_j/n\leq 0$. Equivalently, for any $i\in B_j$ $\hat{\beta}_i=0$ if and only if $|\tilde{\beta}_i|\leq\frac{1}{2}\left(\lambda^\circ/n+\sqrt{(\lambda^\circ/n)^2+4\lambda_j/n}\right)$.

10.4.3 Proof of Proposition ??

Part (i)

By the union bound and by Lemma 10.1 (i),

$$P\left(\hat{S}^{ei} \not\subseteq S\right) \;\leq\; \sum_{j=1}^{b} P\Big(\max_{i \in B_j \setminus S_j} |y_i/\sqrt{n}| > \tau_j\Big) \;\leq\; \sum_{j=1}^{b} \frac{e^{-\frac{n}{2}\left(\tau_j^2 - \frac{2\ln(p_j - s_j)}{n}\right)}}{\sqrt{\pi \ln(p_j - s_j)}}. \qquad \text{(S.106)}$$

By Assumption A4, the numerator on the right is bounded above by 1 for all sufficiently large n. It follows that

$$P\left(\hat{S}^{ei} \not\subseteq S\right) \;\leq\; \sum_{j=1}^{b} (\pi \ln(p_j - s_j))^{-1/2} \to 0 \qquad \text{as } n \to \infty.$$

Part (ii)

By independence, we have

$$P(\hat{S}^{ei} \subseteq S) \;=\; \prod_{j=1}^{b} P(\max_{i \in B_j \setminus S_j} |y_i/\sqrt{n}| \leq \tau_j).$$

Consider j such that $\lim_{n\to\infty} \frac{\sqrt{n}\tau_j}{\sqrt{2\ln(p_j - s_j)}} < 1$. In particular, there exists $c < 1$ such that, for all sufficiently large n, we have that $\tau_j \leq c\sqrt{2\ln(p_j - s_j)/n}$. Then, for any such n,

$$\begin{aligned} P\Big(\max_{i \in B_j \setminus S_j} |y_i/\sqrt{n}| \leq \tau_j\Big) \;&\leq\; P\Big(\max_{i \in B_j \setminus S_j} |y_i| \leq c\sqrt{2\ln(p_j - s_j)}\Big) \\ &\leq\; P\Big(\tfrac{1}{\sqrt{2\ln(p_j - s_j)}} \max_{i \in B_j \setminus S_j} y_i \leq c\Big). \end{aligned}$$

On the other hand, results from extreme value theory, Galambos (1987) Example 4.4.1, show that

$$\frac{1}{\sqrt{2\ln(p_j - s_j)}} \max_{i \in B_j \setminus S_j} y_i \;\xrightarrow{P}\; 1$$

and so

$$P\Big(\frac{1}{\sqrt{2\ln(p_j - s_j)}} \max_{i \in B_j \setminus S_j} y_i \leq c\Big) \;\to\; 0,$$

which implies that $P(\hat{S}^{ei} \subseteq S) \to 0$.

10.4.4 Proof of Lemma ??

Consider the events $A := \{\min_{i\in\{1,\dots,s\}} |\mu_i| - \max_{i\in\{1,\dots,s\}} |y_i - \mu_i| > a\}$ and $B := \{\min_{i\in\{1,\dots,s\}} |y_i| > a\}$. We first show that A implies B. A implies that, for all i, $(\min_{i\in\{1,\dots,s\}} |\mu_i|) - |y_i - \mu_i| > a$. By the triangle inequality, we have that $-|y_i - \mu_i| \leq |y_i| - |\mu_i|$ and therefore that

$$\text{for all } i, \quad (\min_{i\in\{1,\dots,s\}} |\mu_i|) - |y_i - \mu_i| \leq (\min_{i\in\{1,\dots,s\}} |\mu_i|) + |y_i| - |\mu_i| < |y_i|.$$

Then, A implies that, for all i, $|y_i| > a$, that is B. By the monotonicity of the probability P

$$P\Big(\min_{i\in\{1,\dots,s\}} |\mu_i| - \max_{i\in\{1,\dots,s\}} |y_i - \mu_i| > a\Big) \leq P\Big(\min_{i\in\{1,\dots,s\}} |y_i| > a\Big).$$

10.4.5 Proof of Lemma ??

Since the y_i's are independent and identically distributed we have $P(\min_{i\in\{1,\dots,s\}} |y_i| > a) = P(|y_i| > a)^s = \exp\left(s \ln P(|y_i| > a)\right)$ for any $i \in \{1\dots, s\}$. Using that $\ln(1+x) < x$ for $x \in (-1, 0)$, we have

$$P(\min_{i\in\{1,\dots,s\}} |y_i| > a) \leq \exp\left(s(P(|y_i| > a) - 1)\right). \tag{S.107}$$

To get a bound on $P(\min_{i\in\{1,\dots,s\}} |y_i| > a)$ it is then enough to bound $P(|y_i| > a)$ for any i. By Lemma 10.1 (ii), for any $a < |\mu|$ we further have

$$\begin{aligned}
P(|y_i| > a) &= P\Big(z < \frac{|\mu| - a}{\sigma}\Big) + 1 - P\Big(z < \frac{a + |\mu|}{\sigma}\Big) \\
&= P(z < 0) + P\Big(0 < z < \frac{|\mu| - a}{\sigma}\Big) + 1 - P(z < 0) - P\Big(0 < z < \frac{a + |\mu|}{\sigma}\Big) \\
&= 1 + P\Big(0 < z < \frac{|\mu| - a}{\sigma}\Big) - P\Big(0 < z < \frac{a + |\mu|}{\sigma}\Big). \qquad \text{(S.108)}
\end{aligned}$$

From Chu (1955), for any $\delta > 0$, $\frac{1}{2}\left(1 - e^{-\delta^2/2}\right)^{\frac{1}{2}} \leq P(0 < z < \delta) \leq \frac{1}{2}\left(1 - e^{-2\delta^2/\pi}\right)^{\frac{1}{2}}$. Thus, for any $\gamma, \delta > 0$, we have that

$$P(0 < z \leq \gamma) - P(0 < z \leq \delta) \leq \frac{1}{2}\left(1 - e^{-2\gamma^2/\pi}\right)^{\frac{1}{2}} - \frac{1}{2}\left(1 - e^{-\delta^2/2}\right)^{\frac{1}{2}}.$$

Applying the above to (S.108), we get

$$P(|y_i| > a) \leq 1 + \frac{1}{2}\left(\left(1 - e^{-2(|\mu| - a)^2/\pi\sigma^2}\right)^{\frac{1}{2}} - \left(1 - e^{-(a+|\mu|)^2/2\sigma^2}\right)^{\frac{1}{2}}\right).$$

Using that $x^{\frac{1}{2}} - y^{\frac{1}{2}} = \frac{x-y}{x^{\frac{1}{2}}+y^{\frac{1}{2}}}$ for $x, y > 0$, we get

$$P(|y_i| > a) \leq 1 - \frac{e^{-2(|\mu|-a)^2/\pi\sigma^2} - e^{-(a+|\mu|)^2/2\sigma^2}}{2\left((1-e^{-2(|\mu|-a)^2/\pi\sigma^2})^{\frac{1}{2}} + (1-e^{-(a+|\mu|)^2/2\sigma^2})^{\frac{1}{2}}\right)}. \tag{S.109}$$

Finally, inputting in the bound in (S.109) in (S.107) gives the desired inequality.

10.4.6 Proof of Proposition ??

Part (i)

By the union bound,

$$P\left(\hat{S}^{ei} \not\supseteq S\right) \leq \sum_{j=1}^{b} P\left(\min_{i \in S_j} |y_i/\sqrt{n}| \leq \tau_j\right).$$

By Lemma **??**, for each j,

$$P\left(\min_{i \in S_j} |y_i/\sqrt{n}| \leq \tau_j\right) \;\leq\; P\left(\max_{i \in S_j} |y_i/\sqrt{n} - \beta_i| \geq \beta^*_{\min,j} - \tau_j\right).$$

By Lemma 10.1 (i)

$$P\left(\hat{S}^{ei} \not\supseteq S\right) \;\leq\; \sum_{j=1}^{b} \frac{e^{-\frac{n}{2}\left((\beta^*_{\min,j}-\tau_j)^2 - \frac{2\ln(s_j)}{n}\right)}}{\sqrt{\pi \ln(s_j)}}. \tag{S.110}$$

By Assumption A5, the nominator on the right is bounded above by 1 for all sufficiently large n. It follows that

$$P\left(\hat{S}^{ei} \not\supseteq S\right) \;\leq\; \sum_{j=1}^{b} (\pi \ln(s_j))^{-1/2} \to 0 \qquad \text{as } n \to \infty.$$

Part (ii)

Take any $j = 1, \ldots, b$ satisfying $\lim_{n\to\infty} \sqrt{n}(\beta^*_{\min,j} - \tau_j)/\sqrt{(\pi/2)\ln(s_j)} \leq 1$. Consider first the case $\beta^*_{\min,j} \leq \tau_j$. Then $\lim_{n\to\infty} \sqrt{n}(\beta^*_{\min,j} - \tau_j) < \infty$, and by Lemma **??**, $\lim_{n\to\infty} P(\hat{S}^{ei} \supseteq S) < 1$. We now consider the case $\beta^*_{\min,j} > \tau_j$. By Lemma **??**:

$$P\left(\min_{i\in S_j}\left|\frac{y_i}{\sqrt{n}}\right| > \tau_j\right) \;\leq\; \exp\left\{-\frac{s_j}{2}\frac{e^{-\frac{2n}{\pi}(\beta^*_{\min,j}-\tau_j)^2} - e^{-\frac{n}{2}(\tau_j+\beta^*_{\min,j})^2}}{\left(1-e^{-2n(\beta^*_{\min,j}-\tau_j)^2/\pi}\right)^{\frac{1}{2}} + \left(1-e^{-n(\tau_j+\beta^*_{\min,j})^2/2}\right)^{\frac{1}{2}}}\right\}.$$

Let $a_n = 2\left(1 - e^{-2n(\beta^*_{\min,j} - \tau_j)^2/\pi}\right)^{\frac{1}{2}} + 2\left(1 - e^{-n(\tau_j + \beta^*_{\min,j})^2/2}\right)^{\frac{1}{2}}$ and note that $a_n \in (0, 4]$. Thus, to show that $P\left(\min_{i \in S_j} |\frac{y_i}{\sqrt{n}}| > \tau_j\right)$ is bounded away from 1, it is enough to show that

$$\lim_{n\to\infty} \frac{s_j}{4}(e^{-2n(\beta^*_{\min,j} - \tau_j)^2/\pi} - e^{-n(\tau_j + \beta^*_{\min,j})^2/2}) \; > \; 0.$$

Since $\lim_{n\to\infty} \sqrt{n}(\beta^*_{\min,j} - \tau_j)/\sqrt{(\pi/2)\ln(s_j)} \leq 1$, we have that $\lim_{n\to\infty} s_j e^{-2n(\beta^*_{\min,j} - \tau_j)^2/\pi} \geq 1$. To conclude, we show that

$$\lim_{n\to\infty} s_j e^{-n(\tau_j + \beta^*_{\min,j})^2/2} \; \leq \; c \; < \; 1$$

for some $c \in (0, 1)$. Take c such that $\frac{s_j}{p_j} \leq c$. Such c exists by our assumption $\frac{s_j}{p_j} < 1$. We equivalently need $\sqrt{n}(\tau_j + \beta^*_{\min,j}) \geq \sqrt{2\ln(s_j/c)}$ for all n sufficiently large. To show that, note that, by assumption $\tau_j < \beta^*_{\min,j}$, and $s_j/c \leq p_j$, which gives

$$\frac{\tau_j + \beta^*_{\min,j}}{\sqrt{\frac{2\ln(s_j/c)}{n}}} \; \geq \; \frac{2\tau_j}{\sqrt{\frac{2\ln(p_j)}{n}}} \; = \; \frac{\sqrt{\frac{2\ln(p_j - s_j)}{n}}}{\sqrt{\frac{2\ln(p_j)}{n}}} \frac{2\tau_j}{\sqrt{\frac{2\ln(p_j - s_j)}{n}}} \; = \; \sqrt{\frac{\ln(p_j - s_j)}{\ln(p_j)}} \frac{2\tau_j}{\sqrt{\frac{2\ln(p_j - s_j)}{n}}}.$$

Since by assumption τ_j satisfies $\lim_{n\to\infty} \sqrt{n}\tau_j/\sqrt{2\ln(p_j - s_j)} \; \geq \; 1$, the second term on the right converges to something ≥ 2, and it is enough to show that the first term converges to something $> 1/2$. Using that $\frac{s_j}{p_j} < c$, we get

$$\sqrt{\frac{\ln(p_j - s_j)}{\ln(p_j)}} = \sqrt{\frac{\ln(p_j) + \ln(1 - \frac{s_j}{p_j})}{\ln(p_j)}} \; \geq \; \sqrt{\frac{\ln(p_j) + \ln(1 - c)}{\ln(p_j)}} \; \to \; 1,$$

which concludes the proof.

10.4.7 Proof of Lemma ??

We have

$$\frac{\beta^*_{\min,j}}{\sqrt{\frac{2\ln(p_j - s_j)}{n}} + \sqrt{\frac{\pi}{2}\frac{\ln(s_j)}{n}}} = \frac{\beta^*_{\min,j} - \tau_j + \tau_j}{\sqrt{\frac{2\ln(p_j - s_j)}{n}} + \sqrt{\frac{\pi}{2}\frac{\ln(s_j)}{n}}}$$

and (**??**) implies that we have either $\lim_{n\to\infty} \sqrt{n}\tau_j/\sqrt{2\ln(p_j - s_j)} \; < \; 1$ or else $\lim_{n\to\infty} \sqrt{n}(\beta^*_{\min,j} - \tau_j)/\sqrt{(\pi/2)\ln(s_j)} < 1$.

If $\lim_{n\to\infty}\sqrt{n}\tau_j/\sqrt{2\ln(p_j-s_j)} < 1$ by Proposition **??** (ii), $\lim_{n\to\infty} P(\hat{S}^{ei}\subseteq S) < 1$ and recovery is not possible asymptotically. Suppose now that $\lim_{n\to\infty}\sqrt{n}\tau_j/\sqrt{2\ln(p_j-s_j)} \geq 1$, then it holds that $\lim_{n\to\infty}\sqrt{n}(\beta^*_{\min,j}-\tau_j)\sqrt{\pi\ln(s_j)/2} < 1$. By Proposition **??** (ii), $\lim_{n\to\infty} P(\hat{S}^{ei}\supseteq S) < 1$ and recovery is not possible asymptotically.

10.4.8 Proof of Theorem ??

We start by showing the bound in (**??**). By the union bound,

$$P(S\neq \hat{S}^{ei}) \;\leq\; P(\hat{S}^{ei}\not\supseteq S) + P(\hat{S}^{ei}\not\subseteq S) \tag{S.111}$$

By the union bound and by Lemma 10.1 (i), for n large enough,

$$P\left(\hat{S}^{ei}\not\subseteq S\right) \;\leq\; \sum_{j=1}^{b} P\Big(\max_{i\in B_j\setminus S_j}|y_i/\sqrt{n}|>\tau_j\Big) \;\leq\; \sum_{j=1}^{b}\frac{e^{-\frac{n}{2}\left(\tau_j^2-\frac{2\ln(p_j-s_j)}{n}\right)}}{\sqrt{\pi\ln(p_j-s_j)}}. \tag{S.112}$$

where the assumption of Lemma 10.1 (i) is met because Assumption A4 is assumed to hold. Moreover, by the union bound, $P\Big(\hat{S}^{ei}\not\supseteq S\Big)\leq\sum_{j=1}^{b}P\left(\min_{i\in S_j}|y_i/\sqrt{n}|\leq\tau_j\right)$. By Assumption A5, $\beta_{\min,j}>\tau_j$ and by Lemma **??**, for each j,

$$P\left(\min_{i\in S_j}|y_i/\sqrt{n}|\leq\tau_j\right) \;\leq\; P\left(\max_{i\in S_j}|y_i/\sqrt{n}-\beta_i|\geq\beta^*_{\min,j}-\tau_j\right).$$

By Assumption A5, $\beta_{\min,j}-\tau_j\geq\sqrt{2\ln(s_j)/n}$ and by Lemma 10.1 (i)

$$P\left(\hat{S}^{ei}\not\supseteq S\right) \;\leq\; \sum_{j=1}^{b}\frac{e^{-\frac{n}{2}\left((\beta^*_{\min,j}-\tau_j)^2-\frac{2\ln(s_j)}{n}\right)}}{\sqrt{\pi\ln(s_j)}}. \tag{S.113}$$

Inputting the bounds in (S.106) and (S.110) into (S.111) gives

$$P(S\neq\hat{S}^{ei}) \;\leq\; \sum_{j=1}^{b}\frac{e^{-\frac{n}{2}\left(\tau_j^2-\frac{2\ln(p_j-s_j)}{n}\right)}}{\sqrt{\pi\ln(p_j-s_j)}}+\sum_{j=1}^{b}\frac{e^{-\frac{n}{2}\left((\beta^*_{\min,j}-\tau_j)^2-\frac{2\ln(s_j)}{n}\right)}}{\sqrt{\pi\ln(s_j)}}. \tag{S.114}$$

which shows (**??**).

We continue with the second part of the theorem and show that if (**??**) holds then τ_j^* satisfies Assumptions A4 and A5. Under (**??**), there exists $c>1$ such that $\beta^*_{\min,j}=$

$c(\sqrt{\frac{2\ln(p_j-s_j)}{n}}+\sqrt{\frac{2\ln(s_j)}{n}})$. Denote $a=\sqrt{\frac{2\ln(p_j-s_j)}{n}}$ and $b=\sqrt{\frac{2\ln(s_j)}{n}}$. We have

$$\tau_j^* \;=\; \frac{c}{2}(a+b)+\frac{(a^2-b^2)}{2c(a+b)} \;=\; \frac{c}{2}(a+b)+\frac{a-b}{2c} \;=\; \frac{c^2+1}{2c}a+\frac{c^2-1}{2c}b.$$

Since $\frac{c^2+1}{2c}>1$, $\frac{c^2-1}{2c}>0$ and $b\geq 0$, $\tau_j^*>a=\sqrt{\frac{2\ln(p_j-s_j)}{n}}$ and so Assumption A4 holds for the τ_j^*. To show that the upper bound in Assumption A5 also holds, observe that

$$\beta^*_{\min,j}-\tau_j^* \;=\; c(a+b)-\frac{c^2+1}{2c}a-\frac{c^2-1}{2c}b \;=\; \frac{c^2-1}{2c}a+\frac{c^2+1}{2c}b \;>\; b=\sqrt{\frac{2\ln(s_j)}{n}}.$$

We proceed with the proof of the bound in (**??**). Since Assumptions A4 and A5 hold for the τ_j^*'s, for all sufficiently large n, (S.114) holds for these oracle penalties. Moreover, by assumption $p_j-s_j>1$ and $s_j>1$, then $\sqrt{\pi\ln(p_j-s_j)}>1$, and $\sqrt{\pi\ln(s_j)}>1$ and we get the bound:

$$P(S\neq\hat{S}^{ei}) \;\leq\; \sum_{j=1}^{b} e^{-\frac{n}{2}\left(\tau_j^{*2}-\frac{2\ln(p_j-s_j)}{n}\right)}+\sum_{j=1}^{b} e^{-\frac{n}{2}\left((\beta^*_{\min,j}-\tau_j^*)^2-\frac{2\ln(s_j)}{n}\right)}.$$

Simple algebra shows that the τ_j^* satisfies $\tau_j^{*2}-\frac{2\ln(p_j-s_j)}{n}=(\beta^*_{\min,j}-\tau_j^*)^2-\frac{2\ln(s_j)}{n}$ for all j. It follows that

$$P(S\neq\hat{S}^{ei}) \;\leq\; 2\sum_{j=1}^{b} e^{-\frac{n}{2}\left(\tau_j^{*2}-\frac{2\ln(p_j-s_j)}{n}\right)}. \tag{S.115}$$

For convenience, denote $d=\frac{1}{n\beta^*_{\min,j}}\ln(p_j/s_j-1)$ such that $\tau_j^*=\beta^*_{\min,j}/2+d$. Then

$$e^{-\frac{n}{2}\left(\tau_j^{*2}-\frac{2\ln(p_j-s_j)}{n}\right)}=e^{-\left[\frac{n}{8}{\beta^*_{\min,j}}^2+\frac{n}{2}d^2-\frac{1}{2}(\ln(p_j-s_j)+\ln(s_j))\right]}.$$

By considering separately the two possible maxima in $\ln\max\{p_j-s_j,s_j\}$, we get that

$$\frac{e^{-\frac{n}{2}\left(\tau_j^{*2}-\frac{2\ln(p_j-s_j)}{n}\right)}}{e^{-[\frac{n}{8}{\beta^*_{\min,j}}^2-\ln\max\{p_j-s_j,s_j\}]}}=e^{-\left[\frac{n}{2}d^2+\frac{1}{2}|\ln(p_j-s_j)-\ln(s_j)|\right]}<1.$$

From (S.115), we then have

$$P(\hat{S}^{ei}\neq S) \;\leq\; 2\sum_{j=1}^{b} e^{-[\frac{n}{8}{\beta^*_{\min,j}}^2-\ln\max\{p_j-s_j,s_j\}]}.$$

which proves (**??**).

10.4.9 Proof of Corollary ??

Since $\hat{S}^{sd}$ is $\hat{S}^{ei}$ with $b = 1$, the assumptions of Lemma **??** for $\hat{S}^{sd}$ are met and $\lim_{n\to\infty} P(\hat{S}^{sd} = S) < 1$. Since Assumptions A4 and A5 hold, by Propositions **??** (i) and **??** (i), $\lim_{n\to\infty} P(\hat{S}^{ei} \subseteq S) = 1$ and $\lim_{n\to\infty} P(\hat{S}^{ei} \supseteq S) = 1$, and then $\lim_{n\to\infty} P(\hat{S}^{ei} = S) = 1$.

10.4.10 Proof of Proposition ??

By the union bound,

$$P\Big(\hat{S}^{ei} \not\supseteq S\Big) \leq \sum_{j=1}^{b} P\left(\min_{i\in S_j} |y_i/\sqrt{n}| \leq \tau_j\right).$$

By Lemma **??**, for each j,

$$P\left(\min_{i\in S_j} |y_i/\sqrt{n}| \leq \tau_j\right) \;\leq\; P\left(\max_{i\in S_j} |y_i/\sqrt{n} - \beta_i| \geq \beta^*_{\min,j} - \tau_j\right).$$

By Lemma 10.1 (i), we have;

$$P\left(\hat{S}^{ei} \not\supseteq S\right) \;\leq\; \sum_{j=1}^{b} \frac{e^{-\frac{n}{2}\left((\beta^*_{\min,j}-\tau_j)^2 - \frac{2\ln(s_j)}{n}\right)}}{\sqrt{\pi\ln(s_j)}}. \tag{S.116}$$

Under Assumptions A3bis and A5bis, the right-hand side vanishes, which proves part (i).

We now prove part (ii). By Proposition **??** (i), since A4 is assumed to hold, we also have $\lim_{n\to\infty} P\Big(\hat{S}^{ei} \subseteq S\Big) = 1$. This implies that $\lim_{n\to\infty} P\Big(\hat{S}^{ei} = S\Big) = 1$.

10.4.11 Proof of Lemma ??

Take any j, satisfying (**??**). Denote by $i_\circ \in B_j$ an entry such that $|\beta^*_{i_\circ}| = \beta^*_{\min,\,j}$. The event $\hat{S}^{ei} \supseteq S$ implies that $|y_{i_\circ}/\sqrt{n}| > \tau_j$. It follows that

$$P(\hat{S}^{ei} \supseteq S) \leq P(|y_{i_\circ}| > \sqrt{n}\tau_j). \tag{S.117}$$

Consider first the case where $\tau_j > \beta^*_{\min,j}$. Using that $|y_{i_\circ}| = |y_{i_\circ} - \sqrt{n}\beta^*_{\min,\,j} + \sqrt{n}\beta^*_{\min,\,j}| \leq |y_{i_\circ} - \sqrt{n}\beta^*_{\min,\,j}| + \sqrt{n}\beta^*_{\min,\,j}$, we get that

$$P\Big(|y_{i_\circ}| > \sqrt{n}\tau_j\Big) \leq P\Big(|y_{i_\circ} - \sqrt{n}\beta^*_{\min,\,j}| > \sqrt{n}(\tau_j - \beta^*_{\min,\,j})\Big) = 2P\Big(z > \sqrt{n}(\tau_j - \beta^*_{\min,\,j})\Big)$$

where $z \sim N(0,1)$ and the equality follows from the symmetry of the standard Gaussian distribution. Since $\tau_j > \beta^*_{\min,j}$, $P\Big(z > \sqrt{n}(\tau_j - \beta^*_{\min,\,j})\Big) < 1/2$, $P\Big(|y_{i_\circ}| > \sqrt{n}\tau_j\Big) < 1$ and $P(\hat{S}^{ei} \supseteq S) < 1$ too by (S.117).

Consider next the case where $\tau_j \leq \beta^*_{\min,j}$. By Lemma 10.1 (ii), we have

$$P(|y_{i_\circ}| > \sqrt{n}\tau_j) = P\left(z > \sqrt{n}\left(\tau_j - \beta^*_{\min,\,j}\right)\right) + P\left(z < -\sqrt{n}\left(\tau_j + \beta^*_{\min,\,j}\right)\right)$$

where $z \sim N(0,1)$. Since $\tau_j \to \infty$, $\sqrt{n}\left(\tau_j + \beta^*_{\min,\,j}\right) \to \infty$ and we have that $P(z < -\sqrt{n}\Big(\tau_j + \beta^*_{\min,\,j}\Big)) \to 0$. Further, by (**??**) we have that $\lim\limits_{n\to\infty} P\left(z > \sqrt{n}\left(\tau_j - \beta^*_{\min,\,j}\right)\right) < 1$ and hence we get $\lim\limits_{n\to\infty} P(\hat{S}^{ei} \supseteq S) < 1$, as claimed.

10.4.12 Proof of Lemma ??

First, we re-write

$$\sqrt{n}\beta^*_{\min,j} - \sqrt{2\ln(p_j - s_j)} \;=\; \sqrt{n}\beta^*_{\min,j} - \sqrt{n}\tau_j + \sqrt{2\ln(p_j - s_j)}\Big(\tfrac{\sqrt{n}\tau_j}{\sqrt{2\ln(p_j - s_j)}} - 1\Big).$$

Condition (**??**) implies that there exists $c \in \mathbb{R}^+$ such that

$$\lim_{n\to\infty} \sqrt{n}\beta^*_{\min,j} - \sqrt{n}\tau_j + \sqrt{2\ln(p_j - s_j)}\Big(\tfrac{\sqrt{n}\tau_j}{\sqrt{2\ln(p_j - s_j)}} - 1\Big) \leq c \tag{S.118}$$

Consider the case $\lim_{n\to\infty} \frac{\sqrt{n}\tau_j}{\sqrt{2\ln(p_j - s_j)}} - 1 < 0$. Then by Proposition **??** (ii), $\lim_{n\to\infty} P(\hat{S}^{ei} \subseteq S) < 1$ and $\lim_{n\to\infty} P(\hat{S}^{ei} = S) < 1$. Now consider the case $\lim_{n\to\infty} \frac{\sqrt{n}\tau_j}{\sqrt{2\ln(p_j - s_j)}} - 1 \geq 0$. It follows that, by Assumption A2 $\tau_j \to \infty$ and, by (S.118), $\lim_{n\to\infty} \sqrt{n}\beta^*_{\min,j} - \sqrt{n}\tau_j \leq c$. We can then apply Lemma **??**, and we have that $\lim_{n\to\infty} P(\hat{S}^{ei} \supseteq S) < 1$ and $\lim_{n\to\infty} P(\hat{S}^{ei} = S) < 1$.

10.4.13 Proof of Corollary ??

Observe that the conditions of Lemma **??** hold for $\hat{S}^{sd}$ ($\hat{S}^{ei}$ for $b = 1$), and then $\lim_{n\to\infty} P(\hat{S}^{sd} = S) < 1$. Since Assumptions A4 and A5bis hold, by Proposition **??**

(i) and Proposition **??**, $\lim_{n\to\infty} P(\hat{S}^{ei} \subseteq 1) = 1$ and $\lim_{n\to\infty} P(\hat{S}^{ei} \supseteq S) = 1$, and then $\lim_{n\to\infty} P(\hat{S}^{ei} = S) = 1$.